



(12) **United States Patent**
Stubbs et al.

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(45) **Date of Patent: Sep. 16, 2025**

(54) **USER INTERFACE FOR TITRATION ASSIST**

(58) **Field of Classification Search**

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CPC . A61N 1/36167; A61N 1/37247; G16H 20/40
See application file for complete search history.

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(2) Date: **Oct. 1, 2020**

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International Search Report and Written Opinion on PCT/US2019/
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(57) **ABSTRACT**

(65) **Prior Publication Data**

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Systems and methods for a user interface for titration of an
implantable neurostimulator are provided. A method of
managing titration using a management device for a patient
implanted with a neurostimulator includes receiving, from
the neurostimulator, data for the patient relating to stimula-
tion being provided to the patient by the neurostimulator
while the neurostimulator is in communication with the
management device, displaying a user interface on a display
of the management device, the user interface including at
least a portion of the data displayed within a graph, the user
interface allowing a user to modify a parameter of the
titration, receiving a selection of the parameter, based on the
selection, determining a titration schedule for the patient,
and transmitting the titration schedule to the neurostimula-
tor.

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(60) Provisional application No. 62/652,147, filed on Apr.
3, 2018.

(51) **Int. Cl.**

A61N 1/00 (2006.01)

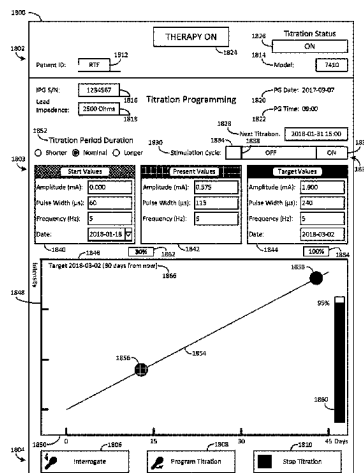
A61N 1/36 (2006.01)

(Continued)

(52) **U.S. Cl.**

CPC **A61N 1/36167** (2013.01); **A61N 1/37247**
(2013.01); **G16H 20/40** (2018.01)

20 Claims, 25 Drawing Sheets

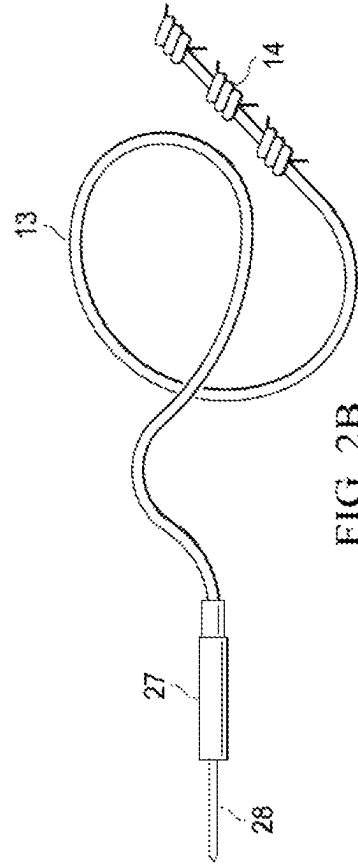
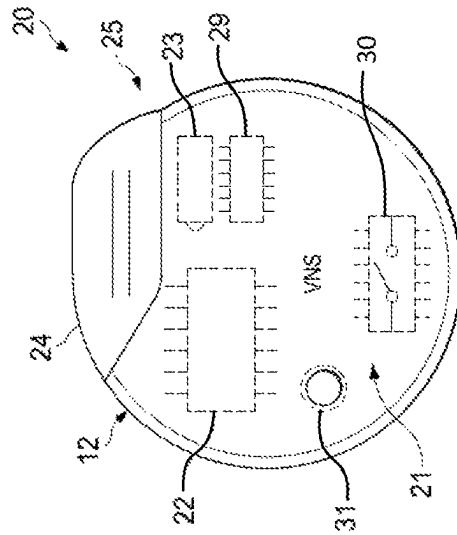
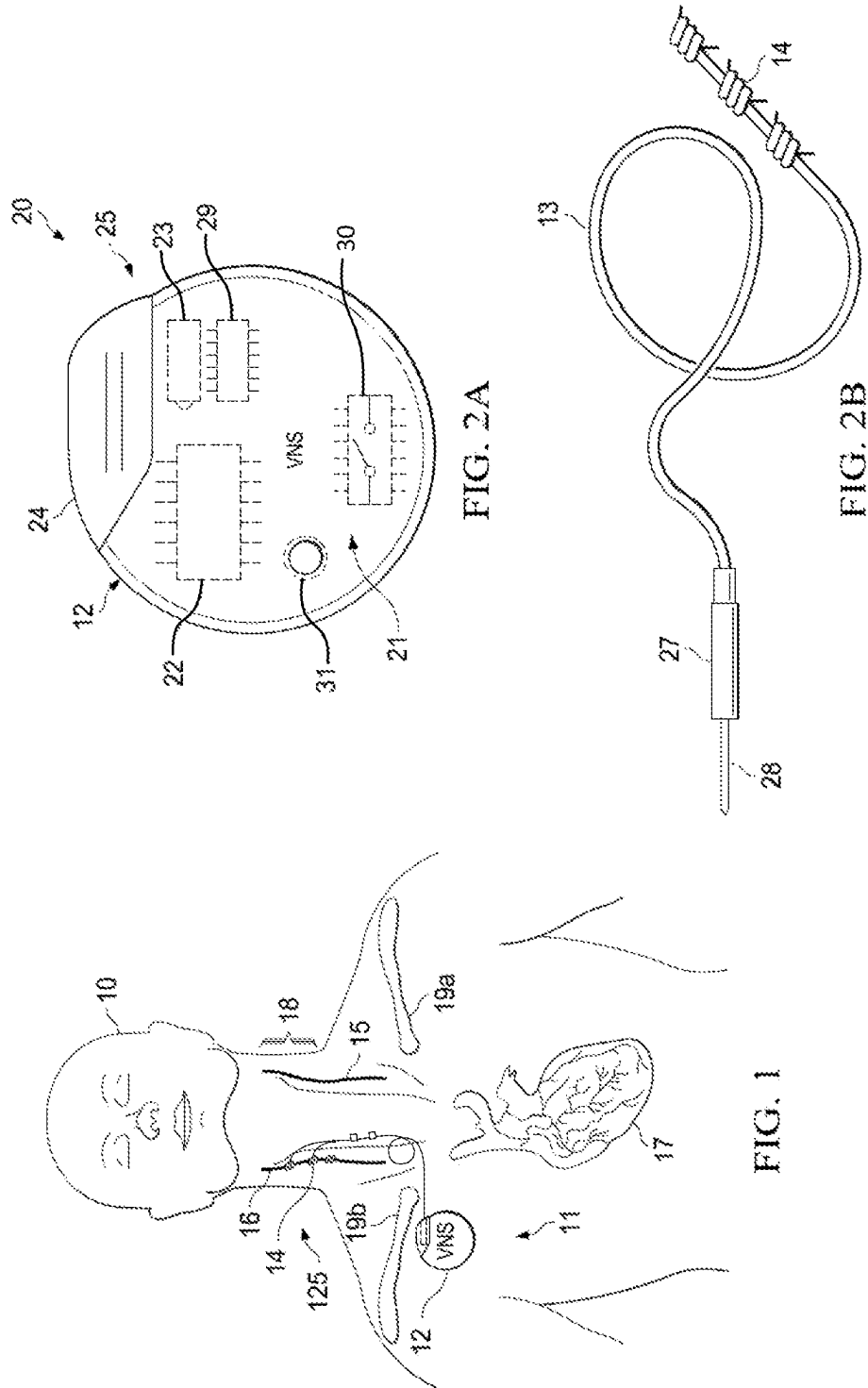


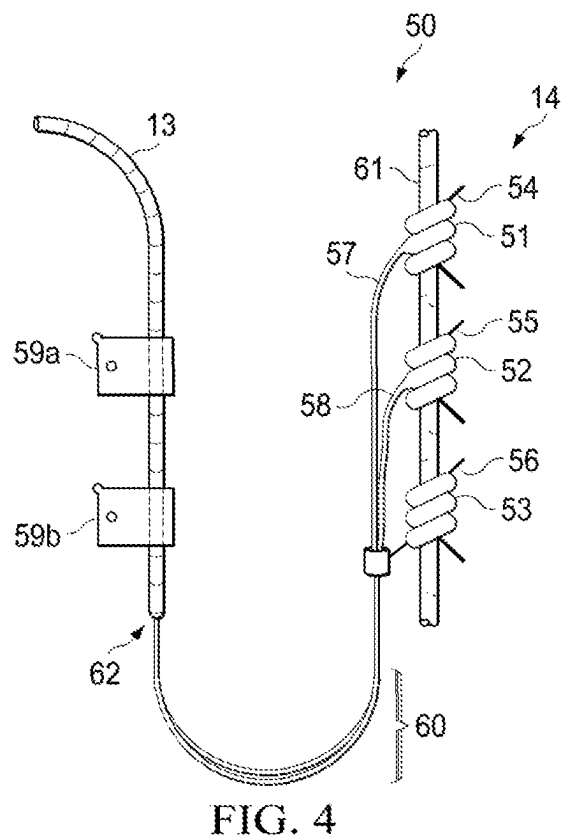
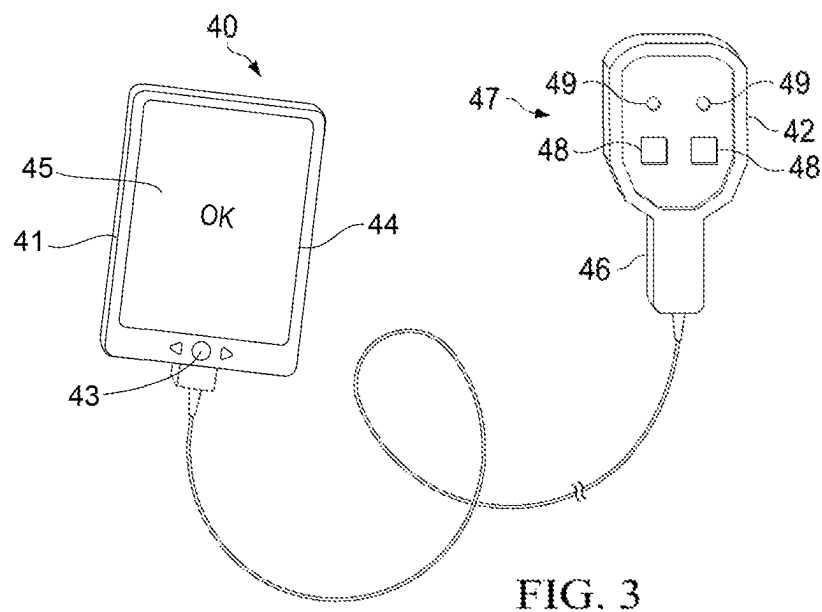
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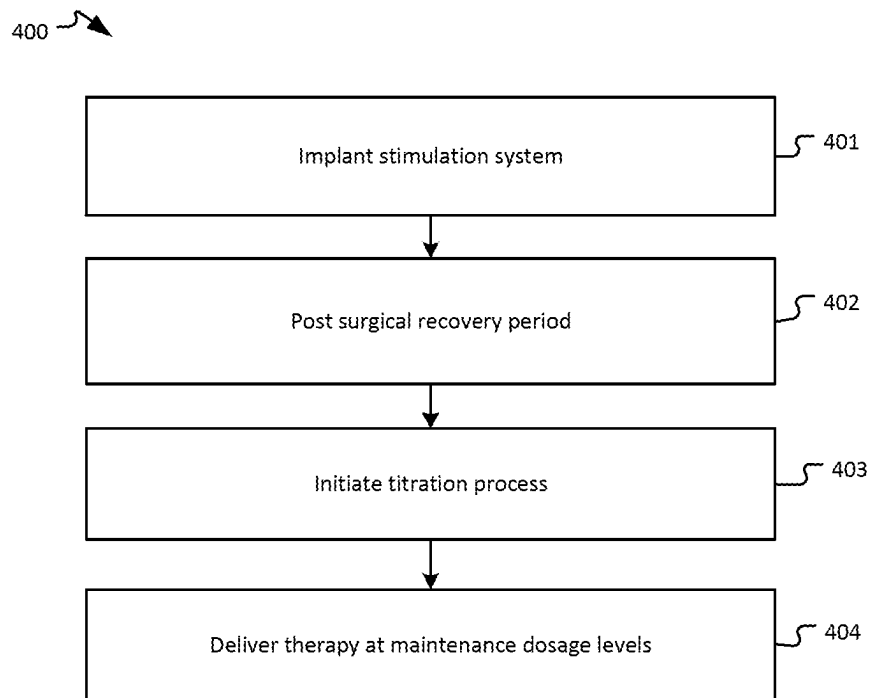


FIG. 5

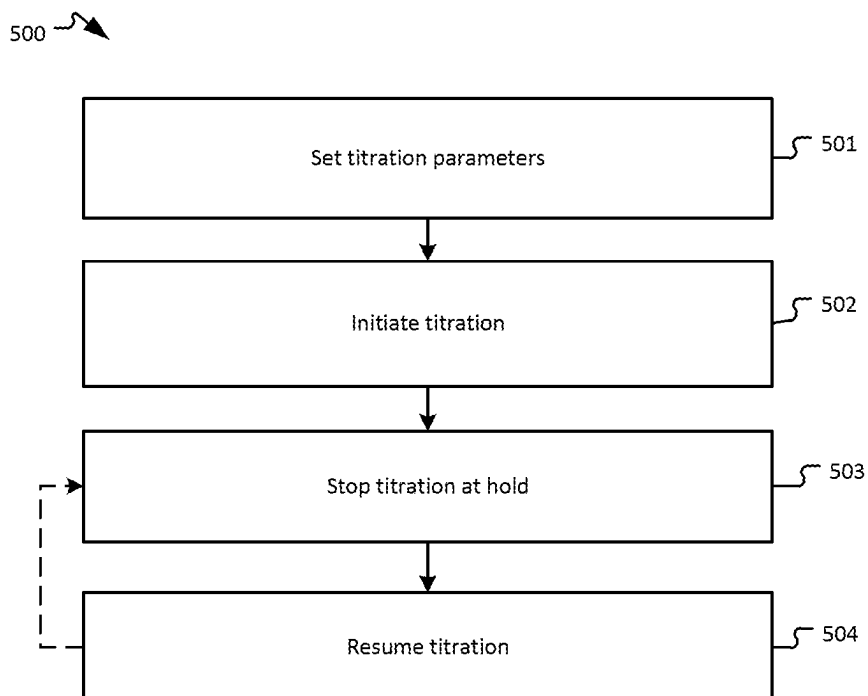


FIG. 6A

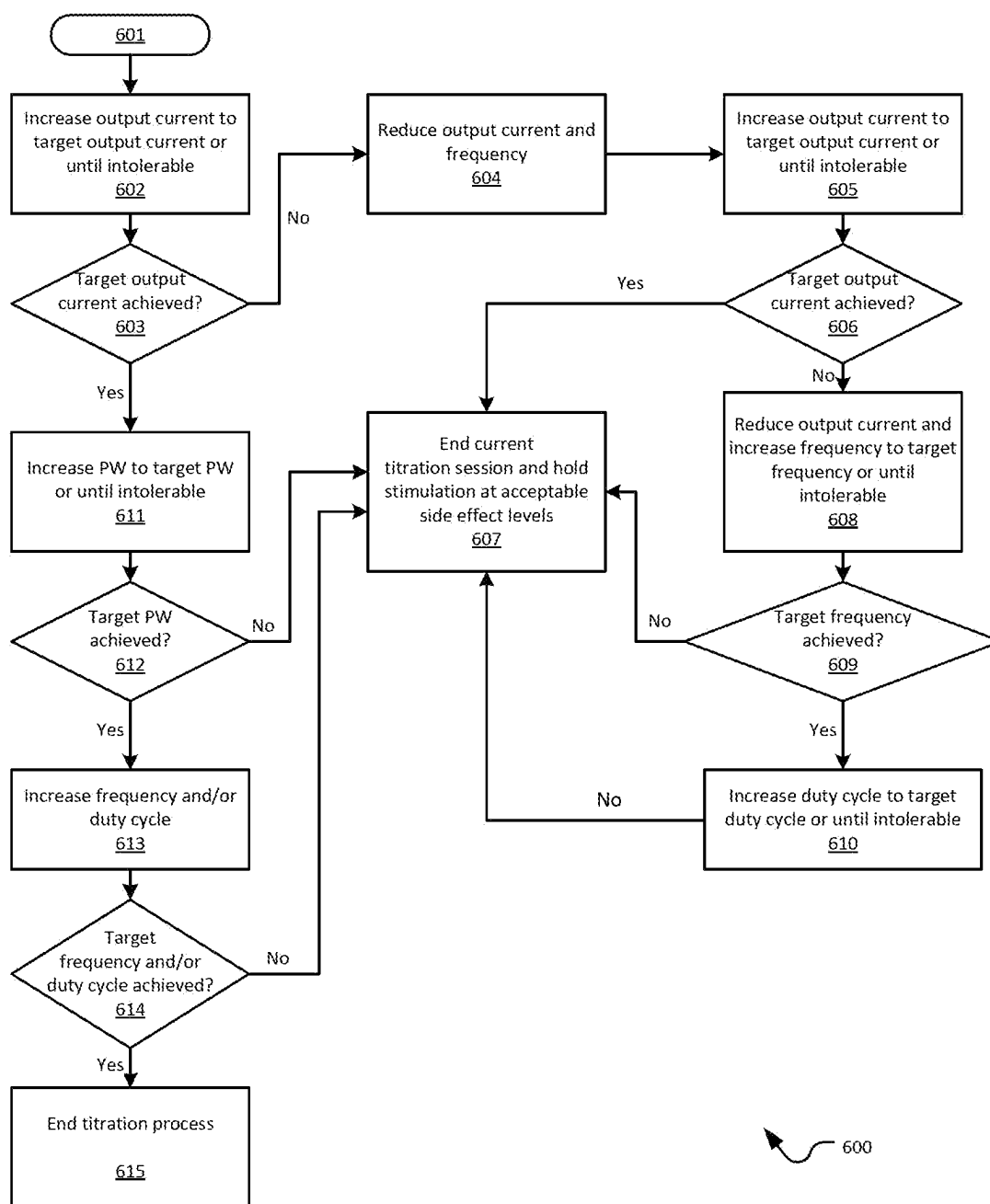


FIG. 6B

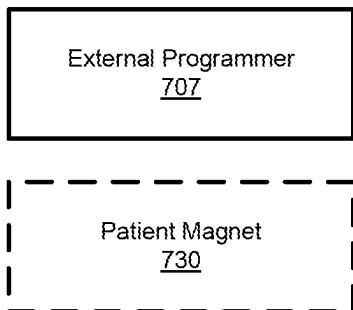
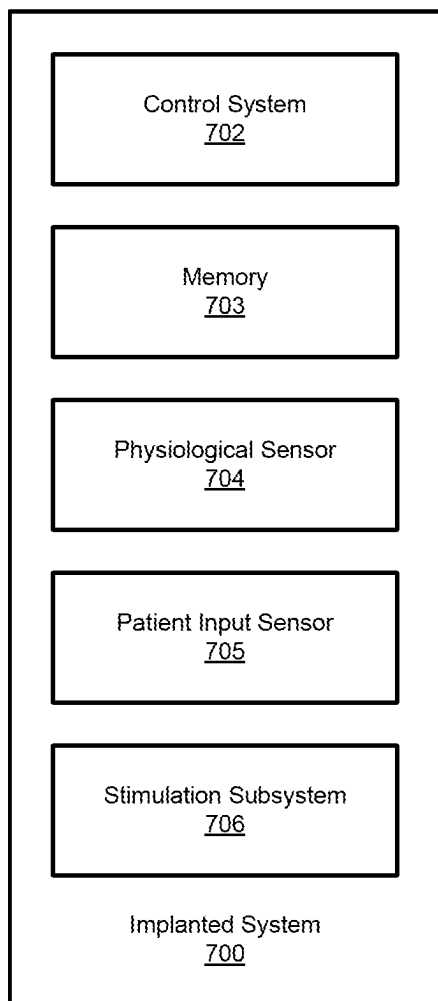


FIG. 7A

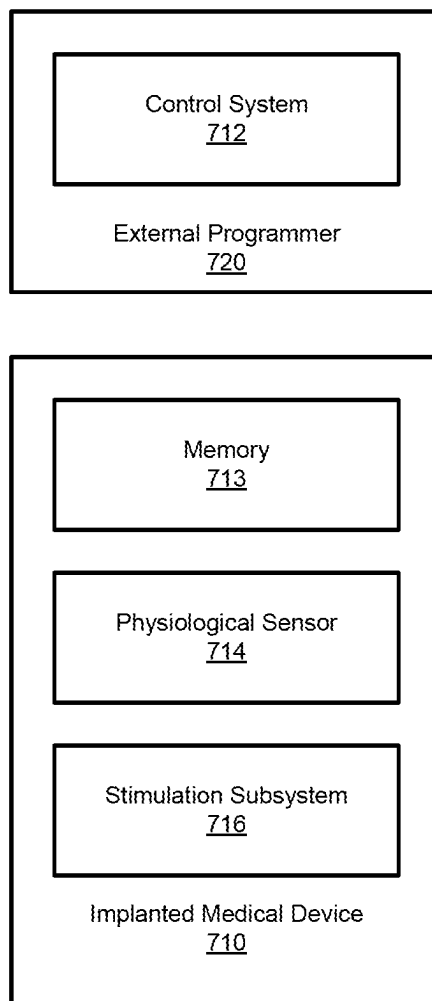


FIG. 7B

800

802

804

Patient (806)	Status (808)	Priority (810)	Patient Notes (812)
Patient 1	HOLD	1	Need to schedule follow up, 0.5 mA stimulation
Patient 2	0.25 mA	2	Parameter settings 1, light profile
Patient 3	1.5 mA	3	Parameter settings 3, aggressive profile
Patient 4	Recovery	4	Week 3 of 5 week recovery

FIG. 8

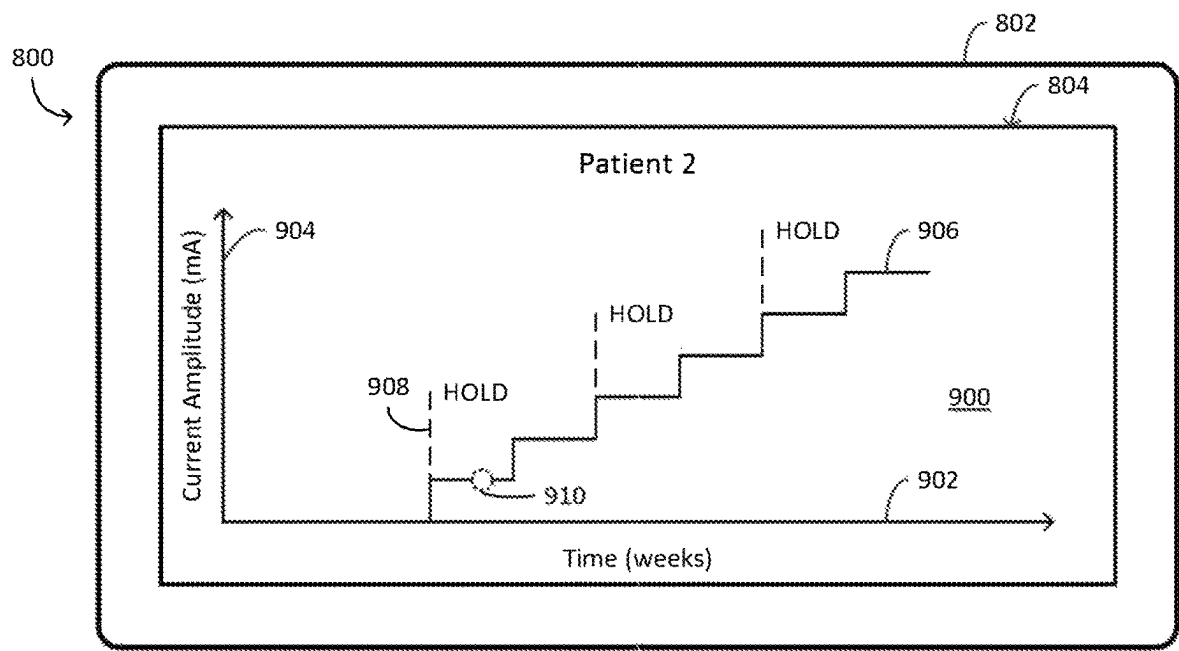


FIG. 9

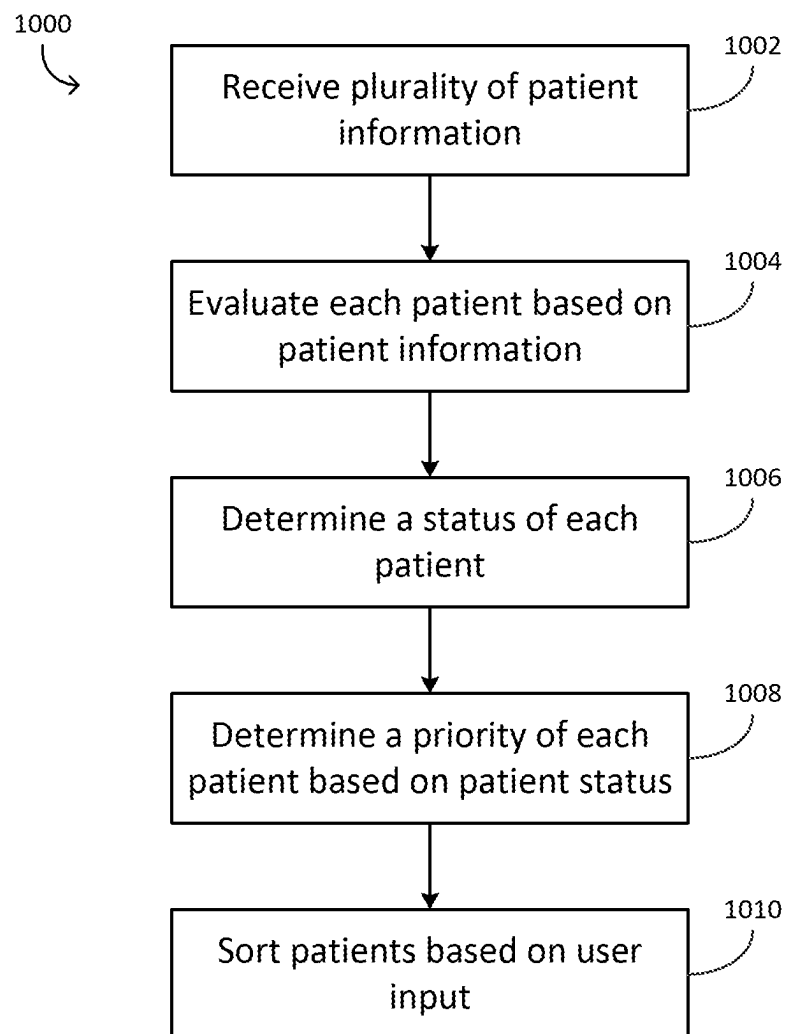


FIG. 10

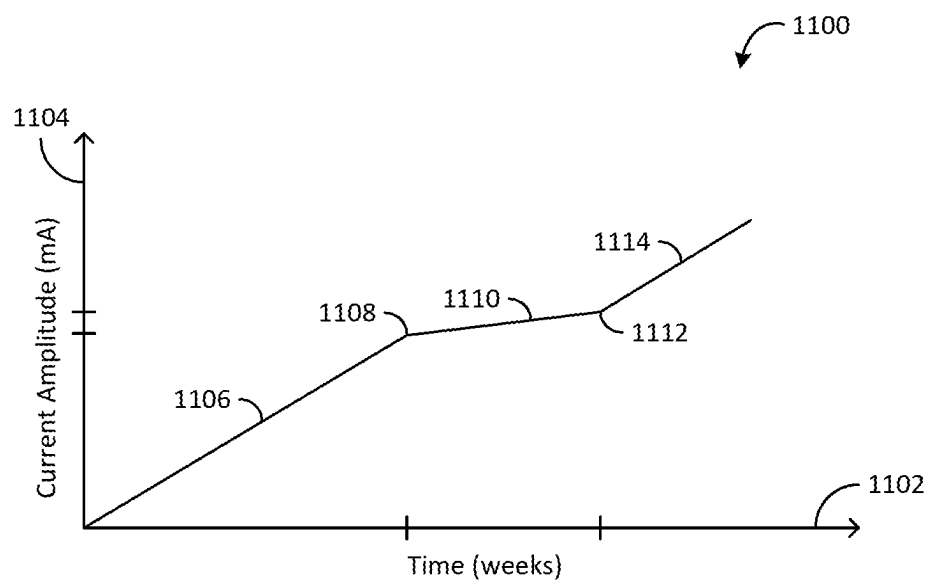


FIG. 11

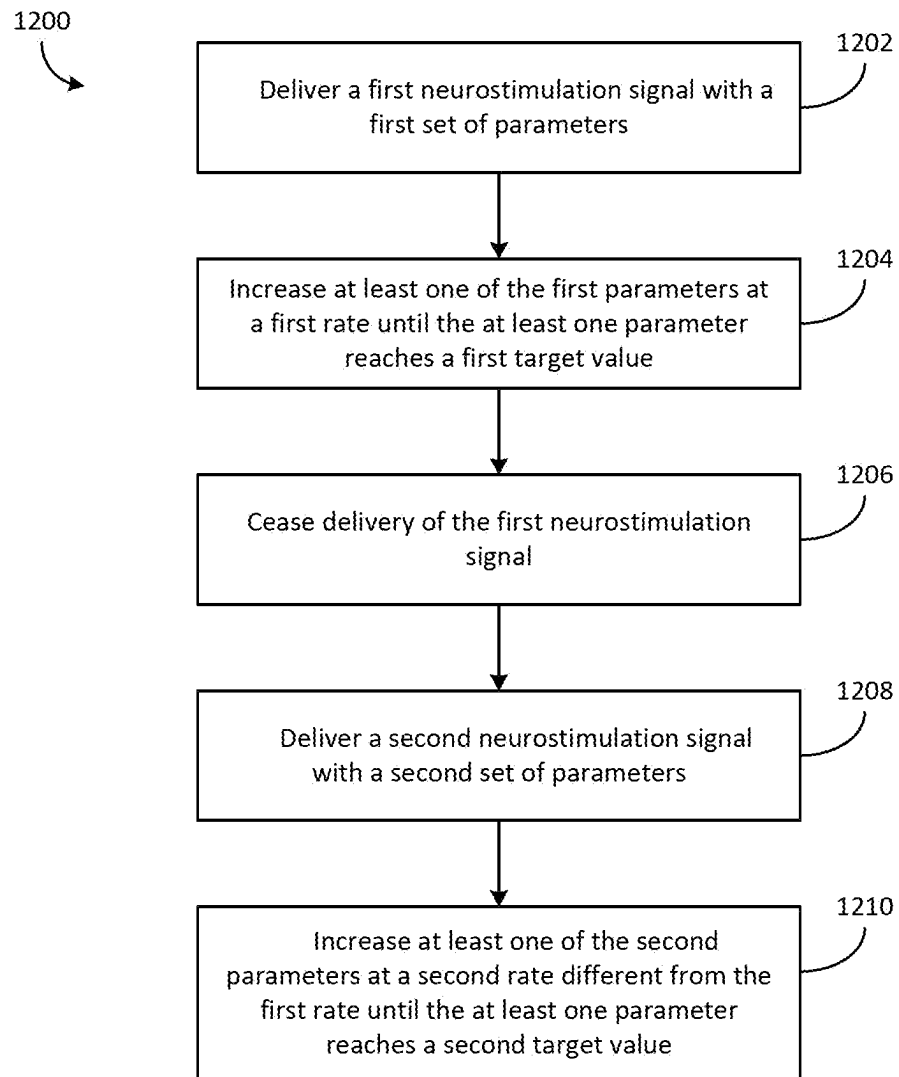


FIG. 12

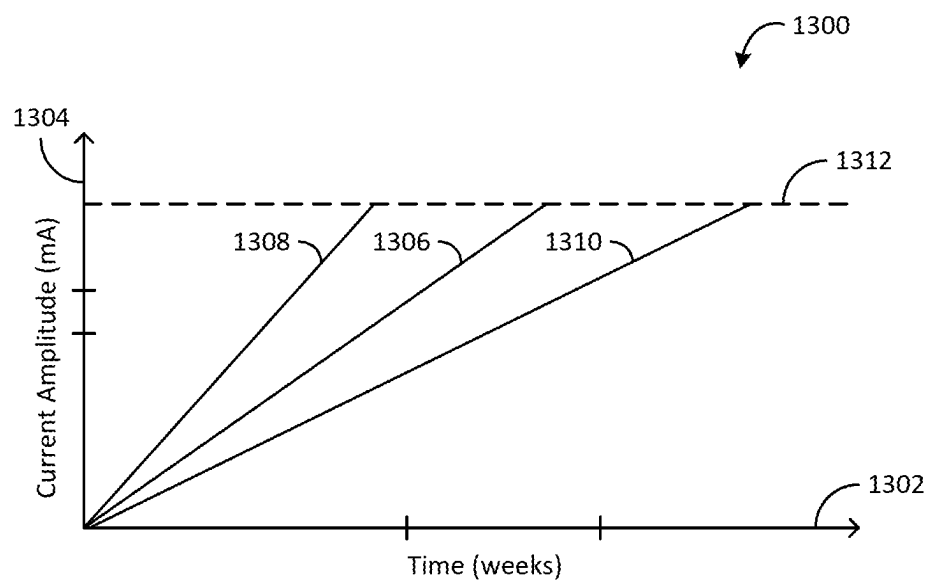


FIG. 13

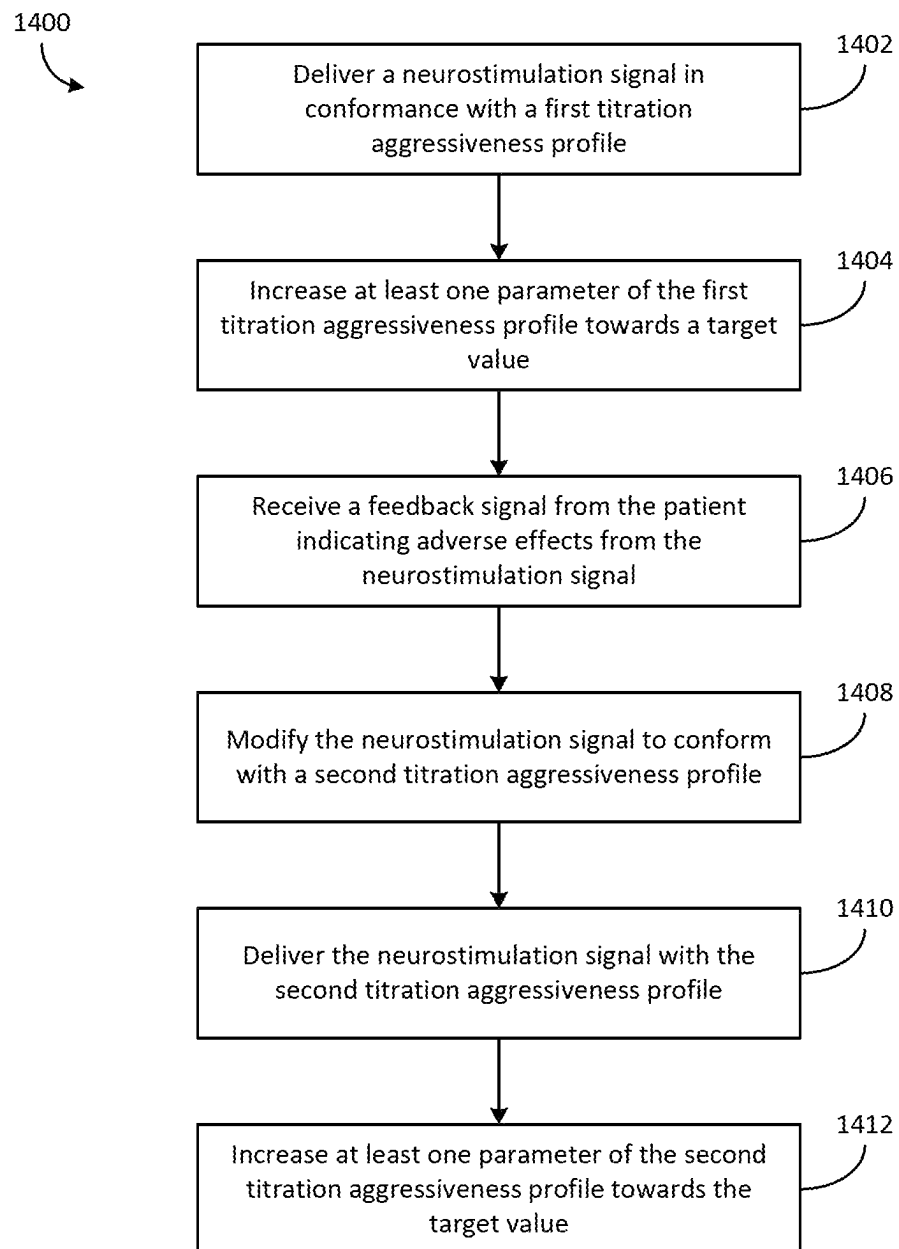


FIG. 14

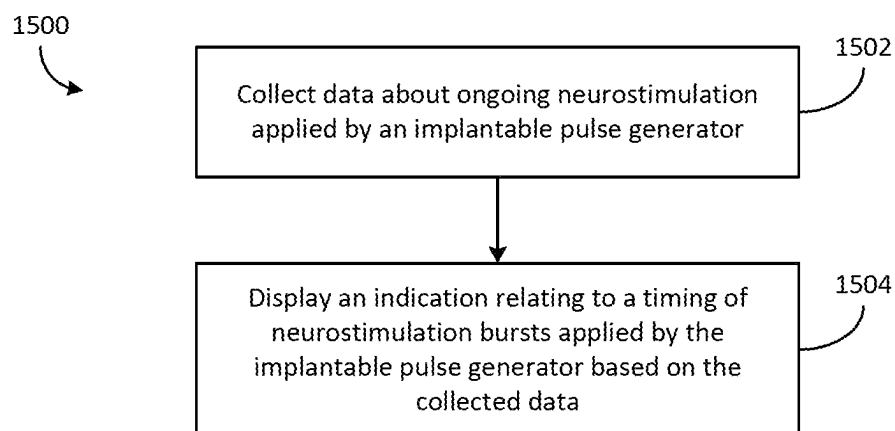


FIG. 15

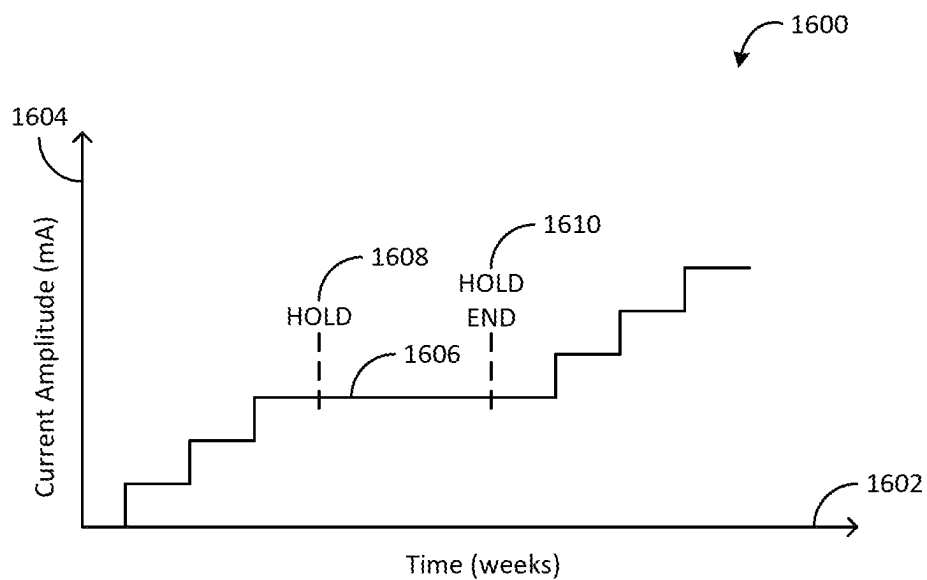


FIG. 16

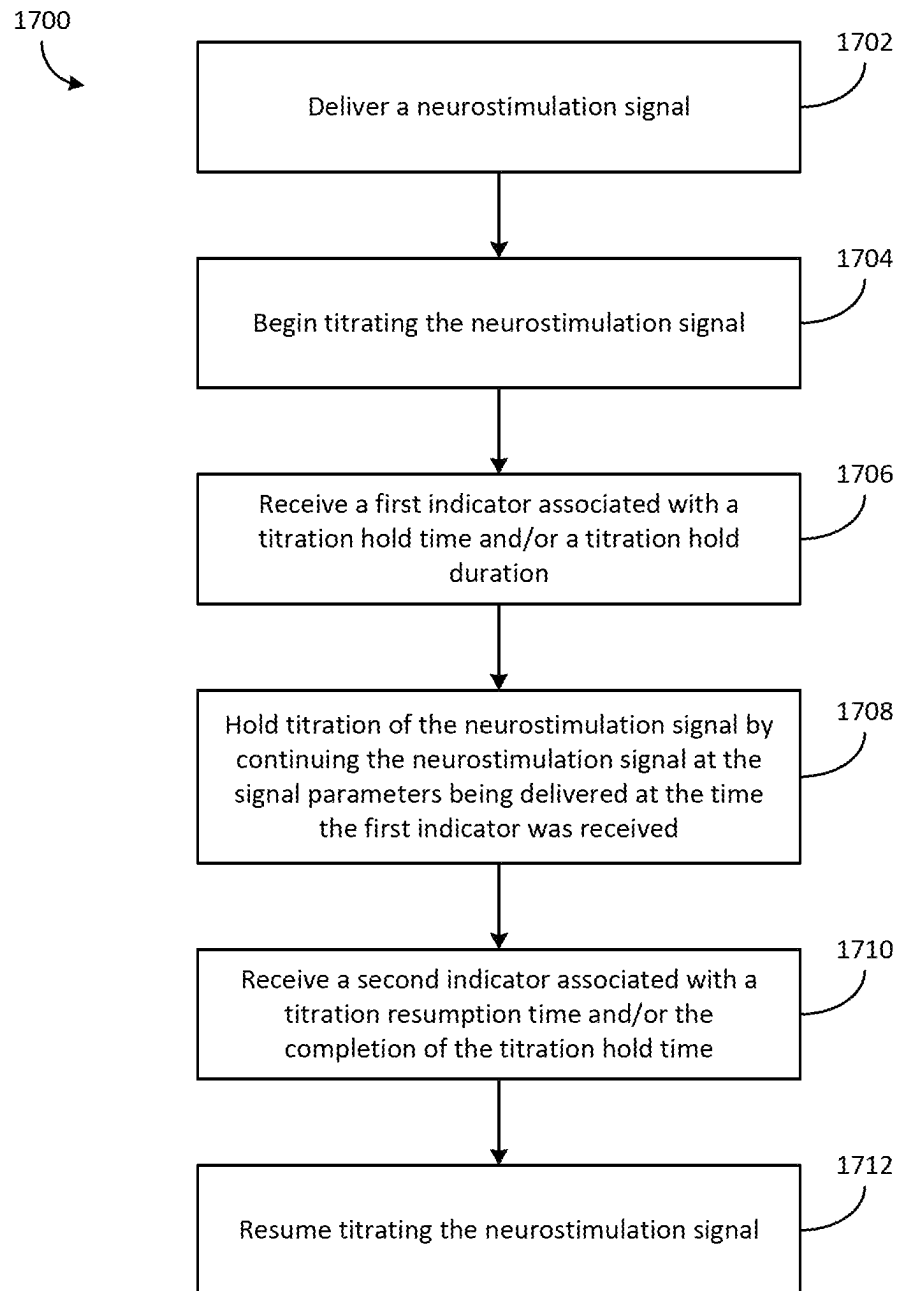


FIG. 17

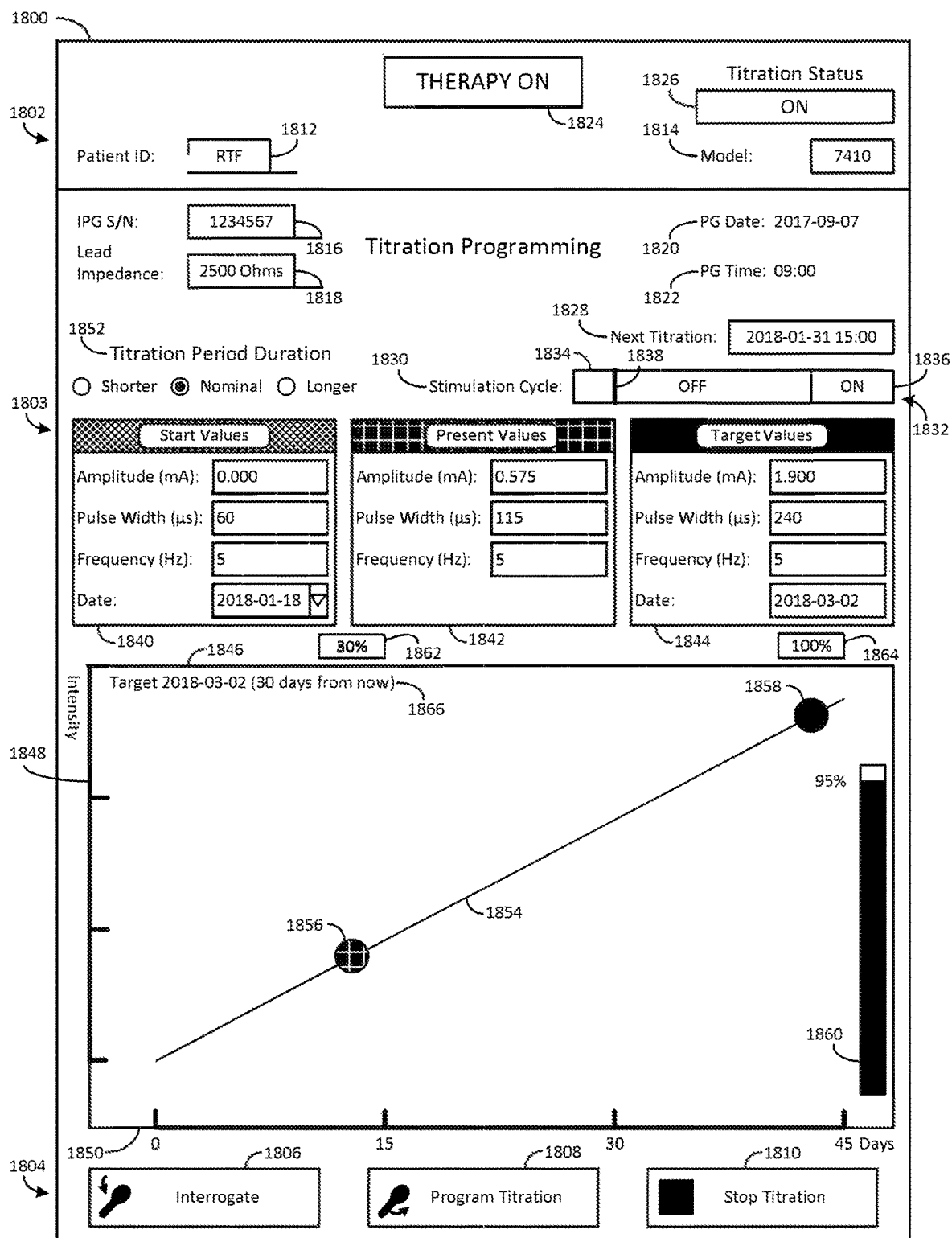


FIG. 18

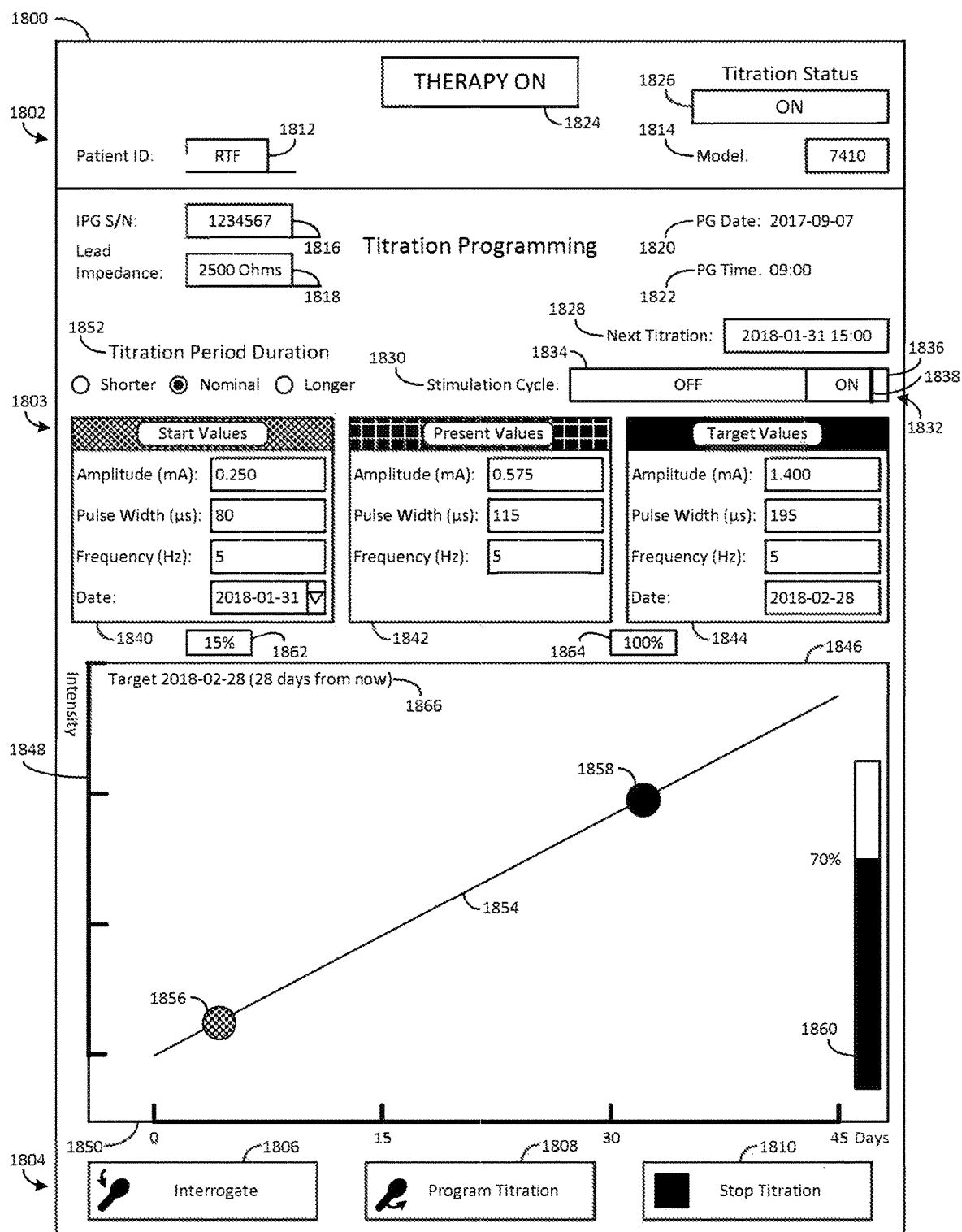


FIG. 19

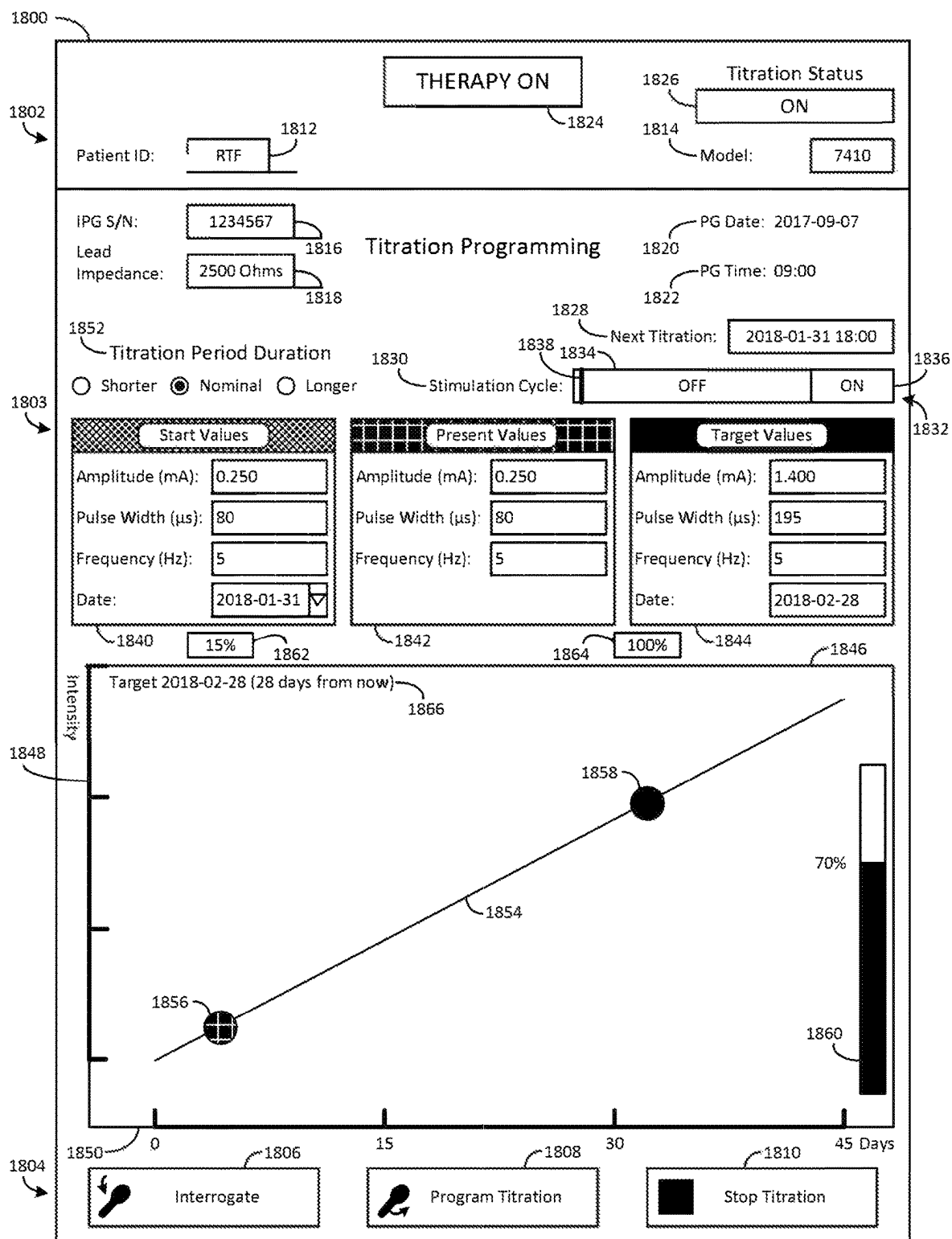


FIG. 20

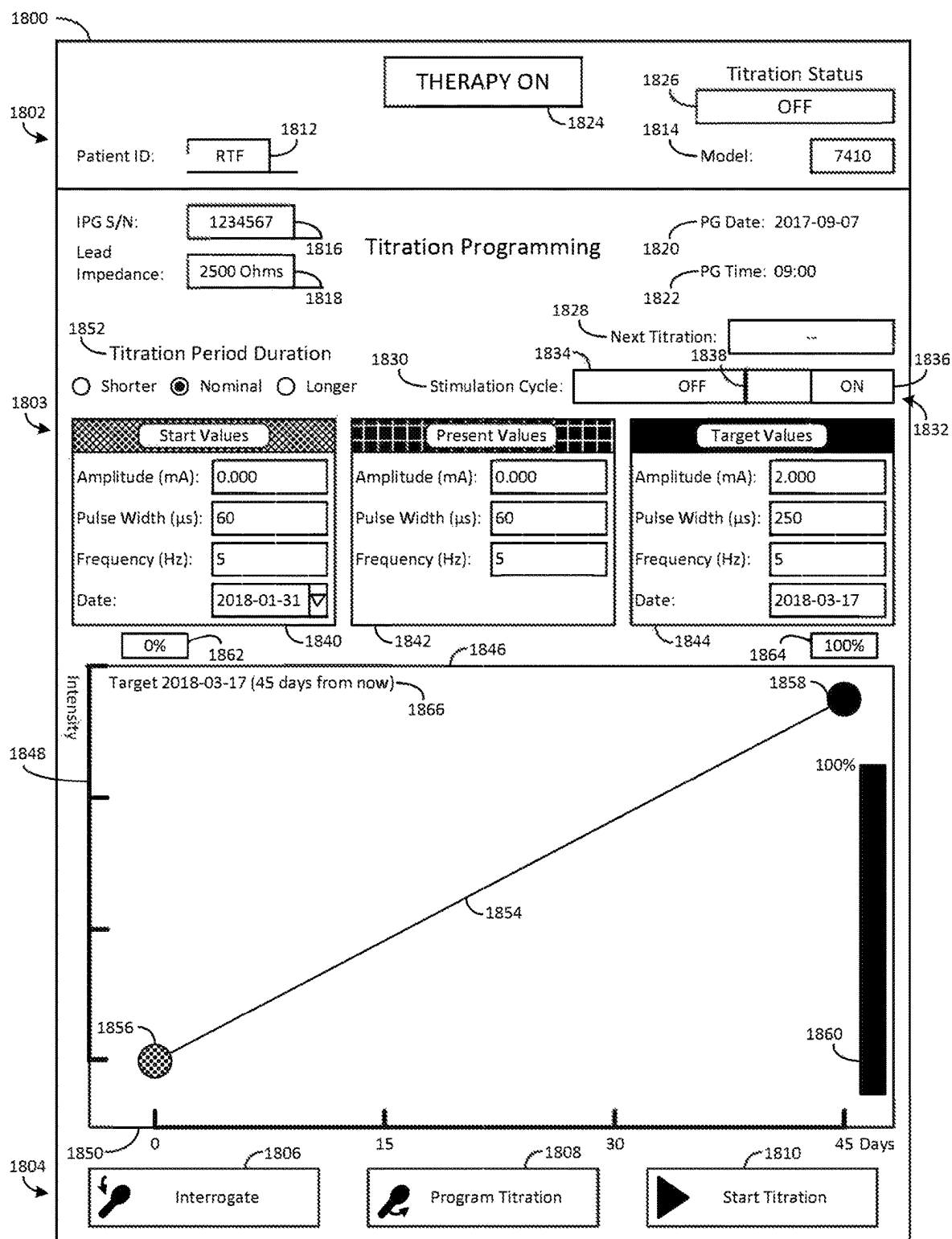


FIG. 21

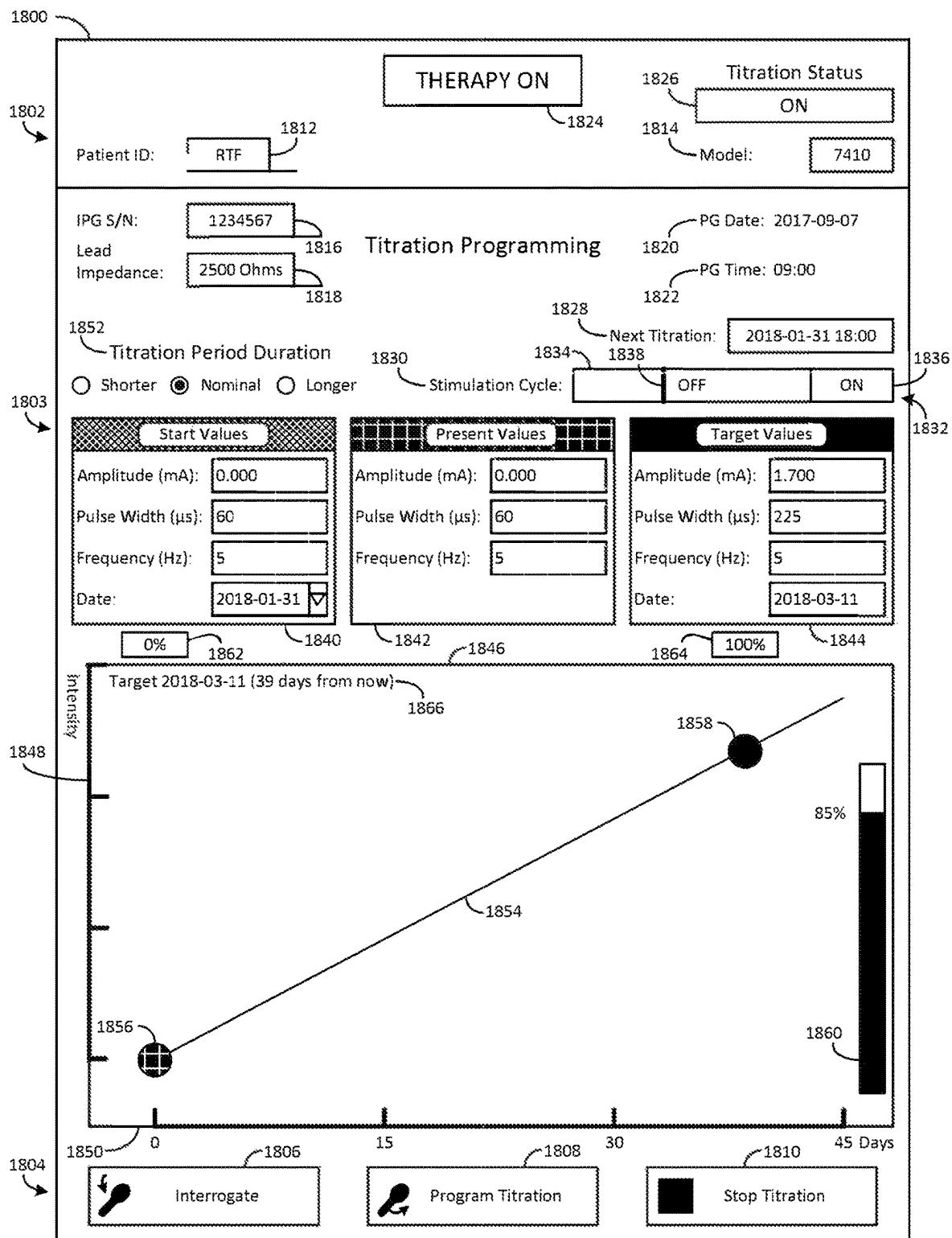


FIG. 22

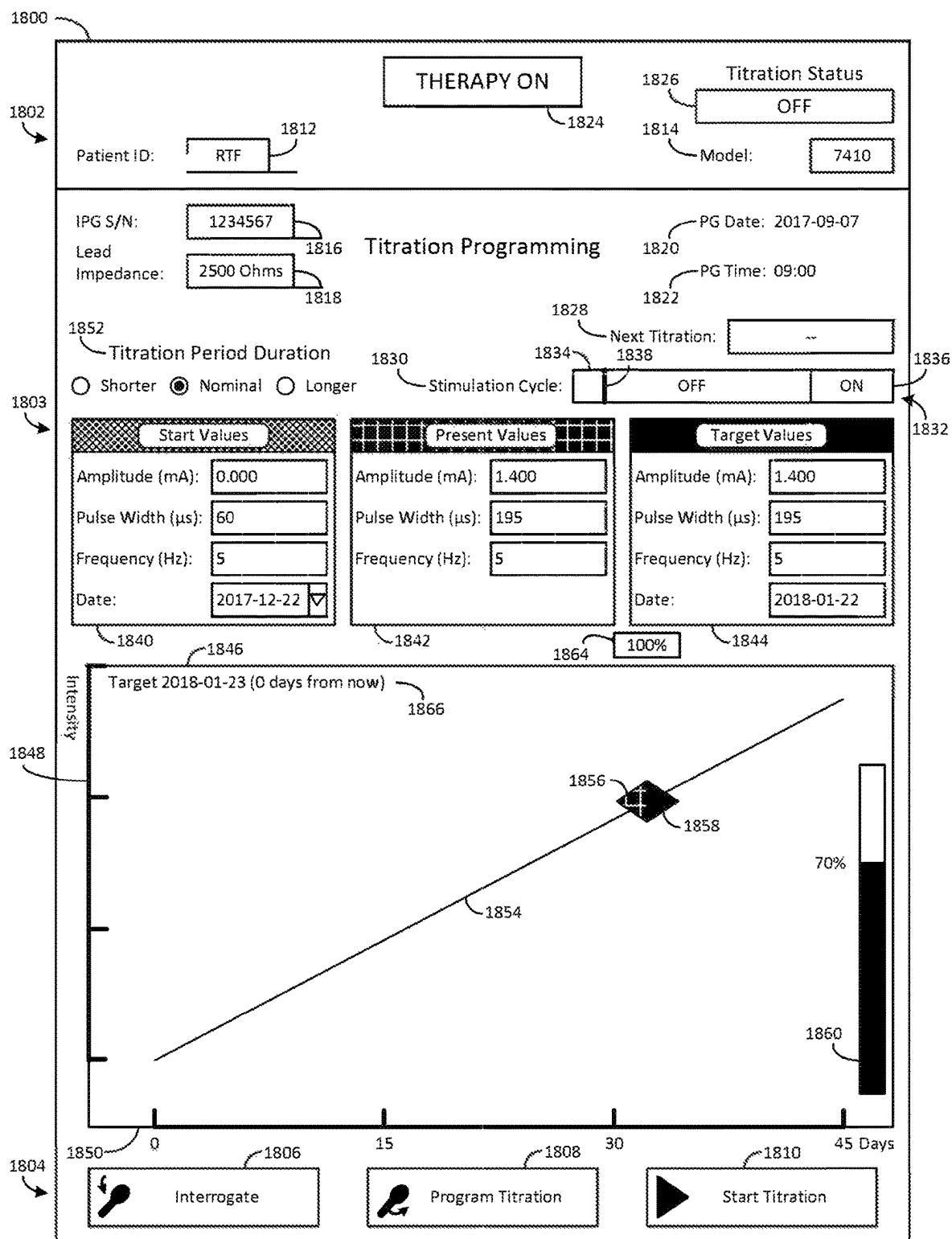


FIG. 23

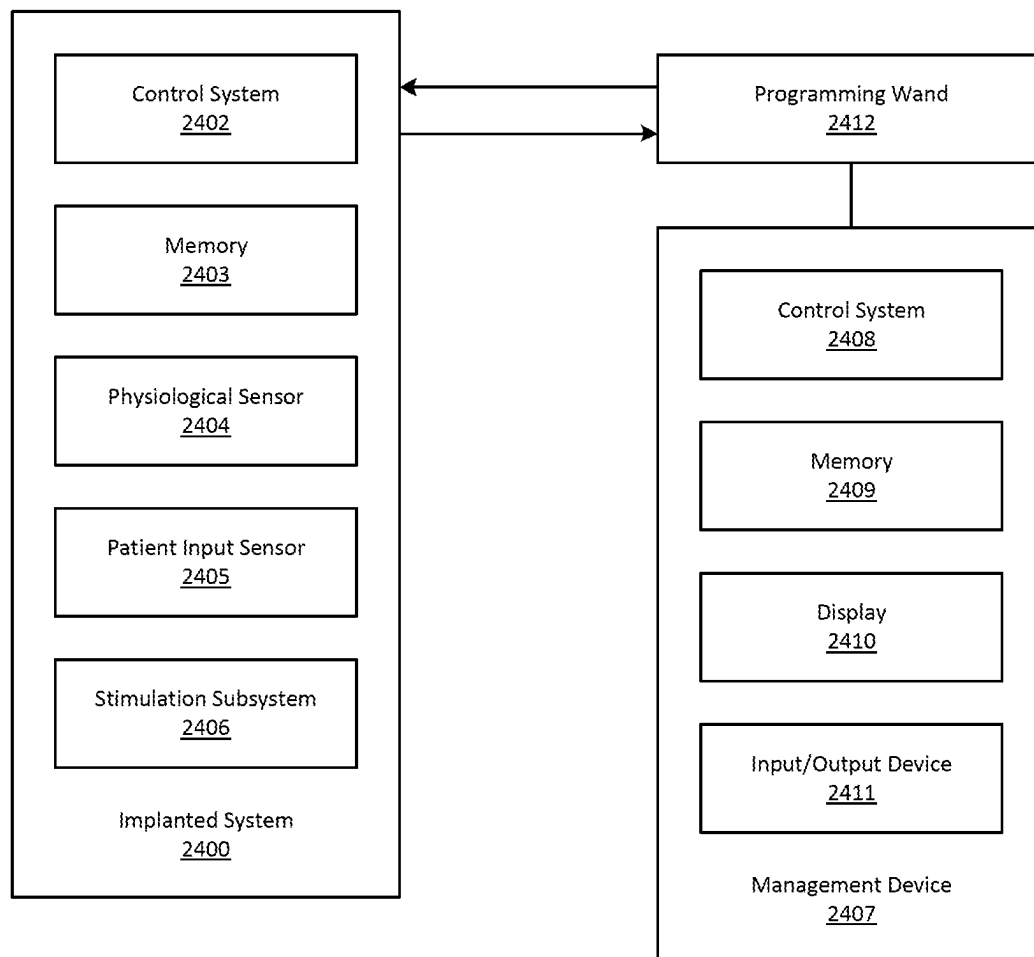


FIG. 24

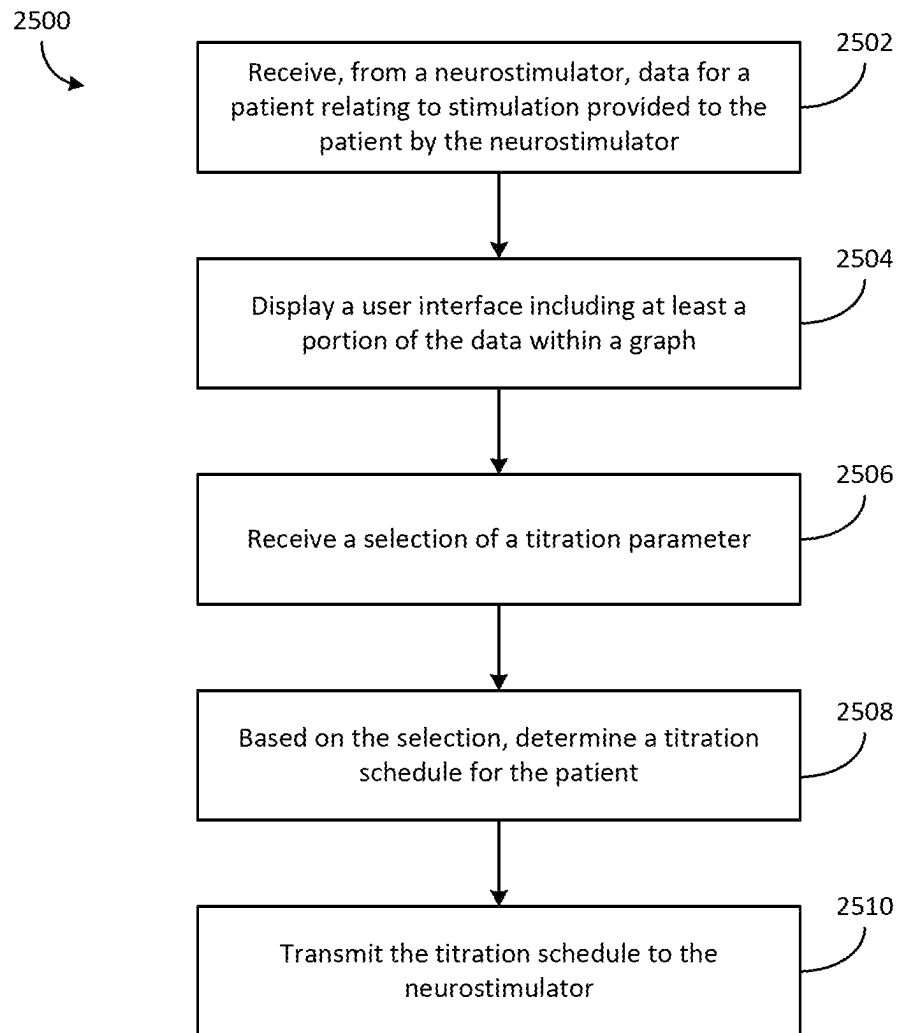


FIG. 25

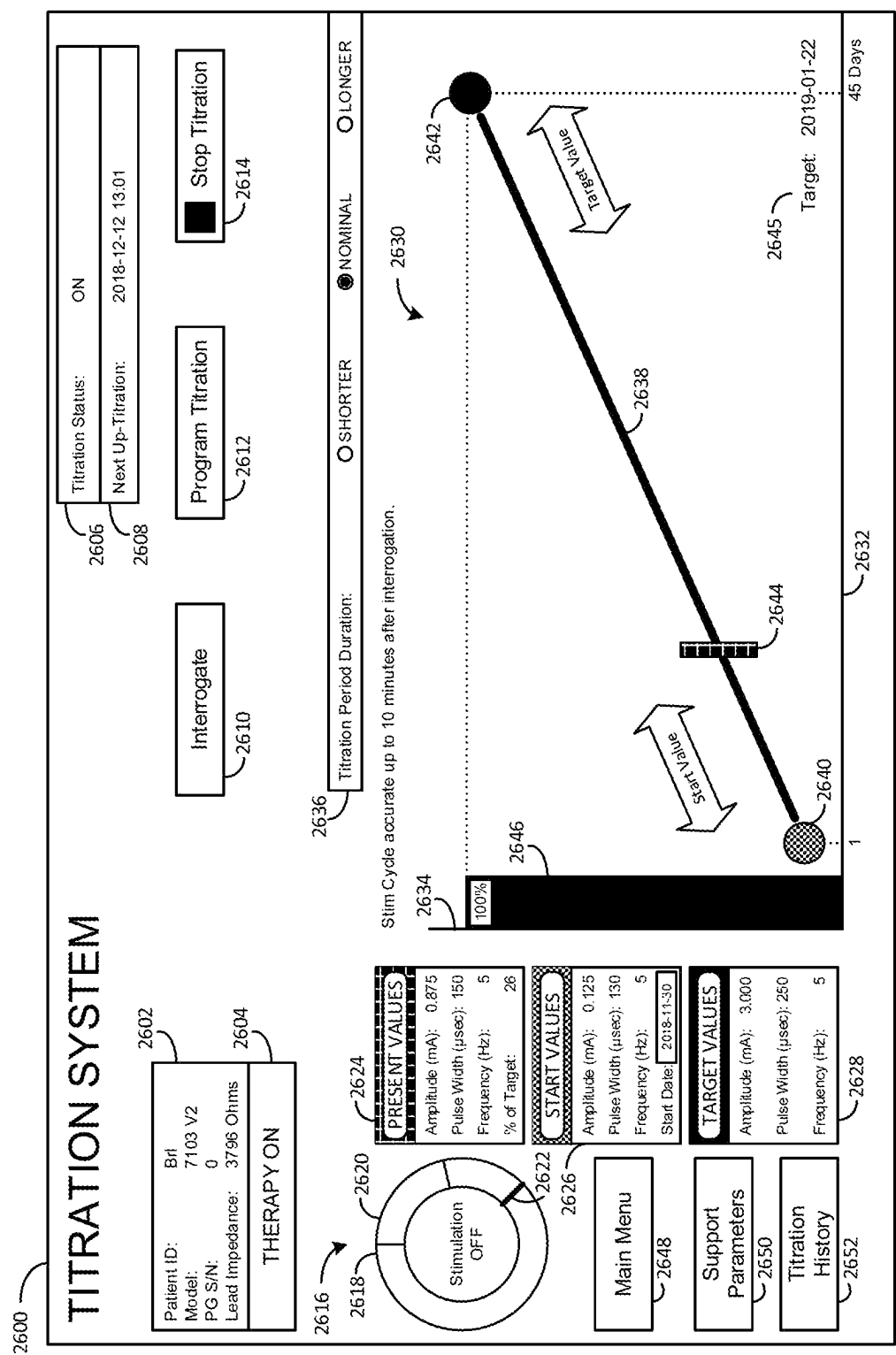


FIG. 26

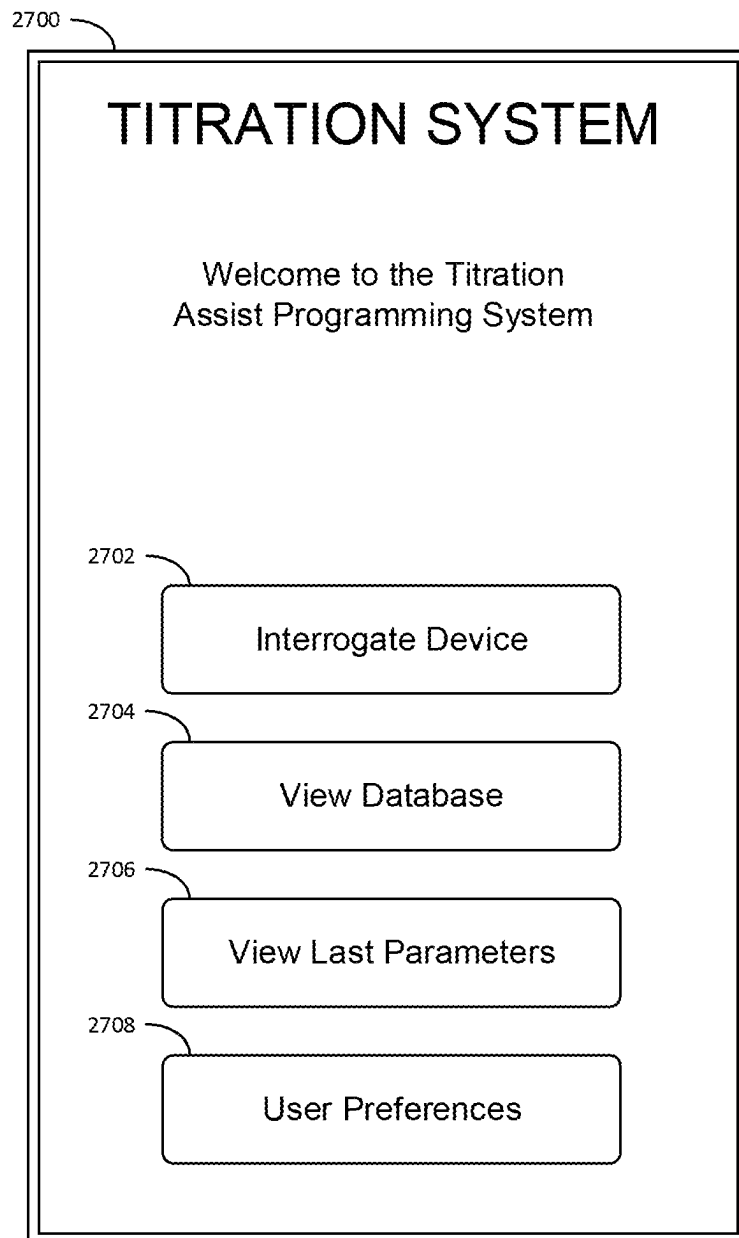


FIG. 27

2800

2802

Support Parameters

TA Signal On Time (sec):

TA Signal Off Time (sec):

TA Rollback:

Auto-Interrogate: ☐ Enable

2804

2806

2808

FIG. 28

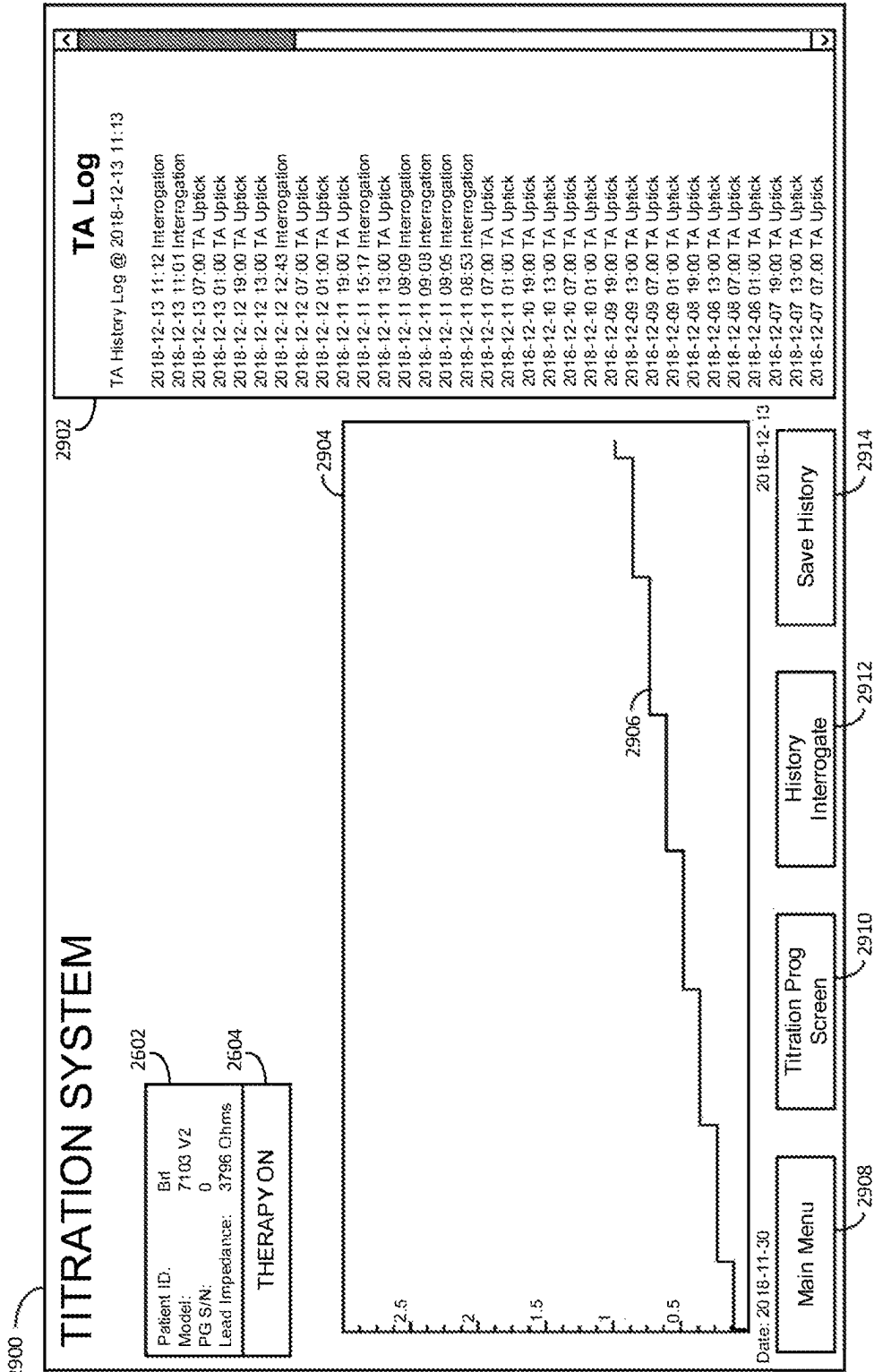


FIG. 29

USER INTERFACE FOR TITRATION ASSIST**CROSS-REFERENCE TO RELATED PATENT APPLICATION**

This application is a U.S. National Stage Application of PCT/US2019/025427, filed Apr. 2, 2019, which claims the benefit of and priority to U.S. Provisional Patent Application No. 62/652,147, entitled "USER INTERFACE FOR TITRATION ASSIST," filed Apr. 3, 2018, incorporated herein by reference in its entirety.

BACKGROUND

The present disclosure relates generally to neurostimulation and, more specifically, to improved systems and methods for managing titration of stimulation.

Chronic heart failure (CHF) and other forms of chronic cardiac dysfunction (CCD) may be related to an autonomic imbalance of the sympathetic and parasympathetic nervous systems that, if left untreated, can lead to cardiac arrhythmogenesis, progressively worsening cardiac function, and eventual patient death. CHF is pathologically characterized by an elevated neuroexcitatory state and is accompanied by physiological indications of impaired arterial and cardiopulmonary baroreflex function with reduced vagal activity.

CHF triggers compensatory activations of the sympathoadrenal (sympathetic) nervous system and the renin-angiotensin-aldosterone hormonal system, which initially helps to compensate for deteriorating heart-pumping function, yet, over time, can promote progressive left ventricular dysfunction and deleterious cardiac remodeling. Patients suffering from CHF are at increased risk of tachyarrhythmias, such as atrial fibrillation (AF), ventricular tachyarrhythmias (ventricular tachycardia (VT) and ventricular fibrillation (VF)), and atrial flutter, particularly when the underlying morbidity is a form of coronary artery disease, cardiomyopathy, mitral valve prolapse, or other valvular heart disease. Sympathoadrenal activation also significantly increases the risk and severity of tachyarrhythmias due to neuronal action of the sympathetic nerve fibers in, on, or around the heart and through the release of epinephrine (adrenaline), which can exacerbate an already-elevated heart rate.

The standard of care for managing CCD in general continues to evolve. For instance, new therapeutic approaches that employ electrical stimulation of neural structures that directly address the underlying cardiac autonomic nervous system imbalance and dysregulation have been proposed. In one form, controlled stimulation of the cervical vagus nerve beneficially modulates cardiovascular regulatory function. Vagus nerve stimulation (VNS) has been used for the clinical treatment of drug-refractory epilepsy and depression, and more recently has been proposed as a therapeutic treatment of heart conditions such as CHF.

VNS therapy commonly requires implantation of a neurostimulator, a surgical procedure requiring several weeks of recovery before the neurostimulator can be activated and a patient can start receiving VNS therapy. Even after the recovery and activation of the neurostimulator, a full therapeutic dose of VNS is not immediately delivered to the patient to avoid causing significant patient discomfort and other undesirable side effects. Instead, to allow the patient to adjust to the VNS therapy, a titration process is utilized in which the intensity is gradually increased over a period of time under a control of a physician, with the patient given time between successive increases in VNS therapy intensity

to adapt to the new intensity. As stimulation is chronically applied at each new intensity level, the patient's tolerance threshold, or tolerance zone boundary, gradually increases, allowing for an increase in intensity during subsequent titration sessions. The titration process can take significantly longer in practice because the increase in intensity is generally performed by a physician or other healthcare provider, and thus, for every step in the titration process to take place, the patient has to visit the provider's office to have the titration adjustments performed. Scheduling conflicts in the provider's office may increase the time between titration sessions, thereby extending the overall titration process, during which the patient in need of VNS does not receive the VNS at the full therapeutic intensity.

For patients receiving VNS therapy for the treatment of epilepsy, a titration process that continues over an extended period of time, such as six to twelve months, may be somewhat acceptable because the patient's health condition typically would not worsen in that period of time. However, for patients being treated for other health conditions, such as CHF, the patient's condition may degrade rapidly if left untreated. As a result, there is a much greater urgency to completing the VNS titration process when treating a patient with a time-sensitive condition, such as CHF.

Accordingly, a need remains for an approach to efficiently titrate neurostimulation therapy for treating chronic cardiac dysfunction and other conditions while minimizing side effects and related discomfort caused by the titration or by the VNS therapy itself.

SUMMARY

One embodiment relates to a method of managing titration using a management device for a patient implanted with a neurostimulator. The method includes receiving, from the neurostimulator, data for the patient relating to stimulation being provided to the patient by the neurostimulator while the neurostimulator is in communication with the management device and displaying a user interface on a display of the management device, the user interface including at least a portion of the data displayed within a graph, the user interface allowing a user to modify a parameter of the titration. The method further includes receiving a selection of the parameter, based on the selection, determining a titration schedule for the patient, and transmitting the titration schedule to the neurostimulator.

Another embodiment relates to a management device for managing titration for a patient implanted with a neurostimulator. The management device includes a display, a processor, and a memory. The memory stores instructions that, when executed by the processor, cause the management device to receive, from the neurostimulator, data for the patient relating to stimulation being provided to the patient by the neurostimulator while the neurostimulator is in communication with the management device and display, by the display, a user interface including at least a portion of the data displayed within a graph, the user interface allowing a user to modify a parameter of the titration. The instructions further cause the management device to receive a selection of the parameter, based on the selection, determine a titration schedule for the patient, and transmit the titration schedule to the neurostimulator.

Another embodiment relates to a non-transitory computer-readable medium comprising instructions executable by a processor. The instructions are executable to receive, from a neurostimulator, real-time data for a patient relating to stimulation being provided to the patient by the neuro-

stimulator and display a user interface on a display, the user interface including at least a portion of the data displayed within a graph, the user interface allowing a user to modify a parameter of the titration. The instructions are further executable to receive a selection of the parameter, based on the selection, determine a titration schedule for the patient, and transmit the titration schedule to the neurostimulator.

BRIEF DESCRIPTION OF THE DRAWINGS

Further features, characteristics, and advantages of the present disclosure will become apparent to a person of ordinary skill in the art from the following detailed description of embodiments of the present disclosure, made with reference to the drawings annexed, in which like reference characters refer to like elements.

FIG. 1 is a front anatomical diagram showing, by way of example, placement of an implantable vagus stimulation device in a male patient, in accordance with one embodiment.

FIGS. 2A and 2B are diagrams respectively showing the implantable neurostimulator and the stimulation therapy lead of FIG. 1, according to an exemplary embodiment.

FIG. 3 is a diagram showing an external programmer for use with the implantable neurostimulator of FIG. 1, according to an exemplary embodiment.

FIG. 4 is a diagram showing electrodes provided as on the stimulation therapy lead of FIG. 2 in place on a vagus nerve in situ, according to an exemplary embodiment.

FIG. 5 is a flowchart of a method for delivering vagus nerve stimulation therapy, according to an exemplary embodiment.

FIGS. 6A and 6B are flowcharts of a titration process, according to an exemplary embodiment.

FIGS. 7A and 7B are block diagrams of neurostimulation systems, according to an exemplary embodiment.

FIG. 8 is a titration assist management dashboard, according to an exemplary embodiment.

FIG. 9 is a patient titration graph of the titration assist management dashboard of FIG. 8, according to an exemplary embodiment.

FIG. 10 is a flowchart of a process for managing patients using the titration assist management dashboard, according to an exemplary embodiment.

FIG. 11 is a patient titration graph incorporating a titration dwell point, according to an exemplary embodiment.

FIG. 12 is a flowchart of a titration process incorporating a titration dwell point, according to an exemplary embodiment.

FIG. 13 is a patient titration graph illustrating different titration aggressiveness profiles, according to an exemplary embodiment.

FIG. 14 is a flowchart of a titration process including different titration aggressiveness profiles, according to an exemplary embodiment.

FIG. 15 is a flowchart of process of displaying an indication of active neurostimulation, according to an exemplary embodiment.

FIG. 16 is a patient titration graph incorporating a titration black-out period, according to an exemplary embodiment.

FIG. 17 is a flowchart of a titration process incorporating a titration black-out period, according to an exemplary embodiment.

FIG. 18 is an image of a graphical user interface that may be used in connection with a titration assist system, according to an exemplary embodiment.

FIG. 19 is an image of a graphical user interface that may be used in connection with a titration assist system, according to an exemplary embodiment.

FIG. 20 is an image of a graphical user interface that may be used in connection with a titration assist system, according to an exemplary embodiment.

FIG. 21 is an image of a graphical user interface that may be used in connection with a titration assist system, according to an exemplary embodiment.

FIG. 22 is an image of a graphical user interface that may be used in connection with a titration assist system, according to an exemplary embodiment.

FIG. 23 is an image of a graphical user interface that may be used in connection with a titration assist system, according to an exemplary embodiment.

FIG. 24 is a block diagram of a neurostimulation system, according to an exemplary embodiment.

FIG. 25 is a flowchart of a process of managing titration using a management device for a patient implanted with a neurostimulator, according to an exemplary embodiment.

FIG. 26 is an image of a graphical user interface that may be used in connection with a titration assist system, according to an exemplary embodiment.

FIG. 27 is an image of a graphical user interface that may be used in connection with a titration assist system, according to an exemplary embodiment.

FIG. 28 is an image of a graphical user interface that may be used in connection with a titration assist system, according to an exemplary embodiment.

FIG. 29 is an image of a graphical user interface that may be used in connection with a titration assist system, according to an exemplary embodiment.

DETAILED DESCRIPTION

Various aspects of the disclosure will now be described with regard to certain examples and embodiments, which are intended to illustrate but not to limit the disclosure. Nothing in this disclosure is intended to imply that any particular feature or characteristic of the disclosed embodiments is essential. The scope of protection is defined by the claims that follow this description and not by any particular embodiment described herein. Before turning to the figures, which illustrate example embodiments in detail, it should be understood that the application is not limited to the details or methodology set forth in the description or illustrated in the figures. It should also be understood that the terminology is for the purpose of description only and should not be regarded as limiting.

CHF and other cardiovascular diseases cause derangement of autonomic control of the cardiovascular system, favoring increased sympathetic and decreased parasympathetic central outflow. These changes are accompanied by elevation of basal heart rate arising from chronic sympathetic hyperactivation along the neurocardiac axis.

The vagus nerve is a diverse nerve trunk that contains both sympathetic and parasympathetic fibers and both afferent and efferent fibers. These fibers have different diameters and myelination and subsequently have different activation thresholds. This results in a graded response as intensity is increased. Low intensity stimulation results in a progressively greater tachycardia, which then diminishes and is replaced with a progressively greater bradycardia response as intensity is further increased. Peripheral neurostimulation therapies that target the fluctuations of the autonomic nervous system have been shown to improve clinical outcomes in some patients. Specifically, autonomic regulation therapy

results in simultaneous creation and propagation of efferent and afferent action potentials within nerve fibers comprising the cervical vagus nerve. The therapy directly improves autonomic balance by engaging both medullary and cardiovascular reflex control components of the autonomic nervous system. Upon stimulation of the cervical vagus nerve, action potentials propagate away from the stimulation site in two directions, efferently toward the heart and afferently toward the brain. Efferent action potentials influence the intrinsic cardiac nervous system and the heart and other organ systems, while afferent action potentials influence central elements of the nervous system.

An implantable vagus nerve stimulator, such as used to treat drug-refractory epilepsy and depression, can be adapted for use in managing chronic cardiac dysfunction (CCD) through therapeutic bi-directional vagus nerve stimulation. FIG. 1 is a front anatomical diagram showing, by way of example, placement of an implantable medical device (e.g., a vagus nerve stimulation (VNS) system 11, as shown in FIG. 1) in a male patient 10, according to an exemplary embodiment. The VNS provided through the stimulation system 11 operates under several mechanisms of action. These mechanisms include increasing parasympathetic outflow and inhibiting sympathetic effects by inhibiting norepinephrine release and adrenergic receptor activation. More importantly, VNS triggers the release of the endogenous neurotransmitter acetylcholine and other peptidergic substances into the synaptic cleft, which has several beneficial anti-arrhythmic, anti-apoptotic, and anti-inflammatory effects as well as beneficial effects at the level of the central nervous system.

The implantable vagus stimulation system 11 comprises an implantable neurostimulator or pulse generator 12 and a stimulating nerve electrode assembly 125. The neurostimulator or pulse generator may be a voltage stimulator or, more preferably, a current stimulator. The stimulating nerve electrode assembly 125, comprising at least an electrode pair, is conductively connected to the distal end of an insulated, electrically conductive lead assembly 13 and electrodes 14. The electrodes 14 may be provided in a variety of forms, such as, e.g., helical electrodes, probe electrodes, cuff electrodes, as well as other types of electrodes.

The implantable vagus stimulation system 11 can be remotely accessed following implant through an external programmer, such as the programmer 40 shown in FIG. 3 and described in further detail below. The programmer 40 can be used by healthcare professionals to check and program the neurostimulator 12 after implantation in the patient 10 and to adjust stimulation parameters during the stimulation titration process. In some embodiments, an external magnet may provide basic controls. For example, an electromagnetic controller may enable the patient 10 or healthcare professional to interact with the implanted neurostimulator 12 to exercise increased control over therapy delivery and suspension. For further example, an external programmer may communicate with the neurostimulation system 11 via other wired or wireless communication methods, such as, e.g., wireless RF transmission. Together, the implantable vagus stimulation system 11 and one or more of the external components form a VNS therapeutic delivery system.

The neurostimulator 12 is typically implanted in the patient's right or left pectoral region generally on the same side (ipsilateral) as the vagus nerve 15, 16 to be stimulated, although other neurostimulator-vagus nerve configurations, including contra-lateral and bi-lateral are possible. A vagus nerve typically comprises two branches that extend from the brain stem respectively down the left side and right side of

the patient, as seen in FIG. 1. The electrodes 14 are generally implanted on the vagus nerve 15, 16 about halfway between the clavicle 19a-b and the mastoid process. The electrodes may be implanted on either the left or right side. The lead assembly 13 and electrodes 14 are implanted by first exposing the carotid sheath and chosen branch of the vagus nerve 15, 16 through a latero-cervical incision (perpendicular to the long axis of the spine) on the ipsilateral side of the patient's neck 18. The helical electrodes 14 are then placed onto the exposed nerve sheath and tethered. A subcutaneous tunnel is formed between the respective implantation sites of the neurostimulator 12 and helical electrodes 14, through which the lead assembly 13 is guided to the neurostimulator 12 and securely connected. Additionally, in various embodiments, the neurostimulator 12 connects to the electrodes 14 as shown in FIG. 4 (e.g., at the top helices).

In one embodiment, the neural stimulation is provided as a low level maintenance dose independent of cardiac cycle. The stimulation system 11 bi-directionally stimulates either the left vagus nerve 15 or the right vagus nerve 16. However, it is contemplated that multiple electrodes 14 and multiple leads 13 could be utilized to stimulate simultaneously, alternatively, or in other various combinations. Stimulation may be through multimodal application of continuously-cycling, intermittent and periodic electrical stimuli, which are parametrically defined through stored stimulation parameters and timing cycles. Both sympathetic and parasympathetic nerve fibers in the vagosympathetic complex are stimulated. Generally, cervical vagus nerve stimulation results in propagation of action potentials from the site of stimulation in a bi-directional manner. The application of bi-directional propagation in both afferent and efferent directions of action potentials within neuronal fibers comprising the cervical vagus nerve improves cardiac autonomic balance. Afferent action potentials propagate toward the parasympathetic nervous system's origin in the medulla in the nucleus ambiguus, nucleus tractus solitarius, and the dorsal motor nucleus, as well as towards the sympathetic nervous system's origin in the intermediolateral cell column of the spinal cord. Efferent action potentials propagate toward the heart 17 to activate the components of the heart's intrinsic nervous system. Either the left or right vagus nerve 15, 16 can be stimulated by the stimulation system 11. The right vagus nerve 16 has a moderately lower (approximately 30%) stimulation threshold than the left vagus nerve 15 for heart rate effects at the same stimulation frequency and pulse width.

The VNS therapy is delivered autonomously to the patient's vagus nerve 15, 16 through three implanted components that include a neurostimulator 12, lead assembly 13, and electrodes 14. FIGS. 2A and 2B are diagrams respectively showing the implantable neurostimulator 12 and the stimulation lead assembly 13 of FIG. 1. In one embodiment, the neurostimulator 12 can be adapted from a VNS Therapy Demipulse Model 103 or AspireSR Model 106 pulse generator, manufactured and sold by Cyberonics, Inc., Houston, TX, although other manufactures and types of implantable VNS neurostimulators could also be used. The stimulation lead assembly 13 and electrodes 14 are generally fabricated as a combined assembly and can be adapted from a Model 302 lead, PerenniaDURA Model 303 lead, or PerenniaFLEX Model 304 lead, also manufactured and sold by Cyberonics, Inc., in three sizes based, for example, on a helical electrode inner diameter, although other manufactures and types of single-pin receptacle-compatible therapy leads and electrodes could also be used.

Referring first to FIG. 2A, the system 20 may be configured to provide multimodal vagus nerve stimulation. In a maintenance mode, the neurostimulator 12 is parametrically programmed to deliver continuously-cycling, intermittent and periodic ON-OFF cycles of VNS. Such delivery produces action potentials in the underlying nerves that propagate bi-directionally, both afferently and efferently.

The neurostimulator 12 includes an electrical pulse generator that is tuned to improve autonomic regulatory function by triggering action potentials that propagate both afferently and efferently within the vagus nerve 15, 16. The neurostimulator 12 is enclosed in a hermetically sealed housing 21 constructed of a biocompatible material, such as titanium. The housing 21 contains electronic circuitry 22 powered by a battery 23, such as a lithium carbon monofluoride primary battery or a rechargeable secondary cell battery. The electronic circuitry 22 may be implemented using complementary metal oxide semiconductor integrated circuits that include a microprocessor controller that executes a control program according to stored stimulation parameters and timing cycles; a voltage regulator that regulates system power; logic and control circuitry, including a recordable memory 29 within which the stimulation parameters are stored, that controls overall pulse generator function, receives and implements programming commands from the external programmer, or other external source, collects and stores telemetry information, processes sensory input, and controls scheduled and sensory-based therapy outputs; a transceiver that remotely communicates with the external programmer using radio frequency signals; an antenna, which receives programming instructions and transmits the telemetry information to the external programmer; and a reed switch 30 that provides remote access to the operation of the neurostimulator 12 using an external programmer, a simple patient magnet, or an electromagnetic controller. The recordable memory 29 can include both volatile (dynamic) and non-volatile/persistent (static) forms of memory, within which the stimulation parameters and timing cycles can be stored. Other electronic circuitry and components are possible.

The neurostimulator 12 includes a header 24 to securely receive and connect to the lead assembly 13. In one embodiment, the header 24 encloses a receptacle 25 into which a single pin for the lead assembly 13 can be received, although two or more receptacles could also be provided, along with the corresponding electronic circuitry 22. The header 24 may internally include a lead connector block (not shown), a setscrew, and a spring contact (not shown) that electrically connects to the lead ring, thus completing an electrical circuit.

In some embodiments, the housing 21 may also contain a heart rate sensor 31 that is electrically interfaced with the logic and control circuitry, which receives the patient's sensed heart rate as sensory inputs. The heart rate sensor 31 monitors heart rate using an ECG-type electrode. Through the electrode, the patient's heartbeat can be sensed by detecting ventricular depolarization. In a further embodiment, a plurality of electrodes can be used to sense voltage differentials between electrode pairs, which can undergo signal processing for cardiac physiological measures, for instance, detection of the P-wave, QRS complex, and T-wave. The heart rate sensor 31 provides the sensed heart rate to the control and logic circuitry as sensory inputs that can be used to determine the onset or presence of arrhythmias, particularly VT, and/or to monitor and record changes in the patient's heart rate over time or in response to applied stimulation signals.

Referring next to FIG. 2B, the lead assembly 13 delivers an electrical signal from the neurostimulator 12 to the vagus nerve 15, 16 via the electrodes 14. On a proximal end, the lead assembly 13 has a lead connector 27 that transitions an insulated electrical lead body to a metal connector pin 28 with a metal connector ring. During implantation, the connector pin 28 is guided through the receptacle 25 into the header 24 and securely fastened in place using the setscrew (not shown) to electrically couple one electrode of the lead assembly 13 to the neurostimulator 12 while a spring contact (not shown) makes electrical contact to the ring connected to the other electrode. On a distal end, the lead assembly 13 terminates with the electrode 14, which bifurcates into a pair of anodic and cathodic electrodes 62 (as further described infra with reference to FIG. 4). In one embodiment, the lead connector 27 is manufactured using silicone, and the connector pin 28 and ring are made of stainless steel, although other suitable materials could be used, as well. The insulated lead body of the lead assembly 13 utilizes a silicone-insulated alloy conductor material.

In some embodiments, the electrodes 14 are helical and placed around the cervical vagus nerve 15, 16 at the location below where the superior and inferior cardiac branches separate from the cervical vagus nerve. In alternative embodiments, the helical electrodes may be placed at a location above where one or both of the superior and inferior cardiac branches separate from the cervical vagus nerve. In one embodiment, the helical electrodes 14 are positioned around the patient's vagus nerve oriented with the end of the helical electrodes 14 facing the patient's head. In an alternate embodiment, the helical electrodes 14 are positioned around the patient's vagus nerve 15, 16 oriented with the end of the helical electrodes 14 facing the patient's heart 17. At the distal end, the insulated electrical lead body of the lead assembly 13 is bifurcated into a pair of lead bodies that are connected to a pair of electrodes. The polarity of the electrodes could be configured into a proximal anode and a distal cathode, or a proximal cathode and a distal anode.

The neurostimulator 12 may be interrogated prior to implantation and throughout the therapeutic period with a healthcare provider-operable control system comprising an external programmer and (shown in FIG. 3) for checking proper operation, downloading recorded data, diagnosing problems, and programming operational parameters. FIG. 3 is a diagram showing an external programmer 40 for use with the implantable neurostimulator 12 of FIG. 1. The external programmer 40 includes a healthcare provider operable programming computer 41 and a programming wand 42. Generally, use of the external programmer is restricted to healthcare providers, while more limited manual control is provided to the patient through "magnet mode."

In one embodiment, the external programmer 40 executes application software 45 specifically designed to interrogate the neurostimulator 12. The programming computer 41 interfaces to the programming wand 42 through a wired or wireless data connection. The programming wand 42 can be adapted from a Model 201 Programming Wand, manufactured and sold by Cyberonics, Inc., and the application software 45 can be adapted from the Model 250 Programming Software suite, licensed by Cyberonics, Inc. Other configurations and combinations of external programmer 40, programming wand 42, and application software 45 are possible.

The programming computer 41 can be implemented using a general purpose programmable computer and can be a personal computer, laptop computer, ultrabook computer,

netbook computer, handheld computer, tablet computer, smart phone, or other form of computational device. For example, in one embodiment, the programming computer **41** is a tablet programmer with a wired or wireless data connection to the programming wand **42**. The programming computer **41** functions through those components conventionally found in such devices, including, for instance, a central processing unit, volatile and persistent memory, touch-sensitive display, control buttons, peripheral input and output ports, and network interface. The computer **41** operates under the control of the application software **45**, which is executed as program code as a series of process or method modules or steps by the programmed computer hardware. Other assemblages or configurations of computer hardware, firmware, and software are possible.

Operationally, the programming computer **41**, when connected to a neurostimulator **12** through wireless telemetry using the programming wand **42**, can be used by a healthcare provider to remotely interrogate the neurostimulator **12** and modify stored stimulation parameters. The programming wand **42** provides data conversion between the digital data accepted by and output from the programming computer and the radio frequency signal format that is required for communication with the neurostimulator **12**. In other embodiments, the programming computer may communicate with the implanted neurostimulator **12** using other wireless communication methods, such as wireless RF transmission. The programming computer **41** may further be configured to receive inputs, such as physiological signals received from patient sensors (e.g., implanted or external). These sensors may be configured to monitor one or more physiological signals, e.g., vital signs, such as body temperature, pulse rate, respiration rate, blood pressure, etc. These sensors may be coupled directly to the programming computer **41** or may be coupled to another instrument or computing device which receives the sensor input and transmits the input to the programming computer **41**. The programming computer **41** may monitor, record, and/or respond to the physiological signals in order to effectuate stimulation delivery in accordance with some embodiments.

The healthcare provider operates the programming computer **41** through a user interface that includes a set of input controls **43** and a visual display **44**, which could be touch-sensitive, upon which to monitor progress, view downloaded telemetry and recorded physiology, and review and modify programmable stimulation parameters. The telemetry can include reports on device history that provide patient identifier, implant date, model number, serial number, magnet activations, total ON time, total operating time, manufacturing date, and device settings and stimulation statistics, and on device diagnostics that include patient identifier, model identifier, serial number, firmware build number, implant date, communication status, output current status, measured current delivered, lead impedance, and battery status. Other kinds of telemetry or telemetry reports are possible.

During interrogation, the programming wand **42** is held by its handle **46** and the bottom surface **47** of the programming wand **42** is placed on the patient's chest over the location of the implanted neurostimulator **12**. A set of indicator lights **49** can assist with proper positioning of the wand and a set of input controls **48** enable the programming wand **42** to be operated directly, rather than requiring the healthcare provider to awkwardly coordinate physical wand manipulation with control inputs via the programming computer **41**. The sending of programming instructions and receipt of telemetry information occur wirelessly through

radio frequency signal interfacing. Other programming computer and programming wand operations are possible.

FIG. **4** is a diagram showing the helical electrodes **14** provided as on the stimulation lead assembly **13** of FIG. **2** in place on a vagus nerve **15**, **16** in situ **50**. Although described with reference to a specific manner and orientation of implantation, the specific surgical approach and implantation site selection particulars may vary, depending upon physician discretion and patient physical structure.

Under one embodiment, helical electrodes **14** may be positioned on the patient's vagus nerve **61** oriented with the end of the helical electrodes **14** facing the patient's head. At the distal end, the insulated electrical lead body of the lead assembly **13** is bifurcated into a pair of lead bodies **57**, **58** that are connected to a pair of electrodes **51**, **52**. The polarity of the electrodes **51**, **52** could be configured into a proximal anode and a distal cathode, or a proximal cathode and a distal anode. In addition, an anchor tether **53** is fastened over or in connection with the lead bodies **57**, **58** that maintains the helical electrodes' position on the vagus nerve **61** following implant. In one embodiment, the conductors of the electrodes **51**, **52** are manufactured using a platinum and iridium alloy, while the helical materials of the electrodes **51**, **52** and the anchor tether **53** are a silicone elastomer.

During surgery, the electrodes **51**, **52** and the anchor tether **53** are coiled around the vagus nerve **61** proximal to the patient's head, each with the assistance of a pair of sutures **54**, **55**, **56**, made of polyester or other suitable material, which help the surgeon to spread apart the respective helices. The lead bodies **57**, **58** of the electrodes **51**, **52** are oriented distal to the patient's head and aligned parallel to each other and to the vagus nerve **61**. A strain relief bend **60** can be formed on the distal end with the insulated electrical lead body of the lead assembly **13** aligned, for example, parallel to the helical electrodes **14** and attached to the adjacent fascia by a plurality of tie-downs **59a-b**.

The neurostimulator **12** delivers VNS under control of the electronic circuitry **22**. The stored stimulation parameters are programmable. Each stimulation parameter can be independently programmed to define the characteristics of the cycles of therapeutic stimulation and inhibition to ensure optimal stimulation for a patient **10**. The programmable stimulation parameters include output current, signal frequency, pulse width, signal ON time, signal OFF time, magnet activation (for VNS specifically triggered by "magnet mode"), and reset parameters. Other programmable parameters are possible. In addition, sets or "profiles" of preselected stimulation parameters can be provided to physicians with the external programmer and fine-tuned to a patient's physiological requirements prior to being programmed into the neurostimulator **12**.

Therapeutically, the VNS may be delivered as a multimodal set of therapeutic doses, which are system output behaviors that are pre-specified within the neurostimulator **12** through the stored stimulation parameters and timing cycles implemented in firmware and executed by the microprocessor controller. The therapeutic doses include a maintenance dose that includes continuously-cycling, intermittent and periodic cycles of electrical stimulation during periods in which the pulse amplitude is greater than 0 mA ("therapy ON") and during periods in which the pulse amplitude is 0 mA ("therapy OFF").

The neurostimulator **12** can operate either with or without an integrated heart rate sensor. Additionally, where an integrated leadless heart rate monitor is available, the neurostimulator **12** can provide autonomic cardiovascular drive evaluation and self-controlled titration. Finally, the neuro-

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stimulator **12** can be used to counter natural circadian sympathetic surge upon awakening and manage the risk of cardiac arrhythmias during or attendant to sleep, particularly sleep apneic episodes.

Several classes of implantable medical devices provide therapy using electrical current as a stimulation vehicle. When such a system stimulates certain organs or body structures like the vagus nerve, therapeutic levels of electrical stimulation are usually not well tolerated by patients without undergoing a process known as titration. Titration is a systematic method or process of incrementally increasing the stimulation parameters employed by an implanted device to deliver a stimulation current to the patient at increasing levels that achieve or improve therapeutic benefit while minimizing side effects that could disrupt the stimulation therapy. Titration in a neuromodulation system may be necessary due to centrally-mediated side effects elicited by large changes in stimulation intensity. For example, the neuromodulation system may be unable to instantly change the intensity of delivered neurostimulation from an inactive state (e.g., stimulation programmed to OFF) to full therapeutic intensity without the patient experiencing adverse effects (e.g., triggering an expiratory cough reflex). That being said, the central processing areas of vagal afferents recruited at low stimulation intensity can often handle small stimulation intensity increases over periods of time without effect. As such, titration usually involves bringing the patient to an initial stimulation level that is tolerable to the patient (i.e., below an initial tolerance threshold), waiting for a period of time for the patient to adjust to the continuing delivery of the initial stimulation level and to define a higher tolerance threshold of the patient, and then increasing the initial stimulation level to a higher stimulation level that is, in some patients, greater than the initial tolerance threshold, and so on. This process is repeated in sequences that progress from a stimulation delivery provided over a waiting period, and then to an increase in a stimulation level that defines the next sequence of the stimulation delivery and the next waiting period. The central neural processors gradually remodel and accommodate the increasing stimulation intensity if given sufficient time between increasing stimulation steps (e.g., function without adverse effects such as triggering the cough reflex).

FIG. 5 is a flow diagram showing a method for delivering vagus nerve stimulation therapy, according to an exemplary embodiment. A titration process **400** is used to gradually increase the stimulation intensity to a desired therapeutic level or maintenance dosage level. If the stimulation intensity is increased too quickly before the patient is fully accommodated to the stimulation signal, the patient may experience undesirable side effects, such as coughing, hoarseness, throat irritation, or expiratory reflex. The titration process gradually increases stimulation intensity within a tolerable level and maintains that intensity for a period of time to permit the patient to adjust to each increase in intensity, thereby gradually increasing the patient's side effect tolerance zone boundary to so as to accommodate subsequent increases in intensity. The titration process continues until adequate adaptation is achieved. In embodiments, the titration process is automated and is executed by the implanted device without manual adjustment of the stimulation intensity by the subject or health care provider. As will be described in greater detail below, adequate adaptation is a composite threshold comprising one or more of the following: an acceptable side effect level, a target intensity level, and a target physiological response. In some embodiments, adequate adaption includes all three objec-

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tives: an acceptable side effect level, a target intensity level, and a target physiological response.

In some embodiments, the titration process is a mix of automation and physician input. As will be described in greater detail below, a physician may use intermediate holds to stop the automated titration at certain thresholds (e.g., a certain number of days or weeks, certain stimulation parameter values, etc.) and evaluate the patient before resuming the automated titration. The physician may receive a graphical titration history to review how the automated titration process has been progressing from one sequence to the next. The graphical titration history may include markers. The markers may represent intermediate holds, when target parameters are reached between adjacent sequences, etc. After the physician has resumed the automatic titration, the next sequence of automated titration may progress until the next intermediate hold is reached.

As described above, it may be desirable to minimize the amount of time required to complete the titration process so as to begin delivery of the stimulation at therapeutically desirable levels, particularly when the patient is being treated for an urgent condition such as CHF. In addition, it is desirable to utilize a maintenance dose intensity at the minimum level required to achieve the desired therapeutic effect. This can reduce power requirements for the neurostimulator and reduce patient discomfort.

It has been observed that a patient's side effect profile is more sensitive to the stimulation output current than to the other stimulation parameters, such as frequency, pulse width, and duty cycle. As a result, accommodation to the stimulation output current is a primary factor in completing the titration process. It has also been observed that if the other stimulation parameters are maintained at a level below the target levels, the output current can be increased to higher levels without eliciting undesirable side effects that would be result when the other parameters are at the target level. As a result, increasing the target output current while maintaining the other stimulation parameters (pulse width in particular) at reduced levels can result in a faster accommodation and shorter overall titration time than would be achieved by attempting to increase the output current while stimulating at the target pulse width.

Referring again to FIG. 5, in step **401**, a stimulation system **11**, including a neurostimulator **12**, a nerve stimulation lead assembly **13**, and a pair of electrodes **14**, is implanted in the patient. In step **402**, the patient undergoes an optional post-surgery recovery period, during which time the surgical incisions are allowed to heal and no VNS therapy occurs. This period may last, e.g., two weeks post-surgery. In step **403**, the stimulation therapy is initiated with the initiation of a titration process. During this titration process, VNS therapy is titrated by adjusting one or more of the stimulation parameters, including output current, pulse width, signal frequency, and duty cycle, as will be described in greater detail below. Completion of the titration process determines the stimulation intensity to be used for subsequent maintenance doses delivered in step **404**. These maintenance doses may be selected to provide the minimum stimulation intensity necessary to provide the desired therapeutic result.

FIG. 6A is a flow diagram illustrating a titration process **500**, according to an exemplary embodiment. Process **500** includes setting titration parameters (step **501**), initiating titration (step **502**), stopping titration at an intermediate hold (step **503**), and resuming titration (step **504**).

In step **501**, a physician sets the titration parameters via programmer **40**, which are received by the implantable

vagus nerve stimulation system 11. In some embodiments, the titration parameters may be defined by one or more titration algorithms that may be selected by the physician, or may be presented to the physician as a preferred or recommended list of titration parameters that the programming physician can adopt. In other embodiments, rather than present the physician with a set titration algorithm with fixed algorithm values, the physician may be presented with default values that could be manually adjusted. The titration parameter starting values, target values, and/or increment values for amplitude, pulse width, frequency, and/or duty cycle may be adjustable, as may the time interval between titration steps. Time of day and delay to therapy start may also be programmable as a titration parameter. The titration parameters may also include one or more intermediate holds that maintain certain parameters until the physician indicates that the automated titration can continue. The physician may be limited so that modification can be made to only a select group of parameters, or some parameters may be considered to be in a locked state until unlocked by the physician. In some embodiments, the physician is able to modify a large number of titration parameters (e.g., 10-12 parameters).

Alternatively, rather than give the physician control over the titration parameter values themselves, the physician's options for the titration process may be presented as a set of "aggressiveness" options to select from, each of which would be used by the system to determine the values to use. For example, the physician may be able to choose from an aggressive profile, a moderate profile, or a light profile (sensitive) that is appropriate for certain types of patients that do not require detailed titration parameter programming. More or fewer aggressiveness profiles could be used, and the aggressiveness profiles may correspond to the overall health status of the patient, the patient's sensitivity to stimulation therapies or titration processes, or the patient's medical history. The aggressiveness profile selected by the physician may result in a predetermined set of titration parameters being selected. The predetermined titration parameters may vary between different aggressiveness profiles, and some titration parameters may remain constant, or similar, between various aggressiveness profiles. For example, the aggressive profile may be suitable for patients that have a high toleration for the titration process and may include shorter time intervals between titration steps, higher intensity target values, and/or larger increment values (e.g., as compared to the moderate or light profiles) that may result in an achievement of a suitable therapy level more quickly as compared to the moderate or light profiles. While some of the parameters may promote a more aggressive titration progression, some of the parameters may be consistent with parameters of other profiles (e.g., titration holds).

In some embodiments, each of the aggressiveness profiles may be mapped by the system to a set of parameters or a range of parameters. For example, if the user selects the aggressive profile, the system may receive the user selection and set the values of one or more parameters (e.g., amplitude, pulse width, frequency, duty cycle, intervals between titration steps, and/or other parameters) to a first set of values. If the user selects the moderate profile, the system may set the values of the parameters to a second predetermined set or range of values that is different than the set associated with the aggressive profile. In some embodiments, the physicians are limited to modification of the parameters within a range of boundary values. The ranges may be for the default parameters, or may be set individually for the aggressiveness options (e.g., the ranges for the aggressive profile and the moderate profile may be different,

but may overlap for some parameters). The physician may be able to customize the parameters in the preset profiles. Titrating according to aggressiveness profiles is described in further detail below with respect to FIGS. 13 and 14. In step 502, the physician initiates titration using the titration parameters defined at 501.

In step 503, titration is stopped at a titration hold. The titration hold may be an intermediate hold set by the physician during step 501. The VNS system 11 may perform automated titration according to process 600, described below. However, the physician is given the option (through the programmer 40) to designate intermediate points at which the titration algorithm would pause and await manual (programmer-based) activation by the physician. These hold points may be either time based (e.g. after 2, 4, 6, and/or 8 weeks of titration) or stimulation based (e.g. once stimulation amplitude reaches 1.0, 1.5, 2.0, and/or 2.5 mA). This would allow the physician to evaluate the patient in the clinic before deciding to continue titration. The physician releases the hold on the titration with the programmer 40 once the patient has been evaluated. The physician may also modify parameters during the clinical evaluations.

The holds may be predefined for the entire titration process during initial set up. Alternatively, the physician may have the option of setting a new intermediate hold when evaluating the patient. The intermediate holds may be consistent throughout the titration process (e.g., every 2 weeks, every 0.5 mA, etc.). In another embodiment, the intermediate holds are different for at least one hold (e.g., 4 weeks to the first hold, 2 weeks for every subsequent hold, etc.). In another embodiment, intermediate holds can be a combination of parameters (e.g., amplitude and pulse width). In some embodiments, the hold may be set to begin when both parameters are met or when one parameter is met. In another embodiment, one parameter cannot exceed the hold value and will remain constant until the second parameter is reached. In some embodiments, both parameters will progress according to the automated titration until both parameters meet the intermediate hold value, but one parameter may exceed the intermediate hold until the second parameter reaches the intermediate hold value. The physician may have the option to set as many or as few intermediate holds as desired.

During the automated titration between intermediate holds, the VNS system 11 may be fully automated or partially automated. In some embodiments, titration is performed without any intervention from either the patient or the healthcare provider. This embodiment also automatically detects patient side effects and intolerance and adjusts stimulation parameters to remain below the side effect threshold, as is described with respect to FIG. 6B. In another embodiment, the VNS system 11 may automatically adjust stimulation parameters slowly over time, without any additional intervention from the healthcare provider. Because the system may not be able to determine if stimulation causes an intolerable side effect, it may be configured to rely on the patient to swipe a magnet to indicate an intolerable level of a side effect. The VNS system 11 may then adjust stimulation parameters in response to patient magnet activation.

For example, patients may require a total of 10 ± 2 clinic visits over a 10-week period to reach the target stimulation intensity. The frequency of required clinic visits is bothersome to both patients and providers and creates a barrier to therapy adoption. In addition, the frequency of required clinic visits extends the time required to titrate patients to the target stimulation intensity. However, the physician may be skeptical of completely automated titration and want to

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ensure the patients are not experiencing intolerable side effects and are adapting to stimulation adequately. By allowing the physician to set the parameters, and evaluate the patient intermediately, but still allow titration to perform automatically between visits, the time period to reach the target stimulation may be reduced, while giving the physicians more control over the titration process. Preferably, the number of clinic visits needed and the overall timeframe of the titration process is reduced by only the use of intermediate holds. Any time penalty related to the intermediate holds is believed to be significantly less than the time penalty resulting from an automated titration process that causes side effect and ultimately requires the patient to undergo a re-titration protocol.

In step **504**, titration is resumed. The physician may resume titration using the programmer **40** after evaluation of the patient. When the physician resumes titration, he or she may have the option to modify stimulation parameters and/or intermediate holds. The titration may resume using automated titration until the next intermediate hold is reached. This process may continue until the therapy parameters are reached.

FIG. **6B** is a flow diagram illustrating a titration process **600** in accordance with exemplary embodiments. When first initiating the titration process, the neurostimulator **12** is configured to generate a stimulation signal having an initial stimulation parameter set. The initial parameter set may comprise an initial output current, an initial frequency, an initial pulse width, and an initial duty cycle. The various initial parameter settings may vary but may be selected so that one or more of the parameters are set at levels below a predefined target parameter set level, such that the titration process is used to gradually increase the intensity parameters to achieve adequate adaptation. In some embodiments, the initial frequency is set at the target frequency level, while the initial output current, initial pulse width, and initial duty cycle are set below their respective target levels. In one embodiment, the target parameter set comprises a 5 Hz frequency, 250 μ sec pulse width, a duty cycle of 14 sec ON and 66 seconds OFF, and an output current of between 1.5 mA-3.0 mA (e.g., 2.5 mA for right side stimulation and 3.0 mA for left side stimulation), and the initial parameter set comprises 5 Hz frequency, 130 μ sec pulse width, a duty cycle of 14 sec ON and 66 seconds OFF, and an output current of between 0.25 mA-0.5 mA. In other embodiments, the target parameter set includes a 10 Hz frequency that is used instead of a 5 Hz frequency. The initial parameter set may also include one or more intermediate holds as discussed with respect to FIG. **6A**. However, this is an exemplary embodiment and these values are not intended to be limiting. Other frequencies, pulse widths, duty cycles and output currents may be implemented. The initial and target parameters may vary from patient to patient based on the patient's sensitivity to stimulation. While the initial parameters are shown to be equal to the target parameters for some of the exemplary parameters (e.g., frequency and duty cycle), some or all of the parameters may have initial parameters that differ from the target parameters.

In step **601**, the stimulation system delivers stimulation to the patient. If this is the first titration session, then the stimulation would be delivered with the initial stimulation parameter set described above. If this is a subsequent titration session, then the stimulation intensity would remain at the same level provided at the conclusion of the previous titration session. Alternatively, the subsequent titration session can start at a level that is set by the physician, e.g., at

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the next titration level that follows the level provided at the conclusion of the previous titration session.

In step **602**, the output current is gradually increased until the stimulation results in an intolerable side effect level, the target output current (e.g., 2.5 mA) is reached, or adequate adaptation is achieved. As described above, adequate adaptation is a composite threshold comprising one or more of the following: an acceptable side effect level, a target intensity level, and a target physiological response. In accordance with some embodiments, the target physiological response comprises a target heart rate change during stimulation. The patient's heart rate may be monitored using an implanted or external heart rate monitor, and the patient's heart rate during stimulation is compared to the patient's baseline heart rate to determine the extent of heart rate change. In accordance with some embodiments, the target heart rate change is a heart rate change of between 4% and 5%. If at any point during the titration process **600** adequate adaptation is achieved, the titration process ends, and the stimulation intensity which resulted in the adequate adaptation is used for ongoing maintenance dose therapy delivery.

The output current may be increased in any desired increment, but small increments, e.g., 0.1 mA or 0.25 mA, may be desirable so as to enable more precise adjustments. In some cases, the output current increments may be determined by the neurostimulator's maximum control capability. During the initial titration sessions, it is likely that the patient's side effect tolerance zone boundary will be reached well before the output current reaches the target level or adequate adaptation is achieved. At decision step **603**, if the target output current has not been achieved but the maximum tolerable side effects have been exceeded, the process proceeds to step **604**.

In step **604**, the output current is reduced one increment to bring the side effects within acceptable levels. In addition, the frequency is reduced. In embodiments in which the initial frequency was 10 Hz, in step **604**, the frequency may be reduced, e.g., to 5 Hz or 2 Hz.

Next, in step **605**, the output current is gradually increased again at the reduced frequency level until the stimulation results in an intolerable side effect level or the target output current (e.g., 2.5 mA) is reached. At decision step **606**, if the target output current has been reached and the maximum tolerable side effects have not been exceeded, the process proceeds to step **607** where the titration session is concluded. The stimulation system may be programmed to continue delivering the stimulation signal at the last parameter settings achieved prior to conclusion of the titration session. After a period of time, another titration session may be initiated and the process returns to step **601**. This can be any period of time sufficient to permit the patient to adjust to the increased stimulation levels. This can be, for example, as little as approximately two or three days, approximately one to two weeks, approximately four to eight weeks, or any other desired period of time.

In some embodiments, the titration sessions are automatically initiated by the stimulation system or initiated by the patient without requiring any intervention by the health care provider. This can eliminate the need for the patient to schedule a subsequent visit to the health care provider, thereby potentially reducing the total amount of time needed for the titration process to complete. In these embodiments, the stimulation system may include a physiological monitor, e.g., an implanted heart rate sensor, that communicates with the stimulation system's control system to enable the control system to detect the patient's physiological response to the

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titration and automatically make adjustments to the titration processes described herein with reduced or no inputs from the patient or health care provider. The monitored signals can also enable the control system to detect when the target physiological response has been achieved and conclude the titration process. The stimulation system could in addition or alternatively include a patient control input to permit the patient to communicate to the control system that the acceptable side effect level has been exceeded. This control input may comprise an external control magnet that the patient can swipe over the implanted neurostimulator or other internal or external communication device that the patient can use to provide an input to the control system. In these automatically-initiated titration sessions, the stimulation system may be configured to wait a period of time after completing one session before initiating the next session. This period of time may be predetermined, e.g., two or three days, or programmable. In another embodiment, the stimulation system is configured to wait until authorization has been received before initiating the next session (i.e., an intermediate hold).

Returning to decision step 606, if the target output current has not been reached but the maximum tolerable side effects have been exceeded, the process proceeds to step 608. In step 608, the output current is reduced one increment to restore an acceptable side effect condition, and the frequency is gradually increased until the stimulation results in an intolerable side effect level or the target frequency (e.g., 5 Hz) is reached. At decision step 609, if the target frequency has not been reached but the maximum tolerable side effects have been exceeded, the frequency is reduced to restore an acceptable side effect level and the process proceeds to step 607. Again, in step 607, the current titration session is concluded, and the stimulation system may be programmed to continue delivering the stimulation signal at the last parameter settings achieved prior to conclusion of the titration session.

At decision step 609, if the target frequency has been reached before the maximum tolerable side effects have been exceeded, the duty cycle is gradually increased until the stimulation results in an intolerable side effect level or the target duty cycle (e.g., 14 sec ON and 66 sec OFF) is reached, at which point the process proceeds to step 607 and the titration session is concluded and ongoing stimulation delivered at the last intensity eliciting acceptable side effect levels.

Returning to decision step 603, if the target output current has been achieved before the maximum tolerable side effects are exceeded, the process proceeds to step 611. In step 611, the pulse width is gradually increased until the stimulation results in an intolerable side effect level or the target pulse width (e.g., 250 μ sec) is reached. In some embodiments, before step 611, the output current is reduced (e.g., by up to 50%), and the pulse width may be increased in step 611 at that reduced output current. After the target pulse width is achieved, the output current may be restored to the target output current. In other embodiments, the output current may be reduced (or may be retained at the reduced level established prior to step 611, as described above), and the frequency and duty cycle are gradually increased in step 613 at that reduced output current. This reduction in output current after achieving the target output current may enable the patient to maintain tolerability with increasing pulse width, frequency, and duty cycle in subsequent titration steps.

At decision step 612, if the target pulse width has not been achieved before the maximum tolerable side effects have

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been exceeded, the pulse width is reduced to restore an acceptable side effect level and the process proceeds to step 607. Again, in step 607, the current titration session is concluded.

If at decision step 612, the target pulse width has been achieved before the maximum tolerable side effects have been exceeded, the process proceeds to step 613. In step 613, the frequency and/or duty cycle are increased until the stimulation results in an intolerable side effect level or the target frequency and target duty cycle are reached. The frequency and duty cycle can be increased in step 612 simultaneously, sequentially, or on an alternating basis.

At decision step 614, if the target frequency and/or target duty cycle have not been achieved before the maximum tolerable side effects have been exceeded, the pulse width and/or frequency are reduced to restore an acceptable side effect level, and the process continues to step 607 and the titration session is concluded. In some embodiments, the conclusion of the titration session represented in step 607 indicates an intermediate hold has been reached. A new titration session could then be initiated after visiting a physician to release the intermediate hold.

At decision step 614, if the target pulse width and target frequency have been achieved before the maximum tolerable side effects have been exceeded, all of the stimulation parameters will have reached their target levels and the titration process concludes at step 615. The stimulation therapy may proceed with the maintenance dose at the target stimulation levels. In some embodiments, the target frequency and duty cycle achieved are for a given titration session with an intermediate hold. In this case, the patient would visit a health care provider or physician for an evaluation. The physician would then release the hold on the titration processes or initiate the beginning of therapy.

In some embodiments, in step 604, instead of reducing the frequency in order to facilitate increase of the output current, the pulse width may be reduced. For example, embodiments where the target pulse width is 250 μ sec, the pulse width may be reduced, e.g., to 150 μ sec or less. Then, the method proceeds to step 605, in which the output current is gradually increased again at the reduced pulse width level until the stimulation results in an intolerable side effect level or the target output current (e.g., 2.5 mA) is reached.

Therapy can also be autonomously titrated by the neurostimulator 12 in which titration progressively occurs in a self-paced, self-monitored fashion. The progression of titration sessions may occur on an autonomous schedule or may be initiated upon receipt of an input from the patient. Ordinarily, the patient 10 is expected to visit his healthcare provider to have the stimulation parameters stored by the neurostimulator 12 in the recordable memory 29 reprogrammed using an external programmer. Alternatively, the neurostimulator 12 can be programmed to automatically titrate therapy by up titrating the VNS through periodic incremental increases using titration sessions as described above. The titration process 600 will continue until the ultimate therapeutic goal is reached.

Following the titration period, therapeutic VNS, as parametrically defined by the maintenance dose operating mode, is delivered to at least one of the vagus nerves. The stimulation system 11 delivers electrical therapeutic stimulation to the cervical vagus nerve of a patient 10 in a manner that results in creation and propagation (in both afferent and efferent directions) of action potentials within neuronal fibers of either the left or right vagus nerve independent of cardiac cycle.

FIG. 7A is a simplified block diagram of an implanted neurostimulation system 700, according to an exemplary embodiment. The implanted neurostimulation system 700 comprises a control system 702 comprising a processor programmed to operate the system 700, a memory 703, an optional physiological sensor 704, and a stimulation subsystem 706. The physiological sensor 704 may be configured to monitor any of a variety of patient physiological signals, and the stimulation subsystem 706 may be configured to deliver a stimulation signal to the patient. In one example, the physiological sensor 704 comprises an ECG sensor or an accelerometer for monitoring heart rate, and the stimulation subsystem 706 comprises a neurostimulator 12 programmed to deliver ON-OFF cycles of stimulation to the patient's vagus nerve. The implanted system 700 may include a patient input sensor 705, described in more detail below.

The control system 702 is programmed to activate the neurostimulator 12 to deliver stimulation signals at varying stimulation intensities to the patient and to monitor the physiological signals in response to those delivered stimulation signals.

The external programmer 707 shown in FIG. 7A may be utilized by a clinician or by the patient for communicating with the implanted system 700 to adjust parameters, activate therapy, retrieve data collected by the system 700, or provide other input to the system 700. In some embodiments, the external programmer 707 may be used remotely from the implanted system 700 (e.g., when the patient is not at a clinic). For example, instead of the patient coming into the clinic for a check-up during a titration hold, the clinician may check on the patient remotely (e.g., phone call, video call, etc.). The clinician could then use the external programmer 707 to activate the next titration session or modify parameters of the titration. In some embodiments, the external programmer 707 may provide an alert indicating the patient has reached a titration hold. In some embodiments, the patient receives an alert indicating a titration hold has been reached (e.g., email, text message, etc.). In some embodiments, the external programmer 707 may include communication circuitry adapted to communicate over a long distance using one or more protocols (e.g., cellular, Internet, etc.). In some embodiments, the external programmer 707 may be configured to program the implanted system 700 with a prescribed time or window of time during which titration sessions may be initiated, as described in further detail below with respect to FIGS. 16 and 17.

Patient inputs to the implanted system 700 may be provided in a variety of ways. The implanted system 700 may include a patient input sensor 705. As described above, a patient magnet 730 may be used to provide external input to the system 700. When the patient magnet 730 is placed on the patient's chest in close proximity to the implanted system 700, the patient input sensor 705 will detect the presence of the magnetic field generated by the patient magnet 730 and provide a control input to the control system 702. The system 700 may be programmed to receive patient inputs to set the time of day during which titration sessions are to be initiated.

In other embodiments, the patient input sensor 705 may comprise a motion sensor, such as an accelerometer, which is configured to detect tapping on the surface of the patient's chest. The patient may use finger taps in one or more predetermined patterns to provide control inputs to the implanted system 700. For example, when the motion sensor detects three rapid taps to the patient's chest, that may trigger an operation on the implanted system 700 (e.g., to

initiate a titration session). Alternatively, if the motion sensor detects a predetermined pattern of taps during a titration session, the implanted system 700 will interpret those taps as a patient input indicating that the patient's tolerance zone boundary has been exceeded.

In other embodiments, the patient input sensor 705 may comprise an acoustic transducer or other sensor configured to detect acoustic signals. The system 700 may be programmed to interpret the detection of certain sounds as patient inputs. For example, the patient may utilize an electronic device, such as a smartphone or other portable audio device, to generate one or more predetermined sequences of tones. The system 700 may be programmed to interpret each of these sequences of tones as a different patient input.

The titration of the stimulation signal delivery and the monitoring of the patient's physiological response (e.g., heart rate) may be advantageously implemented using a control system 702 in communication with both the stimulation subsystem 706 and the physiological sensor 704, such as by incorporating all of these components into a single implantable device 700. In accordance with other embodiments, an external control system 712 may be implemented in a separate implanted device or in an external programmer 720 or other external device, as shown in FIG. 7B to provide control over and communication with an implanted physiological sensor 714 and a stimulation subsystem 716 similar to those describe with regard to FIG. 6A. The external programmer 720 in FIG. 7B may be utilized by a clinician or by the patient for adjusting stimulation parameters. The external programmer 720 may be in wireless communication with the implanted medical device 710, which includes the stimulation subsystem 716 and a memory 713. In the illustrated embodiment, the physiological sensor 714 is incorporated into the implanted medical device 710, but in other embodiments, the sensor 714 may be incorporated into a separate implanted device, may be provided externally and in communication with the external programmer 720, or may be provided as part of the external programmer 720.

When monitoring the patients, the physician uses the external programmer 707 to connect with the implantable medical device 710. However, in some implementations, the physician must manually connect the external programmer 707 to each implantable medical device 710 to perform titration functions (e.g., change parameter settings, titration holds and settings, etc.), which can be burdensome on physicians, as well as the clinic. FIGS. 8-10 address this issue, according to example embodiments, by providing a dashboard capable of monitoring a plurality of patients, even when the patients are not in the clinic.

FIG. 8 is a titration assist management dashboard 800, according to an exemplary embodiment. The titration assist management dashboard 800 operates on a device 802 with a display 804. The display 804 provides the dashboard 800 which includes information relating to patient name 806, patient status 808, patient priority 810 and patient notes 812.

The device 802 includes a processor, memory, a communication circuit, and various input and output circuits. The device 802 may be a programming computer (e.g., programming computer 41, external programmer 707) that is in direct communication with a programming wand (e.g., programming wand 42). In some embodiments, the device 802 may be a computer, a tablet, a handheld device, a wearable, etc. In some embodiments, the device 802 is capable of communicating directly with an implantable medical device (e.g., telemetry). In some embodiments, the device 802 that is capable of communicating with a secondary device (e.g.,

a programming computer) that communicates with an implantable medical device. In some embodiments, the device **802** can communicate with a remote device that is not located at the physician's office, such as a home monitor. The device **802** may communicate with the secondary device via telemetry, a wired connection, or another device/method of communication. In some embodiments, the device **802** is in communication with a website, server, program, etc. that allows the device **802** to access the dashboard **800**. For example, the dashboard **800** may be accessible on multiple devices **802** at the same time.

The dashboard **800** includes information relating to patient name **806**, patient status **808**, patient priority **810**, and/or patient notes **812**. The dashboard **800** compiles patient information when a patient is set up on a titration assist program. Once the patient information is in the dashboard **800**, the dashboard **800** is able to monitor patient status without being in communication with the implantable medical device by knowing the parameters of the titration assist and updating the dashboard according to the titration assist parameters, in some implementations.

For each patient, the dashboard **800** provides patient name **806**, patient status **808**, patient priority **810**, and/or patient notes **812**. In order to use the dashboard **800**, a user (e.g., physician, nurse, medical assistant, etc.) may provide log in credentials. In some embodiments, the amount or detail of information provided may vary based on the log in information provided. For example, the physician may have access to all the information provided on the dashboard **800**, while the information provided to a nurse or medical assistant may be more limited.

The patient name **806** provided on the dashboard **800** may be the actual name of the patient, or a means of identifying the patient while maintaining anonymity of the patient (e.g., patient identification number, etc.). In some embodiments, the patient name **806** may also provide information relating to general patient information (e.g., home address, contact information, medical history, age, gender, insurance information, etc.). While all this information may not be present on a main screen of the dashboard **800** (e.g., as shown in FIG. **8**), the user may be able to access the additional information by selecting a specific patient.

The patient status **808** provided on the dashboard **800** is a status of the progression of the titration based on the titration assist parameters established during interrogation of the implantable medical device. The patient status **808** may include a stimulation parameters (e.g., amplitude, frequency, pulse width, etc.). In some embodiments, the patient status **808** may also indicate if a HOLD is present in the titration progression, an indication of the weeks that have passed since titration has started, or another indication of the time of titration. The dashboard **800** is able to update the patient status **808** based on the settings of titration, such as updating the patient status **808** to indicate a hold is present.

The dashboard **800** may also include patient notes **812**. The patient notes **812** may include a plurality of information relating to the patient, titration, and other information the physician feels may be pertinent. For example, the patient notes **812** may include the aggression profile that was selected for the patient, initial titration parameters, target stimulation parameters, settings for stimulation increases, titration hold settings, a parameter setting profile, patient side effects, etc. The patient notes **812** may be a text box or a plurality of text boxes where the user can insert a variety of notes. In some embodiments, the patient notes **812** are a plurality of check boxes or other selection mechanisms that provides a list of parameters, settings, side effects, etc. that

can be selected. In some embodiments, the patient notes **812** are a combination of check boxes and text boxes to provide diverse means of recording patient notes **812**. The patient notes **812** may all be present on the dashboard **800**. In some embodiments, only a portion of the patients notes **812** are provided on a main screen of the dashboard **800**. In some embodiments, the user may be able to select which patient notes **812** are present on the main screen of the dashboard **800**. In some embodiments, a default set of patient notes **812** are present on the main screen of the dashboard **800**. In some embodiments, the dashboard **800** can create patient notes based on the patient status **808**. For example, if the patient status **808** is updated to indicate a hold has been reached, the patient notes **812** may be updated to indicate a follow up appointment or call needs to be scheduled.

The patient priority **810** indicates a likelihood of the patient needing attention (e.g., most likely to need a clinical visit). The patient priority **810** may be based on a combination of patient status **808**, patient information contained within the patient name **806**, and patient notes **812**. The patient status **808**, patient information contained within the patient name **806**, and patient notes **812** may receive a value based on the information contained within. The patient priority **810** may be a weighted combination of the values of the patient status **808**, patient information contained within the patient name **806**, and patient notes **812**. In some embodiments, the information contained in the patient notes **812** may be individually valued and/or weighted based on the information contained (e.g., aggression profile, target stimulation, side effects, etc.). Some side effects may be indicated as more severe than others, and the dashboard **800** may assign a different value or weight relating to different side effects. In addition, patients may be more prone to side effects during different stages of titration based on the intensity of titration, which again could receive different values or weighting by the dashboard **800**. Patients may also be more or likely to develop various side effects, or experience the side effects more severely based on tolerance; this can be taken into account by the dashboard **800** when determining patient priority **810**.

The patient priority **810** may be selected depending on a value of the weight profile crossing one or more thresholds defining various patient priorities **810**. In some embodiments, the patient priority **810** may be independent of a priority calculated for another patient (i.e., multiple patients may have the same priority level). In some embodiments, the patient may be compared to some or all of the other patients in the dashboard **800** to provide a unique patient priority **810** to each patient. In some embodiments, custom patient priorities **810** may be established by the physician based on physician knowledge that may not be recognized by the dashboard **800**.

The dashboard **800** may also provide addition functions for the physician to interact with and analyze patient information. In some embodiments, the dashboard **800** can provide a log of interactions with the dashboard **800**. In some embodiments, the dashboard **800** provides a log of interactions based on the patient, the person who was logged in, insurance, etc.

In some embodiments, the dashboard **800** provides controls for the physician to collect and analyze physiological data of the patient, modify stimulation parameters, and monitor and modify stimulation holds. In some embodiments, the dashboard **800** collects physiological patient data through remote communication with the implantable medical device of the patient or a home monitoring system of the patient. In some embodiments, the physiological data can be

updated for a patient in real time. In some embodiments, the physiological data is provided to the dashboard **800** periodically (e.g., daily, weekly, etc.).

In some embodiments, the dashboard **800** provides functions allowing a user to modify stimulation parameters. The stimulation parameters can be modified by changing an aggression profile, target parameter settings, titration step settings, etc. In some embodiments, the stimulation parameters can be remotely modified at any time. The patient may be notified to provide an update via a home monitoring unit. In some embodiments, the stimulation parameters are limited to remote modifications during specific times of the titration process (e.g., titration holds). In some embodiments, the remote modification of the stimulation parameters is limited. For example, the parameters can only be modified by certain predefined amounts, maximum amounts, or other limitations. In some embodiments, the modified stimulation parameters are updated in the implantable medical device with a home monitoring device. In some embodiments, the patient is alerted of an update and must take action to update the implantable medical device. In some embodiments, the implantable medical device is automatically updated.

In some embodiments, the physician is able to modify the hold settings of the titration for any given patient. In some embodiments, the physician can initiate a hold, clear a hold, modify the parameter level at which a hold is initiated, add additional holds, or modify the holds in other ways. Therefore, if the physician notices physiological data of the patient is indicating adverse side effects, if the dashboard **800** alerts the physician of adverse side effects, if the patient contacts the physician regarding adverse side effects, etc., then the physician can initiate a hold for the titration settings of the patient using the dashboard **800**. In some embodiments, the physician can initiate a hold with parameters different than the parameters that caused adverse side effects. By initiating the hold, the physician can schedule time to talk to the patient on the phone or in the office without allowing the adverse side effects to continue or worsen.

In some embodiments, the physician can clear a hold remotely using the dashboard **800**. For example, a physician may talk to the patient over the phone to determine if the patient is experiencing any adverse side effects once the dashboard **800** indicates a hold for the patient. The physician can then remotely clear the hold if no adverse side effects are being experienced by the patient.

In some cases, after the initial parameter and titration settings have been established, the physician may determine that a patient is more or less prone to side effects than initially determined. Accordingly, in some embodiments, the physician is able to modify the parameter settings associated with a future hold (e.g., instead of having a hold at 1.5 mA, have a hold at 2.0 mA), without modifying the titration settings, such that the hold occurs sooner or later than initially established. In some embodiments, the physician can add or remove a future hold instead of, or in addition to, modifying the parameter settings associated with a hold.

In some embodiments, the holds are updated in the implantable medical device with a home monitoring device of the patient. In some embodiments, the patient is alerted of an update and must take action to update the implantable medical device. In some embodiments, the implantable medical device is automatically updated.

FIG. 9 is a patient titration graph **900** of the titration assist management dashboard **800** of FIG. 8, according to an exemplary embodiment. The user can select a patient on the dashboard **800** to view in further detail. By selecting a

patient, the user can see the titration graph **900** that is specific to the selected patient. The graph **900** includes an x-axis **902**, a y-axis **904**, a stimulation level **906**, one or more holds **908**, and a current stimulation setting **910**.

The x-axis **902** is a unit of time (e.g., days, weeks, months, etc.), while the y-axis **904** is a parameter of stimulation (e.g., amplitude, frequency, duty cycle, etc.). In some embodiments, the user can change the units of the x-axis **902** and the y-axis **904** to provide alternative views of the titration settings. The stimulation level **906** is shown based on the units set for the x-axis **902** and the y-axis **904**. While the stimulation level **906** is shown in FIG. 9 as increasing uniformly, the stimulation level **906** is based on the titration parameters set forth, which may not increase in a uniform fashion.

The stimulation level **906** also includes the titration holds **908** that were established during set up of titration. The titration holds **908** may be set at equal intervals or may be set at varying intervals, based on the requirements set forth during set up. In some embodiments, the titration graph **900** also includes a marker showing the current stimulation setting **910**. The current stimulation setting **910** indicates the progression of the titration so the user can easily see how soon the next hold **908** will occur and the current stimulation setting **910**, past and future stimulation levels **906**.

FIG. 10 is a flowchart of a process **1000** for managing patients using the titration assist management dashboard **800**, according to an exemplary embodiment. The process **1000** includes receiving a plurality of patient information at **1002**, evaluating each patient at **1004**, determining a status of each patient at **1006**, determining a priority of each patient at **1008**, and sorting the patients based on user input at **1010**.

The titration assist management dashboard **800** receives the plurality of patient information at **1002**. In some embodiments, the titration assist management dashboard **800** receives patient information relating to a single patient one at a time. In some embodiments, the titration assist management dashboard **800** receives patient information relating to multiple patients at once. In some embodiments, the titration assist management dashboard **800** receives patient information by wirelessly communicating with an individual implantable medical device for each patient. In some embodiments, the titration assist management dashboard **800** receives patient information via wired or wireless communication with a programming wand. In some embodiments, the titration assist management dashboard **800** receives patient information via wireless communication with a remote device (e.g., home monitor, etc.). In some embodiments, the titration assist management dashboard **800** receives patient information via communication with another device located in the physician's office or the patient's home. In some embodiments, a user of the titration assist management dashboard **800** must actively prompt the titration assist management dashboard **800** to receive patient information. In some embodiments, the titration assist management dashboard **800** automatically collects patient information when certain criteria are met (e.g., device with patient information connected, device with patient information identified, patient is not currently in titration assist management dashboard, etc.). The patient information may include patient name, address, number, insurance information, titration assist parameters, patient notes **812**, etc.

The titration assist management dashboard **800** evaluates each patient based on the patient information at **1004**. The titration assist management dashboard **800** may evaluate the patient information to determine if any required information

is missing (e.g., name, insurance information, titration settings, etc.). In some embodiments, the titration assist management dashboard **800** prompts a user to enter the missing information (e.g., a pop-up screen, alert, etc.). In some embodiments, the titration assist management dashboard **800** flags a patient as having missing information (e.g., change in color, marking by patient, etc.). In some embodiments, the titration assist management dashboard **800** evaluates the patient information to determine if the patient is likely to obtain side effects from titration or need additional contact with the physician. In some embodiments, the titration assist management dashboard **800** evaluates patient height, weight, gender, titration settings, notes etc. to determine if the patient is likely to obtain side effects from titration or need additional contact with the physician.

The titration assist management dashboard **800** determines a status of each patient at **1006**. As described above, the patient status is the status of the progression of the titration based on the titration assist parameters established during interrogation of the implantable medical device. The patient status may include a stimulation parameters (e.g., amplitude, frequency, pulse width, etc.). In some embodiments, the patient status may also indicate if a HOLD is present in the titration progression, an indication of the weeks that have passed since titration has started, or another indication of the time of titration. The titration assist management dashboard **800** is able to update the patient status based on the settings of titration, such as updating the patient status to indicate a hold is present. The patient status may be updated periodically (e.g., daily, weekly, when a titration setting changes, etc.) without being in communication with the implantable medical device by knowing the parameters of the titration assist and updating the dashboard according to the titration assist parameters that are recorded in the titration assist management dashboard **800**.

The titration assist management dashboard **800** determines a priority of each patient at **1008**, in some implementations. The patient priority indicates a likelihood of the patient needing attention (e.g., most likely to need a clinical visit). The patient priority may be based on a combination of patient status, patient information contained within the patient name, and patient notes. The patient status, patient information contained within the patient name, and patient notes may receive a value based on the information contained within. The patient priority may be a weighted combination of the values of the patient status, patient information contained within the patient name, and patient notes. In some embodiments, the information contained in the patient notes **812** may be individually valued and/or weighted based on the information contained (e.g., aggression profile, target stimulation, side effects, etc.). Some side effects may be indicated as more severe than others, and the dashboard **800** may assign a different value or weight relating to different side effects. In addition, patients may be more prone to side effects during different stages of titration based on the intensity of titration, which again could receive different values or weighting by the dashboard. Patients may also be more or less likely to develop various side effects, or experience the side effects more severely, based on tolerance; this can be taken into account by the titration assist management dashboard **800** when determining patient priority.

The patient priority may be selected depending on a value of the weight profile crossing one or more thresholds defining various patient priorities. In some embodiments, the patient priority may be independent of a priority calculated for another patient (i.e., multiple patients may have the same

priority level). In some embodiments, the patient may be compared to some or all of the other patients in the titration assist management dashboard **800** to provide a unique patient priority to each patient. In some embodiments, custom patient priorities may be established by the physician based on physician knowledge that may not be recognized by the titration assist management dashboard **800**.

The titration assist management dashboard **800** may indicate patient priority in a variety of ways. In some embodiments, the patients are color coded based on patient priority (e.g., red for high priority, green for low priority, etc.). In some embodiments, the patient priority is a number. In some embodiments, the patient priority is a symbol, marker, or other visual indication of patient priority.

The titration assist management dashboard **800** sorts the patients based on user input at **1010**. The user may be able to select a default setting for sorting the patients, such that if no sorting has been selected, the patients will be sorted according to the default setting. The user may be able to sort the patients based on patient name, patient information, patient status, patient priority, etc. In some embodiments, the patients can be sorted in ascending or descending order based on the selected criteria.

By creating a dashboard (e.g., dashboard **800**), a physician can monitor a plurality of patients with a single device. The physician is able to easily view the status of any patient without having to interrogate their implantable medical device. In addition, if a patient calls the physician's office, the physician can determine what the stimulation parameters are for titration and may be able to evaluate the patient over the phone or recommend that the patient come into the office for a check-up based on urgency and severity. The physician can also take notes on the dashboard based on information received from the patient during the call.

As discussed above, the VNS system **11** may perform fully automated or partially automated titration of VNS stimulation parameters. For example, in some arrangements, the VNS system **11** performs automated titration of VNS stimulation parameters between an initial stimulation intensity and a hold intensity prescribed by the patient's physician (e.g., by making small, periodic stimulation intensity increases between the initial and hold intensities). Once the hold intensity is reached, the patient must visit the physician. The physician evaluates the patient for side effects and decides whether to remove the hold and continue titration. This process is continued until the stimulation reaches a physician-prescribed target intensity. In other arrangements, the physician sets an initial stimulation intensity and a target stimulation intensity. The VNS system **11** then performs titration by automatically making small, periodic stimulation intensity increases between the initial and target intensities such that the patient's nervous system is allowed to accommodate to each new intensity. Once the target intensity is achieved, the patient returns to the physician for final intensity adjustments. As such, evaluation of the heart rate effects at higher stimulation intensities occur while the patient is in the clinic environment with appropriate physiological monitoring.

One advantage of performing titration in this manner is that this method of titration reduces or eliminates patient and physician workloads as the patient does not need to visit the clinic for any titration adjustment. The frequency of titration can also occur at a rate of adjustment (e.g., multiple small titration step increases per day) that would otherwise not be practically feasible for patients using existing alternatives of on-site visits for every programmed adjustment. Moreover, this method of titration assures that the patient receives

therapeutic levels of stimulation quickly while simultaneously minimizing the likelihood of serious adverse effects (e.g., minimizing the chances of the patient developing symptomatic bradycardia). To illustrate, a traditional titration method may require 8-12 clinic visits, 12-18 hours of programming time, and 24-48 hours of patient time exposure. Yet, the traditional titration method may only allow for 6-10 therapy adjustments with 10-12 weeks required until the stimulation intensity reaches therapeutic levels. By contrast, the present systems and methods for titration may require only 2 clinic visits, 2 hours of programming time, and 6 hours of patient time exposure, while allowing for 25+ therapy adjustments with only 4-6 weeks required until the stimulation intensity reaches therapeutic levels.

Moreover, in various embodiments, the VNS system **11** may be programmable with high resolution stimulation parameters that enable physicians to use an optimal set of stimulation parameters (e.g., current amplitude, frequency, pulse width, ON-time and OFF-time). As an illustration, physicians may be able to fine-tune therapy around the patient's neural fulcrum (i.e., an operating point formed by a combination of stimulation intensity and duty cycle that gives rise to a small and repeatable reduction in heart rate) using the high resolution parameters. These high resolution stimulation parameters improve the titration experience for the patient by enabling smaller intensity steps, which allows patients to reach a therapeutic range without requiring a clinic visit for every stimulation adjustment.

Additionally, automatic adjustment of the stimulation parameters may occur according to settings programmed by the physician or modified or selected by the physician from factory settings. For example, the physician may be able to select from specific parameters provided by the VNS system **11** or from a parameter range provided by the VNS system **11**. As an illustration, the physician may be able to select between 0.125 mA (for 0.0 to 1.875 mA initial to target intensities) and 0.25 mA (for 2.0 to 3.5 mA target intensities) current amplitude increments; 2, 3, 4, 5, 6, 7, 8, 9, 10, 15, and 20 Hz frequency increments; and 130, 150, 180, 210, 250, 275, 300, 370, and 500 μ s pulse width increments, with the stimulation occurring according to a default duty cycle (e.g., 14 seconds ON and 66 seconds OFF, with a 2 second ramp-up and a 2 second ramp-down). Alternatively, the automatic adjustment may occur based entirely or almost entirely on factory setting, such as on a factory-adjustable fixed time interval (e.g., four steps per day), during fixed time periods (e.g., only during the daytime when the patient is less likely to be asleep, as described below with reference to FIGS. 16 and 17), according to fixed trajectories (e.g., as shown in FIG. 13), and so on. The VNS system **11** may then determine whether any changes should be made to the stimulation parameters and, if so, which changes should be made during the titration process.

As an example, a physician may set (e.g., via the external programmer **40**) the initial stimulation intensity (e.g., zero stimulation, with a 0 mA amplitude, 60 μ s pulse width, and 5 Hz frequency) and the target stimulation (e.g., 1.5-2.0 mA amplitude, 250 μ s pulse width, and 5 Hz frequency). The VNS system **11** may then dictate titration according to an algorithm such that four titration steps (e.g., 0.125 mA, 30 μ sec, and/or 0.1 Hz steps, which result in smooth intensity increases over time) are implemented a day with no changes permitted between 1:00 and 6:00 AM (e.g., to avoid the possibility that the patient may go to sleep with no side effects but later wake up from the side effects, such as a cough). Once the target stimulation is reached, the physician may make further adjustments as the patient will likely

tolerate the adjustments due to the patient's nervous system having become accommodated to the stimulation.

In various embodiments, the VNS system **11** may be programmed (e.g., by the external programmer **40** or the recordable memory **29** can contain certain instructions when the neurostimulator **12** is implanted) to perform the titration according to a stimulation profile adapted to reduce patient side effects during the titration process. As an illustration, FIG. **11** is a patient titration graph **1100** incorporating a "dwell point," according to an exemplary embodiment. The graph **1100** includes an x-axis **1102** and a y-axis **1104**. The x-axis **1102** is a unit of time (e.g., days, weeks, months, etc.), while the y-axis **1104** is a parameter of stimulation (e.g., amplitude, frequency, duty cycle, etc.). In various embodiments, the titration graph **1100** may be incorporated in a user interface displayed to a physician (e.g., on the external programmer **40** or other computing device) during the titration configuration process.

The graph **1100** includes three titration rates at which at least one stimulation parameter (e.g., output current, frequency, pulse width, duty cycle, etc.) is increased. In various embodiments, each of the three titration rates are configured such that the at least one stimulation parameter is gradually increased. The titration according to FIG. **11** initially occurs at a first rate **1106** until the at least one stimulation parameter reaches a first target value **1108**. Subsequently, the titration shifts to a second rate **1110**, which is less than the first rate **1106**. Because the at least one parameter is increased more slowly during this portion of the titration, the second rate **1110** marks a "dwell point" in the graph **1100**. The titration occurs according to the second rate **1110** until the at least one stimulation parameter reaches a second target value **1112**. Once the second target value **1112** is reached, the titration occurs at a third rate **1114**. The third rate **1114** is greater than the second rate **1110** and may be the same as the first rate **1106** (e.g., as shown in FIG. **11**) or different from the first rate **1106**. The at least one stimulation parameter increases according to the third rate **1114** until, e.g., a titration hold is reached or the stimulation is fully titrated.

In some embodiments, the first rate **1106**, second rate **1110**, and third rate **1114** may be continuous rates, as shown in FIG. **11**. In other embodiments, the first rate **1106**, second rate **1110**, and third rate **1114** may be step functions (e.g., with steps corresponding to small stimulation intensity increases that occur several times each day, such as four times a day). In such embodiments, the first rate **1106** and third rate **1114** may be greater than the second rate **1110** because, e.g., the first rate **1106** and third rate **1114** include greater stimulation parameter increases in each step and/or include steps that last for shorter periods of time compared to the steps of the second rate **1110**. Additionally, in certain embodiments, the second rate **1110** may instead be a prolonged hold on the titration such that the at least one stimulation parameter is kept constant for a certain period of time (e.g., 1-7 days), after which the titration resumes at the third rate **1114**. Moreover, in certain embodiments, the second rate may depend on the aggressiveness of a titration profile used to accomplish titration of the neurostimulation, the profile determined either by the physician or as determined by the VNS system **11** itself.

Performing titration according to the graph **1100** may be beneficial for patients because it has been observed that a patient's adaptation to titration is non-linear. Specifically, there is a period in the stimulation intensity progression (e.g., at a stimulation amplitude of 0.75 to 1.0 mA) where the patient's adaptation tends to stall and the patient requires additional time for adaptation to occur. Accordingly, incor-

porating the “dwell point” via the lowered second rate **1110** during this period (e.g., between the stimulation amplitudes of 0.75 to 1.0 mA, where 0.75 mA is the first target value **1108** and 1.0 mA is the second target value **1112**) allows a patient to more smoothly adapt to the stimulation intensity while minimizing deleterious side effects.

In various embodiments, the VNS system **11** implements titration incorporating a dwell point automatically. In other embodiments, however, the VNS system **11** may incorporate a dwell period based on external input (e.g., from a physician using the programmer **40**). For example, a physician may program the VNS system **11** to titrate according to the second rate for a certain number of days, until a certain stimulation amplitude is reached, and so on.

FIG. **12** is a flowchart of a titration process **1200** (e.g., implemented by the VNS system **11**) that incorporates a titration dwell point, according to an exemplary embodiment. First, the VNS system **11** delivers a first neurostimulation signal with a first set of parameters at **1202**. The first set of parameters has a first value for, e.g., at least one of an output current, frequency, pulse width, or duty cycle. The VNS system **11** then increases at least one of the first parameters at a first rate (e.g., the first rate **1106**) until the at least one parameter reaches a first target value (e.g., the first target value **1108**) at **1204**. Once the first target value is reached, the VNS system **11** ceases delivery of the first neurostimulation signal at **1206**.

Subsequently, the VNS system **11** delivers a second neurostimulation signal with a second set of parameters at **1208**. The second set of parameters has a second value for, e.g., at least one of output current, frequency, pulse width, or duty cycle, where the second value is equal to the first target value. For the purposes of this disclosure, a first value may be considered “equal” to a second value if the first value is exactly equal to the second value or if the first and second values are within a threshold of each other (e.g., five percent). The VNS system **11** increases at least one of the second parameters at a second rate different from the first rate (e.g., the second rate **1110**) until the at least one parameter reaches a second target value (e.g., the second target value **1112**) at **1210**. For example, as discussed above, the second rate may be less than the first rate, or the second rate may be an application of a substantially constant neurostimulation signal for a period of time. As another example, as also discussed above, the first and second rates may be stepwise functions, where the second steps are applied for greater periods of time than the first steps.

Further, as shown in FIG. **11**, once the second target value is reached, the VNS system **11** may cease delivery of the second neurostimulation signal and apply a third neurostimulation signal with a third set of parameters having a third value equal to the second target value. The VNS system **11** may then increase at least one of the third parameters at a third rate that is equal to the first rate or, alternatively, different from the first rate and the second rate. In some embodiments, the VNS system **11** may also implement a hold between ceasing delivery of the second neurostimulation signal and applying the third neurostimulation signal.

As another example of titration adapted to reduce adverse side effects, FIG. **13** is a patient titration graph **1300** illustrating different titration aggressiveness profiles, according to an exemplary embodiment. In various embodiments, the titration graph **1300** may be incorporated in a user interface displayed to a physician (e.g., on the external programmer **40** or other computing device) during the titration configuration process. The aggressiveness profiles shown in graph **1300** are a medium aggressiveness profile **1306**, a high

aggressiveness profile **1308**, and a low aggressiveness profile **1310**. Each of the aggressiveness profiles **1306**, **1308**, and **1310** is associated with a rate of titration (e.g., an increment value between initial stimulation parameters and a target value **1312**), where the rate of titration for the high aggressiveness profile **1308** is greater than the rate of titration for the medium aggressiveness profile **1306**, which in turn is greater than the rate of titration for the low aggressiveness profile **1310**. Accordingly, patients using the high aggressiveness profile **1308** transition to a target value **1312** from initial stimulation parameters for within a shortest time interval (e.g., 30 days), while patients using the low aggressiveness profile **1310** transition to the target value **1312** from initial stimulation parameters within a longest time interval (e.g., 60 days), and patients using the medium aggressiveness profile **1306** transition to the target value **1312** from initial stimulation parameters within an interval sometime in between the high aggressiveness profile **1308** and the low aggressiveness profile **1310** (e.g., 45 days). Alternatively, or additionally, the low aggressiveness profile **1310**, the medium aggressiveness profile **1306**, and the high aggressiveness profile **1308** may be associated with increasingly greater target intensity values.

With each profile **1306**, **1308**, and **1310**, at least one stimulation parameter (e.g., stimulation current amplitude, pulse width, frequency, duty cycle, etc.) is increased (e.g., gradually increased) according to the rate of titration of the profile **1306**, **1308**, or **1310**. In some embodiments, the rate of titration may be linear and continuous for each of the aggressiveness profiles **1306**, **1308**, and **1310**, as shown in FIG. **13**. In other embodiments, each rate of titration may instead be implemented as a step function where, e.g., the high aggressiveness profile **1308** includes steps that are the closest together in time and the low aggressiveness profile **1310** includes steps that are the furthest apart in time. For example, the titration may occur according to a fixed time interval during the daytime hours, where the number of steps implemented during that fixed time interval depends on which profile **1306**, **1308**, or **1310** is used.

The profiles **1306**, **1308**, and **1310** may also be linear, or the profiles **1306**, **1308**, and **1310** may be non-linear (e.g., incorporating a dwell period, as shown in FIG. **11** above, or steeper after the stimulation amplitude exceeds a threshold after which intolerance is less likely, such as 2.0 mA). Additionally, in certain embodiments, the shape, step frequency, step sizes, etc. of the profiles **1306**, **1308**, and **1310** may depend on which stimulation parameter is being increased according to the profile. For example, for the medium aggressiveness profile **1306** where a final stimulation with a current amplitude of 3.5 mA, a frequency of 20 Hz, and a pulse width of 500 μ s is desired, the profile **1306** may include steps of 0.125 mA increments for current amplitude, 0.2 Hz for frequency, and 30 μ s for pulse width. If the profiles **1306**, **1308**, and **1310** are step functions, at each step the increase amount and type of the stimulation increase is determined by the trajectory of the profile **1306**, **1308**, and **1310**. As an example, at each step, amplitude, pulse width, and frequency may all increase, only one or two parameters may increase, or none of the parameters may increase depending on the trajectory of the profile **1306**, **1308**, and **1310** according to which the stimulation is being delivered. Together, the steps for each of the stimulation parameters create a titration curve with a smooth increase in intensity.

In various embodiments, the profiles **1306**, **1308**, and **1310** have the same shape and only differ on the scale of the titration. For example, the profiles **1306**, **1308**, and **1310** are

each stepwise functions using the same sizes of steps. The profiles **1306**, **1308**, and **1310** instead differ based on how much time is allowed to pass between moving to the next step. Accordingly, the steps of the high aggressiveness profile **1308** are closer together (e.g. more compressed) than the medium aggressiveness profile **1306**, which in turn has steps that are closer together than the low aggressiveness profile **1310**. However, in other embodiments, the profiles **1306**, **1308**, and **1310** may instead have differently sized and/or spaced steps, the profiles **1306**, **1308**, and **1310** may have different shapes, and so on. Additionally, in certain embodiments, the profiles **1306**, **1308**, and **1310** may implement the same titration for a certain period of time (e.g., a standardized portion of the titration) and then branch out into their different, respective titration aggressiveness functions, for example, after a certain stimulation intensity threshold is reached.

For VNS systems **11** implementing different titration aggressiveness profiles, such as shown in graph **1300**, the physician inputs the final stimulation parameters that the titration is designed to reach, and the default titration profile for reaching those parameters is the medium aggressiveness profile **1306**. The medium aggressiveness profile **1306** may be the factory-adjustable default for the VNS system **11**, or the medium aggressiveness profile **1306** may be the recommended profile for the physician to select (e.g. via the programmer **40**) for the titration. The physician may then be able to customize the stimulation for the patient. For example, the physician may modify the duty cycle from the default (e.g., 14 seconds ON, 66 seconds OFF, with an amplitude ramp up at the beginning of each cycle and an amplitude ramp down at the end of each cycle), after which the duty cycle will be constant during the titration unless modified again by the physician.

Once the VNS system **11** is thus initialized, stimulation will occur according to the medium aggressiveness profile **1306**. However, if the patient experiences unwanted side effects as the titration progresses, the patient can provide a feedback signal indicating that the patient is experiencing unwanted side effects to the VNS system **11**, for example, via the patient magnet **730**. Those of skill in the art will appreciate, however, that while reference is made herein to the patient magnet **730**, the patient may be able to signal unwanted side effects to the VNS system **11** through another mechanism, such as by an external patient programmer. Once the patient signals to the VNS system **11** that the patient is experiencing unwanted side effects, the VNS system **11** modifies the titration accordingly. The VNS system **11** may also transmit, via an implantable pulse generator (e.g., the neurostimulator **12**), a receipt of the feedback signal to an external device.

In one embodiment, if the patient places the patient magnet **730** over the implanted system **700** for at least 10 seconds but less than 60 seconds, the implanted system **700** automatically decrements the stimulation intensity along the profile **1306**, **1308**, or **1310** being used for the patient. If the titration is decremented three times, the implanted system **700** automatically switches the patient to a less aggressive titration profile, if possible. For example, the implanted system **700** switches the patient from the high aggressiveness profile **1308** to the medium aggressiveness profile **1306** or switches the patient from the medium aggressiveness profile **1306** to the low aggressiveness profile **1310**. In switching to a less aggressive titration profile, the implanted system **700** identifies and moves to a location on the less aggressive titration profile that has a stimulation intensity less than or equal to the stimulation intensity currently being

used on the current profile. As an example, the implanted system **700** moves to the highest location on the less aggressive titration profile that has a amplitude, frequency, and pulse width less than or equal to the amplitude, frequency, and pulse width currently being applied to the patient according to the more aggressive titration profile. Alternatively, if the patient places the patient magnet **730** over the implanted system **700** for at least 60 seconds, the implanted system **700** inhibits stimulation until the patient magnet **730** is removed. Once the patient magnet **730** is removed, stimulation resumes, and the stimulation intensity is not decremented. In some arrangements, if the patient uses the patient magnet **730** to decrement the stimulation intensity, move to a less aggressive profile, and/or pause the stimulation, the implanted system **700** keeps a record of the magnet activation. Additionally, the dashboard **800** may factor the magnet activation record into the determination of patient priority **810**.

Alternatively, in another embodiment, the patient informs the physician that the patient has experienced unwanted side effects, and the physician can switch the patient to a less aggressive titration profile in response to the patient feedback. Alternatively, if the patient is experiencing no side effects, the physician can switch the patient to a more aggressive profile (e.g., from the low aggressiveness profile **1310** to the medium aggressiveness profile **1306** or from the medium aggressiveness profile **1306** to the high aggressiveness profile **1308**). The physician may be able to switch the profiles in person and/or remotely.

In some arrangements, once the patient is switched from a higher aggressiveness profile to a lower aggressiveness profile, the patient cannot be switched back to the higher aggressiveness profile. Conversely, in other arrangements, the patient may be switched back to a higher aggressiveness profile (e.g., either automatically by the VNS system **11** or by the physician using the programmer **40**) if the patient experiences no further subsequent side effects. Additionally, in various embodiments, the VNS system **11** keeps a record of the timing of all stimulation parameter changes throughout the titration period. The log includes a record of, e.g., the timing and duration of all magnet **730** activations and can be downloaded from the implanted system **700** for viewing by the physician.

Additionally, those of skill in the art will appreciate that graph **1300** is merely exemplary of different aggressiveness profiles. A VNS system **11** may implement additional or fewer aggressiveness profiles. For example, a VNS system **11** may be adapted to implement a titration aggressiveness profile that is even less aggressive than the low aggressiveness profile **1310**. Thus, after being moved from the medium aggressiveness profile **1306** to the low aggressiveness profile **1301**, a patient may indicate to the VNS system **11** (e.g., via the patient magnet **730**) that the patient is still experiencing adverse side effects with the low aggressiveness profile **1310**. Accordingly, the VNS system **11** may modify the neurostimulation signal to conform to the even less aggressive titration profile, deliver the modified neurostimulation signal, and titrate (e.g., gradually increase) the neurostimulation according to the even less aggressive titration profile.

FIG. **14** is a flowchart of a titration process **1400** (e.g., implemented by the VNS system **11**) including different titration aggressiveness profiles, according to an exemplary embodiment. First, the VNS system **11** delivers a neurostimulation signal in conformance with a first titration aggressiveness profile at **1402**. For example, the first titration aggressiveness profile may be the medium aggressiveness profile **1306** shown in FIG. **13**. Next, the VNS system

11 increases at least one parameter of the first titration aggressiveness profile (e.g., output current, frequency, pulse width, or duty cycle) towards a target value (e.g., the target value 1312) at 1404. The VNS system 11 then receives a feedback signal from the patient indicating adverse effects from the neurostimulation signal at 1406. The patient may provide the feedback signal to the VNS system 11 via a patient magnet, as described above, or through any other feedback mechanism, such as by an external patient programmer. In response to the feedback signal, the VNS system 11 modifies the neurostimulation signal to conform with a second titration aggressiveness profile at 1408. For example, as discussed above, the VNS system 11 may modify the neurostimulation signal to match the low aggressiveness profile 1310 shown in FIG. 13. The VNS system 11 delivers the neurostimulation signal with the second titration aggressiveness profile at 1410. The VNS system 11 then increases at least one parameter of the second titration aggressiveness profile towards the target value at 1412.

As described above, stimulation may be applied by the VNS system 11 according to a duty cycle, where the duty cycle includes an ON period, an OFF period, an amplitude ramp up at the beginning of each cycle, and an amplitude ramp down at the end of each cycle. Accordingly, in many situations, it would be beneficial for the physician in a clinic setting to be able to determine when therapy is actually being applied by the VNS system 11. As such, referring back to FIG. 3, in addition to allowing the physician to interrogate the neurostimulator 12 and set parameters, the external programmer 40 may display indicate to the user whether the VNS system 11 is currently delivering therapy to the patient (e.g., the “therapy active” status) based on interrogation of the neurostimulator 12. In various embodiments, the external programmer 40 may display the therapy active status on the visual display 44 of the programming computer 41. For example, in one embodiment, the visual display 44 may include an on-off therapy status indicator and show via the indicator whether the neurostimulator 12 is ON or OFF. In another embodiment, in response to an interrogation of the neurostimulator 12, the external programmer 40 may display a countdown time to the next stimulation burst.

The external programmer 40 may display the therapy active status in real time. Alternatively, the external programmer 40 may display the therapy active status asynchronously from the interrogation response from the neurostimulator 12 such that the external programmer 40 accounts for the transport delay between the external programmer 40 and the neurostimulator 12 and the display of the therapy active status coincides with the stimulation burst being applied to the patient. In another arrangement, when interrogating the neurostimulator 12, the external programmer 40 may query the neurostimulator 12 to receive a block of timing data that enables the programmer 40 to sync its therapy active displays to the stimulation delivered by the VNS system 11. Additionally, in yet another arrangement, the external programmer 40 may display the therapy active status in real time or accounting for the transport delay while in communication with the neurostimulator 12 and also receive a block of timing data that enables the programmer 40 to continue displaying the therapy active status once communication ceases between the external programmer 40 and the neurostimulator 12.

FIG. 15 is a flowchart of a process 1500 (e.g., implemented by the programmer 40) of displaying an indication of active neurostimulation, according to an exemplary embodiment. The external programmer 40 first communicates with an implantable pulse generator (e.g., the neuro-

stimulator 12) to collect data about ongoing neurostimulation applied by the neurostimulator 12 at 1502. For example, as discussed above, the external programmer 40 interrogates the neurostimulator 12 and receives back information on whether the neurostimulator 12 is currently providing active stimulation, information relating to the timing to the next active stimulation, information relating to the timing of future stimulation bursts, and so on. The external programmer 40 then displays (e.g., via the visual display 44) an indication relating to a timing of neurostimulation bursts applied by the neurostimulator 12 at 1504. As an example, the external programmer 40 may display a therapy active status such as an on-off therapy status indicator, a countdown time to the next stimulation burst, and so on. Additionally, the external programmer 40 may display the therapy active status in real time, based on a transport delay between the external programmer 40 and the neurostimulator 12 such that the display coincides with the stimulation burst actually being applied, based on a block of timing data from the neurostimulator 12, and so on.

As yet another example of titration configured to reduce adverse side effects in the patient, FIG. 16 is a patient titration graph 1600 incorporating a titration black-out period, according to an exemplary embodiment. The graph 1600 includes an x-axis 1602 and a y-axis 1604. The x-axis 1602 is a unit of time (e.g., days, weeks, months, etc.), while the y-axis 1604 is a parameter of stimulation (e.g., amplitude, frequency, pulse width, duty cycle, etc.). In various embodiments, the titration graph 1600 may be incorporated in a user interface displayed to a physician (e.g., on the external programmer 40 or other computing device) during the titration configuration process.

As shown in the graph 1600, titration may be accomplished as a step function (e.g., with multiple small stimulation intensity increase steps implemented per day, such as four times a day) incorporating one or more black-out periods 1606, shown in graph 1600 as the hold period between a start hold time 1608 and a stop hold time 1610. During the black-out period 1606, stimulation is kept constant, and no titration steps occur. Black-out periods 1606 may be implemented on titration to help ensure that the patient does not observe adverse side effects and thereby experience discomfort during the titration. For example, black-out periods 1606 may be implemented during a time of day or a duration of time during which the patient is more likely to experience side effects of the titration. Accordingly, in some embodiments, black-out periods 1606 may be implemented during the night because the increased stimulation intensity as a result of increased titration steps causes the patient to awaken. Conversely, in other embodiments, black-out periods 1606 may be implemented during the daytime, as the patient is less likely to notice an adverse side effect from the titration when the patient is sleeping.

Black-out periods 1606 may be programmed into a patient's titration schedule through various methods. In one embodiment, the timing and duration of black-out periods 1606 are preset as part of the firmware design (e.g., programmed to occur from 10 pm to 8 am, during when most individuals sleep). In another embodiment, the timing and duration of black-out periods 1606 are programmable by technicians for the VNS system 11. In a third embodiment, a physician can program a preset timing and duration of black-out periods 1606 (e.g., by using the programmer 40) based on the patient's schedule (e.g., program black-out periods 1606 during time periods when the patient is usually awake or usually asleep). In a fourth embodiment, the patient can program a preset timing and duration of black-

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out periods **1606** (e.g., by using the external programmer **707**). In a fifth embodiment, the patient can use a user device (e.g., a handheld unit, the patient magnet **730**, etc.) to indicate to the VNS device **11** when the patient is going to bed and when the patient has woken up, and the VNS device begins or suspends black-out periods **1606** accordingly. Alternatively, the patient can use the user device to delay a default black-out period **1606** or start a default black-out period **1606** early.

FIG. **17** is a flowchart of a titration process **1700** (e.g., implemented by the VNS system **11**) incorporating a titration black-out period (e.g., the black-out period **1606**), according to an exemplary embodiment. The VNS system **11** first delivers a neurostimulation signal at **1702**. The neurostimulation signal is delivered in conformance with a set of parameters (e.g., output current, frequency, pulse width, duty cycle, etc.). The VNS system **11** begins titrating the neurostimulation signal at **1704**. For example, the VNS system **11** increases (e.g., gradually increases) at least one stimulation parameter of the neurostimulation signal according to a stepwise function. The VNS system **11** then receives a first indicator associated with a titration hold time and/or a titration hold duration (e.g., the start hold **1606** instruction shown in FIG. **16**) at **1706**. As discussed above, the first indicator may be a preset black-out start hold time **1608** (e.g., fixed as part of the firmware design or programmed by a technician, a physician, or the patient) or may be an indicator sent by a user device to the VNS system **11** (e.g., indicating that the patient is planning on going to bed or that the patient has awoken).

In response, the VNS system **11** holds titration of the neurostimulation signal by continuing the neurostimulation signal at the signal parameters being delivered at the time the first indicator was received at **1708**. Subsequently, the VNS system **11** receives a second indicator associated with a titration resumption time and/or the completion of the titration hold time (e.g., the stop hold time **1610**) at **1710**. For example, as discussed above, the second indicator may be the expiration of the preprogrammed black-out period or may be an indicator sent by the user device to the VNS system **11** (e.g., indicating that the patient has awoken or that the patient is planning on going to bed). In response to the second indicator, the VNS system **11** resumes titrating the neurostimulation signal at **1712**. Additionally, while FIG. **16** illustrates a resumption of the titration after the black-out period **1606** occurring at the same level as before the black-out period **1606**, in some embodiments, the VNS system **11** may increase the neurostimulation after the black-out period **1606** and/or modify the rate of the titration after the black-out period **1606**. As such, through the titration process **1700**, a temporary titration hold may be applied to the neurostimulation provided to the patient to reduce adverse effects observed by the patient during the titrating.

In various embodiments, the VNS system **11** may also be configured to provide "titration training" to the patient under the supervision of a physician. The goal of the titration training is to provide increasing stimulation intensity to the patient at a controlled rate over a relatively short period of time (e.g., 10 minutes) so that the patient can experience the sensation of stimulation. For example, at the beginning of the titration period for a patient, the physician interacts with the patient in a titration training session. During the session, the VNS system **11** provides stimulation levels that are (1) imperceptible, (2) perceptible but tolerable, and/or (3) perceptible and slightly intolerable to the patient (e.g., in response to an indication from the physician via the external programmer **40** to conduct titration training). By educating

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the patient, and well as the patient's caregiver in certain arrangements, through the titration training session on the expected levels of stimulation during the titration period, patients may become acclimated to the sensation of mild stimulation in the midst of rapid accommodation to stimulation. In this way, the training may improve the timeliness and sensitivity of magnet interventions from the patient in response to the titration (e.g., by teaching the patient what side effects feel like so that the patient only uses the magnet interventions when a side effect is actually present). As such, these training sessions may help the titration and patient accommodation processes to proceed smoothly without unnecessary interruptions. These sessions may also allow the physician to assess the patient and/or caregiver motivation, cognitive ability, and physical abilities before the physician activates the titration.

In various embodiments, the physician of a patient implanted with the neurostimulator **12** of the VNS system **11** is provided with a user interface designed to facilitate the physician in configuring titration for the patient. FIG. **18** shows a user interface **1800** for the VNS system **11**, according to an exemplary embodiment. The user interface **1800** may be displayed to a user, such as the physician, on a display of a titration management device (e.g., on the visual display **44** of the external programmer **40**, on a display of a computing device coupled to an external programmer, etc.). As shown in FIG. **18**, the user interface **1800** includes a top portion **1802** with information for the user about the VNS system **11**, a middle portion **1803** configured to aid the user in setting up or modifying titration for the patient, and a bottom portion **1804** for communicating with the neurostimulator **12** of the VNS system **11**.

Looking first to the bottom portion **1804**, the user interface **1800** includes three communication buttons: an interrogate button **1806**, a program titration button **1808**, and a start/stop titration button **1810**. The interrogate button **1806** is configured to interrogate the neurostimulator **12** to populate the user interface **1800** with information relating to the VNS therapy provided by the neurostimulator **12**. For example, as shown in FIG. **18**, the interrogate button **1806** includes an image of a programming wand, such as programming wand **42**. In some embodiments, when the user presses the interrogate button **1806**, the user is shown a pop-up box on the user interface **1800** instructing the user to place the programming wand **42** near the patient (e.g., place the programming wand **42** on the patient's chest, wave the programming wand **42** in predetermined pattern near the patient, etc.). For example, the pop-up box may instruct the user to hold the programming wand **42** steady. The programming wand **42** interrogates the neurostimulator **12** and receives information relating to the VNS therapy being provided to the patient by the neurostimulator **12**. The received information is then displayed on the user interface **1800**, as discussed in further detail below. In other arrangements, however, the interrogation occurs through a different mechanism in response to the user pressing the interrogate button **1806**, such as by the external programmer **40** connecting to the neurostimulator **12** via Bluetooth or RFID to interrogate the neurostimulator **12**.

The program titration button **1808** is configured to transmit a titration program configured by the user via the user interface **1800**, as described in further detail below, to the neurostimulator **12**. As an example, as shown in FIG. **18**, the interrogate button **1806** also includes an image of a programming wand, such as programming wand **42**. When the user presses the program titration button **1808**, the user is shown a second pop-up box on the user interface **1800**

instructing the user to use the programming wand **42** near the patient (e.g., place the programming wand **42** on the patient's chest, wave the programming wand **42** in a predetermined pattern different from the interrogation predetermined pattern, etc.). The programming wand **42** then transmits the titration program to the neurostimulator **12**, which may start the titration immediately, wait for a programmed starting date for the titration, or wait for a second signal from the user indicating that the neurostimulation should be started. It should be understood, however, that in other arrangements, interrogating and/or programming the neurostimulator **12** may occur through a different mechanism, such as by the external programmer **40** connecting to the neurostimulator **12** via Bluetooth or RFID to program the neurostimulator **12**.

The start/stop titration button **1810** is configured to allow the user to start and stop the titration being applied to the patient. If titration is currently being applied to the patient, the start/stop titration button **1810** is configured to show "stop titration" (e.g., along with a square, as shown in FIG. **18**) such that the user knows that pressing the button **1810** will stop the titration being applied. Conversely, if titration is not currently being applied to the patient, the start/stop titration button **1810** is configured to show "start titration" (e.g., along with a green triangle). The user then knows that pressing the button **1810** will start or resume the titration according to the programmed schedule. For example, the patient's physician may decide to temporarily suspend the titration for the patient by pressing the button **1810**, which shows "stop titration." Later, the physician may resume the titration by pressing the button **1810**, which shows "start titration," again. In some embodiments, pressing the start/stop titration button **1810** will cause the user interface **1800** to show a pop-up message instructing the user to place a programming wand, such as the programming wand **42**, near the patient. The programming wand **42** then transmits the start/stop instruction to the neurostimulator **12**. In other embodiments, pressing the start/stop titration button **1810** causes the external programmer to transmit the start/stop instruction through a different mechanism, such as Bluetooth or RFID.

Looking next to the top portion **1802** and the middle portion **1803**, the user interface **1800** includes a number of fields with information about the VNS system **11**. In various embodiments, the fields are updated in real-time or near real-time (e.g., within a threshold amount of time from when changes occur, such as no greater than one second lag) to reflect the actual therapy and titration status of the neurostimulator **12** (e.g., in response to the user interrogating the neurostimulator **12** after pressing the interrogate button **1806**). For example, the user may press the interrogate button **1806** and hold, or have the patient hold, the programming wand **42** over the neurostimulator **12** to constantly receive real-time updates that reflect the actual therapy and titration status of the neurostimulator **12**. Alternatively, in other embodiments, the fields are shown in near real-time. As an example, the user may press the interrogate button **1806** and hold the programming wand **42** over the neurostimulator **12** to receive information about the actual therapy and titration status of the neurostimulator **12**, as well as predicted future therapy and titration information for the neurostimulator **12** (e.g., the future titration schedule and duty cycle). The user interface **1800** then displays therapy and titration information for the neurostimulator **12** that should align with the actual therapy and titration information of the neurostimulator **12** based on the predicted future information.

Accordingly, FIG. **18** illustrates a number of example fields that may be displayed to the user as part of the user interface **1800** and updated to include real-time, or near real-time, information. In the embodiment of FIG. **18**, the top portion **1802** includes a patient identification ("ID") field **1812** and a model field **1814**. The patient ID field **1812** includes alphanumeric characters that identify the patient being interrogated, such as the patient's name or an ID number associated with the patient. The model field **1814** includes an identification of the model of the neurostimulator **12** implanted in the patient.

The top portion **1802** also includes an implantable pulse generator ("IPG") serial number ("S/N") field **1816**, a lead impedance field **1818**, a pulse generator date field **1820**, and a pulse generator time field **1822**. The IPG S/N field **1816** includes a unique serial number corresponding to the neurostimulator **12** implanted in the patient. The lead impedance field **1818** identifies the impedance of the leads, such as the lead assembly **13**, implanted into the patient as part of the VNS system **11**. In some embodiments, the lead impedance field **1818** may identify the original lead impedance of the lead assembly **13**. In other embodiments, the neurostimulator **12** may be configured to test the current lead impedance of the lead assembly **13** (e.g., as the lead impedance may change over time due to fibrous tissue buildup around the lead assembly **13**), and the lead impedance field **1818** may accordingly identify the current impedance of the lead assembly **13** determined based on the testing. In various arrangements, the pulse generator date field **1820** identifies the date and the pulse generator time field **1822** identifies the time that the neurostimulator **12** was implanted and initially configured in the patient.

The user, such as a physician, may use these various fields to identify information about the VNS system **11**, and more specifically the neurostimulator **12** implanted in the patient. For example, the physician may use the serial number of the neurostimulator **12** implanted in the patient to determine the maximum VNS dose that the neurostimulator **12** is capable of delivering.

The top portion **1802** additionally includes a therapy on/off indicator **1824** and a titration on/off status indicator **1826**. The therapy on/off indicator **1824** is configured to display, based on the interrogated data from the neurostimulator **12**, whether therapy is currently being delivered to the patient by the neurostimulator **12**. For example, when therapy is on, the therapy on/off indicator **1824** is green and displays a "THERAPY ON" message. When the therapy is off, the therapy on/off indicator **1824** is red and displays a "THERAPY OFF" message. As an illustration, the therapy on/off indicator **1824** may be beneficial because it allows the physician to quickly verify that therapy is currently being delivered to the patient. If the therapy on/off indicator **1824** shows that the neurostimulator **12** is not delivering therapy to the patient, it may be an indication that something is wrong with the VNS system **11**, that the physician has forgotten to turn the therapy on, and so on. Accordingly, the physician may take steps to remedy the lack of therapy.

Similarly, the titration on/off status indicator **1826** is configured to display, based on the interrogated data from the neurostimulator **12**, whether the patient is currently being delivered VNS therapy according to a titration schedule. For example, when the titration is on, the titration on/off status indicator **1826** is green and displays an "ON" message, in some embodiments. When the titration is off, the titration on/off status indicator **1826** is red and displays an "OFF" message, in some embodiments. Similar to the therapy on/off indicator **1824**, the titration on/off status

indicator **1826** may be beneficial to a physician because it allows the physician to quickly verify whether the patient is currently on a titration schedule. If the titration on/off status indicator **1826** indicates that titration is off, depending on the patient's circumstances, the physician knows or can identify that titration has not yet started, that titration is completed, that the patient has reached a hold in the titration schedule, that there is an issue with the titration schedule programming, that the physician has pressed the stop titration button **1810**, and so on.

In some embodiments, the user interface **1800** includes a next titration field **1828**. As shown in FIG. **18**, the next titration field **1828** includes the date and time that the next titration step, as determined by the titration schedule for the patient, will be applied. In some embodiments, the next titration field **1828** may additionally include, for example, the size of the titration step and/or the amount of the VNS dose that will be applied to the patient once the next titration step occurs. In various arrangements, the next titration field **1828** is configured to reflect the date, time, dose, etc. of the next titration step automatically set by the presently selected titration period length and target VNS dose, as described in further detail below. Accordingly, the next titration field **1828** is not customizable by the user and instead reflects when the next stimulation parameter increase (e.g., a stimulation increase in amplitude, pulse width, and/or frequency increase) will occur according to the set schedule, in some implementations.

As shown in FIG. **18**, the middle portion **1803** includes a stimulation cycle indicator **1830**. As discussed above, the VNS system **11** delivers therapy to the patient according to a duty cycle including an ON period during which stimulation is delivered and an OFF period during which stimulation is suspended. As such, the stimulation cycle indicator **1830** displays where in the duty cycle the therapy currently being delivered to the patient is. In the example of FIG. **18**, the stimulation cycle indicator **1830** is configured to be a bar **1832** with an "OFF" section **1834** and an "ON" section **1836**. A marker **1838** slides along the bar **1832** from a left end to a right end of the bar, starting over at the left end once the marker **1838** reaches the right end of the bar **1832**, such that the marker **1838** is within the OFF section **1834** when the stimulation is suspended and within the ON section **1836** when the stimulation is being delivered. The movement of the indicator **1830** along the bar **1832** is shown across FIGS. **18-23**. However, in other embodiments, the stimulation cycle indicator **1830** may be configured differently from the indicator shown in FIGS. **18-23**. For example, the stimulation cycle indicator **1830** may be configured as a circle with a slice of the circle representing the ON period for the stimulation and the rest of the circle representing the OFF period for the stimulation. A marker then rotates around the circle to represent where in the stimulation cycle the VNS therapy currently being delivered to the patient is.

The middle portion **1803** also includes several sections displaying values relevant to the titration of the VNS therapy being delivered to the patient. In particular, in the embodiment of FIG. **18**, the user interface **1800** includes a start values section **1840**, a present values section **1842**, and a target values section **1844**. The start values section **1840** is configured to display values related to the starting VNS dose according to the titration program currently being delivered to the patient. The present values section **1842** is configured to display values related to the present VNS dose being delivered to the patient. The target values section **1844** is configured to display values related to the final VNS dose to be delivered to the patient once the titration is complete. In

the embodiment of FIG. **18**, each section **1840**, **1842**, and **1844** includes the amplitude (mA) of the VNS dose, the pulse width (μ s) of the VNS dose, and frequency (Hz) of the VNS dose. Additionally, the start values section **1840** includes the date the starting VNS dose was delivered or will start being delivered, as described in further detail below, and the target values section **1844** includes the date the titration will be completed.

In addition, the sections **1840**, **1842**, and **1844** are tied to a titration graph **1846** also included in the middle portion **1803**. An x-axis **1848** of the titration graph **1846** is marked in days, and a y-axis **1850** of the titration graph **1846** is marked in intensity. In various arrangements, the number of days for the therapy, as shown on the x-axis **1848**, depends on the length of the titration therapy scheduled for the patient. The user may schedule, or modify the schedule of, the length of the titration therapy using a titration period duration section **1852**. In the embodiment of FIG. **18**, the titration period duration section **1852** includes three radial buttons: a "shorter" button, a "nominal" button, and a "longer" button. The titration is automatically set to nominal, but a user may decrease the titration period (e.g., the amount of time over which the titration occurs) by pressing the shorter button or lengthen the titration period by pressing the longer button. As an example, the shorter button may correspond with a titration period of 30 days, the nominal button may correspond with a titration period of 45 days, and the longer period may correspond with a titration period of 60 days. In various embodiments, the shorter, nominal, and longer periods correspond with the high aggressiveness profile **1308**, the medium aggressiveness profile **1306**, and the low aggressiveness profile **1310**, respectively, discussed above with reference to FIG. **13**. As shown in FIG. **18**, the nominal button has been selected, and therefore the x-axis **1848** of the titration graph **1846** includes a 45-day period.

It should be understood that, in other embodiments, the titration period duration section **1852** may include additional or fewer options for the titration period. Further, in various arrangements, the user may only change the length of the titration by the set periods defined by the buttons or other indicators in the titration period duration section **1852**. However, the factory may be able to customize the lengths of the titration periods, as well as the titration schedule associated with each period.

The graph **1846** also includes a titration path **1854** with a first titration marker **1856** and a second titration marker **1858**. In various embodiments, the first titration marker **1856** and the second titration marker **1858** are configured to slide along a titration path **1854** representing a titration schedule, thereby allowing a user to set or modify the titration schedule for the patient. As described in further detail below, the first titration marker **1856** is tied with the start values section **1840** or the present values section **1842**, and the second titration marker **1858** is tied with the target values section **1844**.

As an illustration, FIG. **18** shows an example user interface **1800** populated with data interrogated from the neuro-stimulator **12** in the process of delivering therapy according to a titration schedule. As shown in FIG. **18**, the first marker **1856** corresponds to the present values section **1842** (e.g., the present values of the VNS dose being delivered to the patient) and thus is filled with a color or pattern corresponding to a color or pattern of the present values section **1842**. Similarly, the second marker **1858** corresponds to the target values section **1844** (e.g., the target values of the VNS dose

being delivered to the patient) and is filled with a color or pattern corresponding to a color or pattern of the target values section **1844**.

However, the user may modify the titration therapy by dragging the first marker **1856** and/or the second marker **1858** along the titration path **1854**, as shown in FIG. **19**. The user may drag the first marker **1856** along the titration path **1854** to increase or decrease the intensity of the VNS dose being currently delivered to the patient. For example, the patient may suffer from an unacceptable number of side effects at the present dose being delivered to the patient such that the patient's physician decides that the patient needs to be moved to a lower dose and have titration progress from the lower dose. Once the user drags the first marker **1856** along the titration path **1854**, the first marker **1856** changes to correspond with the start values section **1840**. Accordingly, as shown in FIG. **19**, the fill of the first marker **1856** changes to a color or pattern corresponding to a color or pattern of the start values section **1840**, and the start values section **1840** changes to display values related to the new dose selected by the user by dragging the first marker **1856**. In some arrangements, the user interface **1800** may also warn the user if the new selected start values represent a large VNS dose change from the present VNS dose. For example, the graph **1846** may display a warning to the user if the new selected start values represent more than a 10% increase in the VNS dose. As another example, the graph **1846** may display a pop-up warning to the user if the user selects the program titration button **1808** to implement new selected start values that represent more than a 10% increase in the VNS dose.

The user may also drag the second marker **1858** along the titration path **1854** to increase or decrease the intensity of the final VNS dose to be delivered to the patient. Once the user drags the second marker **1858** along the titration path **1854**, the target values section **1844** changes to display values related to the new final dose selected by the user by dragging the second marker **1858**. Additionally, the user may also modify the length of titration period by selecting a different radial button in the titration period duration section **1852**, as described above.

Once the user has modified the titration therapy, the user may implement the therapy by pressing the program titration button **1808**. For example, the monitoring device displaying the user interface **1800** determines a modified titration schedule for the patient based on the values input by the user via the graph **1846** and/or the titration period duration section **1852**, and the monitoring device provides the neurostimulator **12** with a titration program (e.g., via the programming wand **42**) including the modified titration schedule.

In some arrangements, upon receiving the titration program, the neurostimulator **12** immediately changes the present dose to the new starting dose. Accordingly, as shown in FIG. **20**, pressing the program titration button **1808** causes the first marker **1856** to revert to the color or pattern of the present values section **1842**. Additionally, the present values section **1842** is updated to reflect the values shown in the start values section **1840**, and the next titration field **1828** changes to reflect the next titration dose in the modified titration schedule. Alternatively, the date shown in the start values section **1840** may be a calendar function whereby the user may select a start date for the titration that is in the future. As such, if the user selects a future start date for the modified titration schedule, pressing the program titration button **1808** will not cause the first marker **1856**, present

values section **1842**, and next titration field **1828** to update to reflect the modified titration schedule, as described above, until the future date.

As another illustration, FIG. **21** shows an example user interface **1800** populated with data interrogated from the neurostimulator **12** not delivering therapy according to a titration schedule. Accordingly, the titration on/off status indicator **1826** reflects that the titration is off, and the next titration field **1828** is blank. Additionally, the graph **1846** is in an initial configuration, with the first marker **1856** and the second marker **1858** on the opposite ends of the titration path **1854** of the graph **1846**. To set up the titration, the user may drag the first marker **1856** and/or the second marker **1858** and/or select a different radial button in the titration period duration section **1852** to create a titration schedule. The user may then press the program titration button **1808** to transmit a program with the titration schedule to the neurostimulator **12**, as discussed above. As shown in FIG. **22**, the titration on/off status indicator **1826** then changes to reflect that the titration is on, and the next titration field **1828** changes to reflect the next titration step in the newly programmed titration schedule. Additionally, the start/stop titration button **1810** changes to show "stop titration."

As shown in FIGS. **18-22**, the titration graph **1846** additionally includes a therapy percentage indicator **1860**. The therapy percentage indicator **1860** illustrates the percentage of the current target VNS dose to the maximum VNS dose that the neurostimulator **12** may deliver to the patient (e.g., as prescribed by the factory settings for the neurostimulator **12**). The top right end of the titration path **1854** represents the maximum VNS dose such that if the second marker **1858** is on the top right end of the titration path **1854**, the therapy percentage indicator **1860** is at 100% as shown in FIG. **21**. Accordingly, if the user drags the second marker **1858** up or down along the titration path **1854**, the therapy percentage indicator **1860** updates to reflect the percentage of the currently selected target VNS dose to the maximum VNS dose, with the top right end of the titration path **1854** representing the maximum VNS dose. As such, the user may easily observe via the therapy percentage indicator **1860** whether the target VNS dose may be increased for the patient and by how much.

The titration graph **1846** also includes a first marker percentage **1862** and a second marker percentage **1864**. As shown in FIGS. **18-21**, the second marker percentage **1864** remains statically at 100% to reflect that the second marker **1858** corresponds with the final VNS dose to be delivered to the patient. However, the first marker percentage **1862** changes over time and/or changes if the user drags the first marker **1856** along the titration path **1854** to reflect the percentage of the present VNS dose/currently selected VNS starting dose to the target VNS dose. Further, the titration graph **1846** also include a completion date **1866** that reflects the day the titration will be completed according to the currently selected titration therapy schedule, as well as the number of days to completion.

Referring now to FIG. **23**, an example user interface **1800** populated with data interrogated from the neurostimulator **12** having completed the titration schedule is shown, according to an exemplary embodiment. As shown in FIG. **23**, the titration on/off status indicator **1826** reflects that the titration is off because the titration schedule has been completed, although in some arrangements, the titration on/off status indicator **1826** may instead show that the titration has been completed and/or change colors to indicate that the titration is complete. Similarly, the next titration field **1828** is blank. Additionally, as shown in FIG. **23**, the first marker **1856** and

the second marker **1858** have changed shape to indicate that the titration is complete. In the embodiment of FIG. **23**, the first marker **1856** and the second marker **1858** are shown as back-to-back triangles resting on the same point on the titration path **1854**. The completion date **1866** on the graph **1846** also indicates that the titration is complete. However, even after titration is complete, the user is still able to modify the VNS therapy by dragging the first marker **1856** and/or the second marker **1858**, as discussed above. For example, if the patient's physician decides that the patient needs a larger VNS dose, or the physician set the original target VNS dose as a hold, the physician may drag the second marker **1858** further up the titration path **1854** to modify the target VNS dose. Once the physician does so, the first marker **1856** and the second marker **1858** will revert to the non-completion shapes (e.g., the circular first marker **1856** and second marker **1858** shown in FIGS. **18-22**).

It should be understood, however, that FIGS. **18-23** are example embodiments of the user interface **1800**. Accordingly, in other embodiments, the style, shapes, layout, etc. of the components of the user interface **1800** may be altered. As an example, instead of the graph **1846**, the user interface **1800** may include input fields, bars that the user can raise or lower, and so on for the user to modify the titration schedule. Furthermore, in other embodiments, the user interface **1800** may include additional or fewer components. For example, in some arrangements, the graph **1846** of the user interface **1800** may include three separate markers, one of which corresponds to the start values section **1840**, one of which corresponds to the present values section **1842**, and one of which corresponds to the target values section **1844**.

Additionally, it should be understood that the arrangement of elements shown in FIGS. **18-23** is merely illustrative. Elements described, for example, as appearing within the top portion **1802**, middle portion **1803**, or bottom portion **1804** may appear on any portion of the user interface **1800** without departing from the teachings of the present disclosure.

FIG. **24** is a block diagram of an implanted neurostimulation system **2400** and management device **2407**, according to an exemplary embodiment. The implanted neurostimulation system **2400** includes a control system **2402** comprising a processor programmed to operate the system **2400**, a memory **2403**, an optional physiological sensor **2404**, a patient input sensor **2405**, and a stimulation subsystem **2406**. In various embodiments, the implanted neurostimulation system **2400** is configured similarly to the implanted neurostimulation system **700** shown in FIG. **7A** and described above.

As further shown in FIG. **24**, the management device **2407** includes a control system **2408** comprising a processor programmed to operate the management device **2407**, a memory **2409**, a display **2410**, and an input/output device **2411**. The control system **2408** operably coupled to the memory **2409**, which, in various embodiments, stores instructions that are executable by the processor of the control system **2408** to perform various functions described herein, including creating, managing, and displaying the user interface **1800** shown in FIGS. **18-23**. Additionally, the management device **2407** is further coupled to a programming wand **2412**. In various embodiments, the programming wand **2412** is configured similarly to the programming wand **42** described above with reference to FIG. **3** and may be physically (e.g., through a wired connection) or communicably (e.g., through a wireless connection such as RFID or Wi-Fi) coupled to the management device **2407**. Accordingly, the control system **2408** is further configured to

communicate with the implanted neurostimulation system **2400** via the programming wand **2412**. For example, the control system **2408** may interrogate and receive current stimulation parameters from the neurostimulation system **2400** via the programming wand **2412**. As another example, the control system **2408** may transmit a titration program to the neurostimulation system **2400** via the programming wand **2412**.

The display **2410** may include any system capable of presenting visual information to a user. For example, the display **2410** may be a screen, a touchscreen, and the like. Similarly, the input/output device **2411** may include any device capable of receiving information from a user and/or outputting information to a user. As an example, the input/output device **2411** may be or include a keyboard, a keypad, a mouse, a microphone, a speaker, a joystick, a printer, and the like. In some embodiments, the input/output device **2411** may include the display **2410**. As an illustration, the display **2410** may be a touchscreen whereby the management device **2407** may present visual information, such as the user interface **1800**, to the user and receive inputs from the users.

It should be understood, however, that the arrangement of an implanted neurostimulation system and a management device shown in FIG. **24** is an example arrangement. Those of skill in the art will appreciate that other arrangements may alternatively be used. For example, in other embodiments, the implanted neurostimulation system **2400** may be configured similarly to the implanted medical device **710** and the management device **2407** may be configured similarly to the external programmer **720**, which are described above with reference to FIG. **7B**. Additionally, in some embodiments, the management device **2407** may be configured similarly to the external programmer **40**, which is described above with reference to FIG. **3**.

FIG. **25** is a flowchart of a process **2500** of managing titration using a management device, such as management device **2407**, for a patient implanted with a neurostimulator, such as the implanted neurostimulation system **2400**. The management device **2407** receives, from the system **2400**, data for the patient relating to stimulation being provided to the patient by the system **2400** (e.g., while the system **2400** is in communication with the management device **2407**) at **2502**. In some embodiments, the management device **2407** receives the data via the programming wand **2412**, though in other embodiments, the management device **2407** may receive the data through a different method. For example, the management device **2407** may communicate with the system **2400** directly (e.g., via Wi-Fi) through a network interface. Additionally, in various embodiments, the data is received in real-time or near real-time (e.g., due to a delay in the communication between the management device **2407** and the system **2400**). Moreover, the data may be received automatically (e.g., in response to a user placing the programming wand **2412** near the patient) or in response to a command from a user (e.g., in response to the user pressing the interrogate button **1806**).

The management device **2407** then displays a user interface, such as the user interface **1800**, including at least a portion of the data displayed within a graph, such as the graph **1846**, at **2504**. Additionally, in various embodiments, the user interface **1800** allows a user to modify a parameter of the titration being applied by the system **2400**. For example, the user interface **1800** may allow a user to modify the titration by manipulating one or more markers included in the graph **1846**, as described above with reference to FIGS. **18-23**.

The management device **2407** receives a selection of a titration parameter at **2506**. As an illustration, the user moves the first marker **1856** and/or the second marker **1858** to modify the titration start values and/or target values, as described above with reference to FIGS. **18-23**. Based on the selection, the management device **2407** determines a titration schedule for the patient at **2508**. For example, in some embodiments, the management device **2407** matches the selected parameter(s) to a preprogrammed titration schedule from a set of preprogrammed titration schedules. In other embodiments, the management device **2407** generates a titration schedule based on the selected parameter(s).

Finally, the management device **2407** transmits the titration schedule to the neurostimulator at **2510**. In some embodiments, the management device **2407** waits to transmit the titration schedule until a user instructs the management device **2407** to transmit the schedule (e.g., by pressing the program titration button **1808**). In other embodiments, the management device **2407** transmits the titration schedule automatically (e.g., in response to the user selecting the titration parameter).

FIG. **26** shows another user interface **2600** for the VNS system **11**, according to an exemplary embodiment. Similar to the user interface **1800**, the user interface **2600** may be displayed to a user, such as a physician, on a display of a titration management device.

In the upper left corner, the user interface **2600** includes a generator and patient information box **2602** that includes information about the VNS system **11** and/or the patient implanted with the VNS system **11**. For example, in the embodiment of FIG. **26**, the box **2602** includes a patient ID and the model number, serial number, and lead impedance of the VNS system **11**. A titration on/off status indicator **2604** is positioned adjacent to the box **2602**. Similar to the titration on/off status indicator **1826**, the indicator **2604** is configured to display, based on interrogated data from the neurostimulator **12** of the VNS system **11**, whether the patient is currently being delivered VNS therapy according to a titration schedule. For example, the indicator **2604** is green and displays a "THERAPY ON" message when the titration is on.

In the upper right corner, the user interface **2600** includes a titration assist status indicator **2606**. The titration assist status indicator **2606** shows the status of a titration assist process, with the titration assist process facilitating the automatic up-titration of the therapy delivered by the neurostimulator **12** as discussed above. For example, the titration assist status indicator **2606** may display "ON" to indicate that the titration assist process is ongoing (e.g., as shown in FIG. **26**), "OFF" to indicate that the titration assist process is off, or "COMPLETE" to indicate that the titration assist process has been completed (e.g., the stimulation therapy being delivered to the patient has reached a target set of values). If the titration assist process is on, a next up-titration indicator **2608** beneath the titration assist status indicator **2606** displays the time and date of the next therapy increase (e.g., Dec. 12, 2018 at 13:01, in the embodiment of FIG. **26**).

Several buttons are located underneath the titration assist status indicator **2606** and the next up-titration indicator **2608**. These buttons include an interrogate button **2610**, a program titration button **2612**, and a stop titration button **2614**. The interrogate button **2610** is configured similarly to the interrogate button **1806** and is configured to interrogate the neurostimulator **12** to populate the user interface **2600** with information relating to the VNS therapy provided by the neurostimulator **12** in response to being pressed. The

program titration button **2612** is configured similarly to the program titration button **1808** and is configured to transmit a titration program (e.g., configured by the user via the user interface **2600**) to the neurostimulator **12** in response to being pressed. The stop titration button **2614** is configured to turn off the titration assist process in response to being pressed. In some embodiments, the stop titration button **2614** is also configured similarly to the start/stop titration button **1810** and therefore is also configured to show "start titration" (e.g., along with a green triangle) if titration is not currently being applied to a patient. In such embodiments, the user may also select the button **2614** to resume titration therapy. Alternatively, in other embodiments, the user may resume stopped titration by selecting the program titration button **2612**, which provides the titration schedule currently shown in the user interface **2600** to the neurostimulator **12**, with the neurostimulator **12** resuming titration. In some embodiments, in addition to being turned off in response to the user pressing the stop titration button **2614**, the titration assist process may also be automatically turned off after parameters are manually programmed into the neurostimulator **12** or a diagnostics process is performed for the VNS system **11**.

A stimulation cycle indicator **2616** is shown underneath the box **2602** and the titration on/off status indicator **2604**. The stimulation cycle indicator **2616** is configured similarly to the stimulation cycle indicator **1830**, displaying where in the duty cycle the therapy currently being delivered to the patient is, although the stimulation cycle indicator **2616** is configured as a circle instead of a bar. The stimulation cycle indicator **2616** includes an "OFF" section **2618** (e.g., colored in gray) and an "ON" section **2620** (e.g., colored in magenta). A marker **2622** (e.g., colored in blue to contrast with the gray and magenta) rotates around the circle (e.g., in a clockwise or counterclockwise direction) such that the marker **2622** is within the OFF section **2618** when the stimulation is suspended according to the duty cycle and within the ON section **2620** when stimulation is being delivered according to the duty cycle. Additionally, the stimulation cycle indicator **2616** indicates with text whether the stimulation is currently on or off. For example, the stimulation cycle indicator **2616** displays "Stimulation OFF" within the circle when the stimulation is off, as shown in the embodiment of FIG. **26**, and displays "Stimulation ON" within the circle when the stimulation is on. In some embodiments, the marker **2622** is accurately synchronized to the neurostimulator **12** for a certain period of time after the last interrogation of the neurostimulator **12** (e.g., up to ten minutes after the last interrogation).

The user interface **2600** also includes several sections displaying values relevant to the VNS therapy being delivered to the patient. Similar to the user interface **1800**, the user interface **2600** includes a present values section **2624**, a start values section **2626**, and a target values section **2628**. The present values section **2624** is configured similarly to the present values section **1842** and displays values related to the present VNS dose being delivered to the patient. The start values section **2626** is configured similarly to the **1842** and display values related to the starting VNS dose according to the titration program currently being delivered to the patient. The target values section **2628** is configured similarly to the target values section **1844** and displays values related to the final VNS dose to be delivered to the patient once the titration is complete. In the embodiment of FIG. **26**, the present values section **2624** includes the amplitude (mA), pulse width (μsec), frequency (Hz), and percentage of target values for the present dose; the start values section

2626 includes the amplitude (mA), pulse width (μ sec), frequency (Hz), and start date for the starting dose; and the target values section 2628 includes the amplitude (mA), pulse width (μ sec), and frequency (Hz) for the target dose.

Additionally, similar to the sections 1840, 1842, and 1844, the present values section 2624, start values section 2626, and target values section 2628 are tied to a titration graph 2630. As shown in FIG. 26, the titration graph 2630 is configured similarly to the titration graph 1846. An x-axis 2632 of the titration graph 2630 is marked in days (or other units of time), and a y-axis 2634 of the titration graph 2630 is marked in intensity. Moreover, similar to the user interface 1800, the user interface 2600 further includes a titration period duration section 2636 with a shorter button, a nominal button, and a longer button that the user can select to changes the length of the titration. In some embodiments, the nominal setting is default, but the user can modify the length of the titration at any time by selecting a different button. For example, the nominal setting may complete the titration in approximately 50 days, the shorter setting may complete the titration in approximately 35 days, and the longer setting may complete the titration in approximately 70 days. Selecting a different titration period duration in the titration period duration section 2636 may also change the number of days that are shown on the x-axis 2632 of the graph 2630 (e.g., as the titration period duration), as well as a target titration completion date 2645 shown in the graph 2630.

The graph 2630 includes a titration path 2638 with a first titration marker 2640, a second titration marker 2642, and third titration marker 2644. As shown in FIG. 26, the first marker 2640 corresponds to the start values section 2626 and thus is filled with a color or pattern corresponding to a color or pattern of the start values section 2626 (e.g., a yellow color). The second titration marker 2642 corresponds to the target values section 2628 and thus is filled with a color or pattern corresponding to a color or pattern of the target values section 2628 (e.g., a purple color). The third titration marker 2644 corresponds to the present values section 2624 and thus is filled with a color or pattern corresponding to a color or pattern of the present values section 2624 (e.g., a blue color). The third titration marker 2644 progressively increases along the titration path 2638 until the third titration marker 2644 reaches the target intensity (e.g., the second titration marker 2642).

Similar to the first marker 1856 and the second marker 1858, the first marker 2640 and the second marker 2642 may be dragged by the user along the titration path 2638 to modify the titration therapy. For example, the user may drag the first marker 2640 along the titration path 2638 to change the starting intensity of the titration therapy (e.g., which modifies the present values of the titration therapy if the modified titration program is transmitted to the neurostimulator 12 using the program titration button 2612). The user may also drag the second marker 2642 along the titration path 2638 to increase or decrease the intensity of the final VNS dose to be delivered to the patient. Dragging the second marker 2642 may also cause a change in a therapy percentage indicator 2646. Similar to the therapy percentage indicator 1860, the therapy percentage indicator 2646 illustrates the percentage of the current target VNS dose to the maximum dose that the neurostimulator 12 may deliver to the patient.

Additionally, in some embodiments, the automatic titration may be programmed to start on a future date. As an example, by default, the start date may be set to the current date, which results in immediate activation of the titration

assist process once the neurostimulator 12 is programmed using the program titration button 2612. However, the user may manually program the start date of the automatic titration using the start values section 2626. More specifically, as noted above, the start values section 2626 includes a start date for the titration, and the user can modify this start date by entering a new date into the start date box of the start values section 2626.

In the lower left corner, the user interface 2600 also includes several navigation buttons. In the embodiment of FIG. 26, the user interface 2600 includes a main menu button 2648, a support parameters button 2650, and a titration history button 2652. The user may select the main menu button 2648 to navigate to a main menu for the VNS system 11. FIG. 27 illustrates an example user interface 2700 displaying a main menu for the VNS system 11 that may be shown to the user in response to selection of the main menu button 2648, according to one embodiment. The user interface 2700 includes several buttons that the user can press to navigate to various functionalities of the VNS system 11, including an interrogate device button 2702, a view database button 2704, a view last parameters button 2706, and a user preferences button 2708. In some embodiments, the user can press the interrogate device button 2702 to establish communication with the neurostimulator 12 such that, once an interrogation is performed, the user may perform other tasks such as programming new parameters or performing diagnostics testing. As an example, the interrogate device button 2702 may function similarly to the interrogate button 2610. In some embodiments, the view database button 2704 allows the user to view all information from a previously recorded interrogation, programming, and diagnostic events performed by a titration management device (e.g., the external programmer 40). This option may not require real-time interaction with the neurostimulator 12. In some embodiments, the view last parameters button 2706 may be selected to view the most recent parameters selected for a specific patient or the VNS system 11 and also does not require real-time interaction with the neurostimulator 12. Finally, in some embodiments, the user preferences button 2708 allows the user to select values for a number of user settings. For example, the user settings may include the number of minutes that the titration management device may be idle before a new interrogation becomes necessary, the current date and time, the maximum number of records viewable at a time, a language selection (e.g., when available), a computer screen brightness, and a computer screen calibration.

Referring back to FIG. 26, the user may select the support parameters button 2650 to navigate to a support parameters screen. FIG. 28 illustrates an example user interface 2800 displaying a support parameters screen for the VNS system 11 that the user can navigate to by selecting the support parameters button 2650, according to one embodiment. The user interface 2800 includes a support parameters section 2802 including a number of support parameters that the user may adjust to modify the stimulation therapy, the titration process, and/or the operation of the titration management device. In the embodiment of FIG. 28, the support parameters section 2802 includes (1) a "TA Signal On Time (sec)" parameter that the user can adjust to modify the stimulation on time of the titration assist process duty cycle (e.g., signal ON time), (2) a "TA Signal Off Time (sec)" parameter that the user can also adjust to modify the stimulation off time of the titration assist process duty cycle (e.g., signal OFF time), (3) a "TA Rollback" parameter that the user may adjust to enable and adjust the rollback of the titration assist process,

where the rollback allows the patient to temporarily or permanently reduce stimulation intensity with an external device (e.g., the patient magnet **730**), and (4) an “Auto-Interrogate” parameter that the user may check to enable auto-interrogation, which causes the titration management device to automatically update the displayed titration assist values shown on the user interface **2600** after a titration assist programming event.

In addition, the user interface **2800** includes several navigation buttons: an “OK” button **2804**, an “Advanced” button **2806**, and a “Cancel” button **2808**. For example, the user may press the OK button **2804** to accept any changes made to the parameters in the support parameters section **2802** and return to the user interface **2600**. Conversely, the user may press the Cancel button **2808** to reject any changes made to the parameters in the support parameters section **2802** and return to the user interface **2600**. Finally, the user may press the Advanced button **2806** to navigate to an advanced parameters screen. In various arrangements, the advanced parameters screen is configured similarly to the user interface **2800** but instead includes a section showing advanced parameters that the user may adjust to modify the stimulation therapy, the titration process, and/or the operation of the titration management device. As an illustration, an advanced parameters screen may include (1) a “Sleep Inhibit Threshold (mA)” parameter that the user may adjust to set the amplitude of stimulation above which the neurostimulator **12** suspends the stimulation during the night, (2) a “To Bed Time (hr:min)” parameter that the user may adjust to set the clock time at which sleep inhibition (e.g., suspension or lowering of stimulation intensity while the patient is sleeping) begins, (3) a “Rollback Refractory (min)” parameter that the user may adjust to set the time after an external patient device activation (e.g., an activation by the patient magnet **730**) before the next external patient device activation can result in a rollback of the titration assist process, and (4) a “Days per Rollback” parameter that the patient may adjust to set the number of days that a titration assist rollback will reverse the titration to decrease the stimulation intensity.

Referring back to FIG. **26**, the user may select the titration history button **2652** to navigate to a titration history screen. As an illustration, FIG. **29** shows an example user interface **2900** displaying a titration history screen for the VNS system **11** that the user can navigate to by selecting the titration history button **2652**, according to one embodiment. In the upper left corner, the user interface **2900** includes the generator and patient information box **2602** and the titration on/off status indicator **2604**. On the right side, the user interface **2900** includes a titration history log **2902** showing a detailed log of events relating to the titration assist process. For example, the log **2902** includes the date and time of each interrogation of the neurostimulator **12** and each increase in the stimulation therapy intensity according to the titration schedule. The user interface **2900** also includes a progressive increase graph **2904**, which charts a line **2906** of progressive increase in stimulation amplitude (mA) over time.

At the bottom, the user interface **2900** includes several navigation buttons. In the embodiment of FIG. **29**, the user interface **2900** includes a main menu button **2908**, a titration progression screen button **2910**, a history interrogate button **2912**, and a save history button **2914**. In some embodiments, the main menu button **2908** is configured similarly to the main menu button **2648** (e.g., such that pressing the main menu button **2648** navigates the user to the user interface **2700** shown in FIG. **27**). In some embodiments, the titration progression screen button **2910** is configured to navigate the

user back to the user interface **2600** in response to being selected. In some embodiments, the history interrogate button **2912** is configured to navigate the user to a more detailed screen showing a history of interrogations with the neurostimulator **12** in response to being selected. Finally, in some embodiments, the save history button **2914** is configured to save the log of events to a flash drive in response to being selected.

It should be understood that FIGS. **26-29** are example embodiments of user interfaces. In other embodiments, the style, shapes, layout, etc. of the components of the user interfaces shown may be altered. Further, in other embodiments, the user interfaces may include additional or fewer components, as well as additional or fewer functionalities.

While embodiments have been particularly shown and described, those skilled in the art will understand that the foregoing and other changes in form and detail may be made therein without departing from the spirit and scope. For example, in various embodiments described above, the stimulation is applied to the vagus nerve. Alternatively, spinal cord stimulation (SCS) may be used in place of or in addition to vagus nerve stimulation for the above-described therapies. SCS may utilize stimulating electrodes implanted in the epidural space, an electrical pulse generator implanted in the lower abdominal area or gluteal region, and conducting wires coupling the stimulating electrodes to the generator.

The present disclosure contemplates methods, systems, and program products on any machine-readable media for accomplishing various operations. The embodiments of the present disclosure may be implemented using existing computer processors, or by a special purpose computer processor for an appropriate system, incorporated for this or another purpose, or by a hardwired system. Embodiments within the scope of the present disclosure include program products comprising machine-readable media for carrying or having machine-executable instructions or data structures stored thereon. Such machine-readable media can be any available media that can be accessed by a general purpose or special purpose computer or other machine with a processor. By way of example, such machine-readable media can comprise RAM, ROM, EPROM, EEPROM, CD-ROM or other optical disk storage, magnetic disk storage or other magnetic storage devices, or any other medium which can be used to carry or store desired program code in the form of machine-executable instructions or data structures and which can be accessed by a general purpose or special purpose computer or other machine with a processor. When information is transferred or provided over a network or another communications connection (either hardwired, wireless, or a combination of hardwired or wireless) to a machine, the machine properly views the connection as a machine-readable medium. Thus, any such connection is properly termed a machine-readable medium. Combinations of the above are also included within the scope of machine-readable media. Machine-executable instructions include, for example, instructions and data which cause a general purpose computer, special purpose computer, or special purpose processing machines to perform a certain function or group of functions.

Although the figures may show a specific order of method steps, the order of the steps may differ from what is depicted. Also two or more steps may be performed concurrently or with partial concurrence. Such variation will depend on the software and hardware systems chosen and on designer choice. All such variations are within the scope of the disclosure. Likewise, software implementations could be

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accomplished with standard programming techniques with rule based logic and other logic to accomplish the various connection steps, processing steps, comparison steps and decision steps.

Also, the term “or” is used in its inclusive sense (and not in its exclusive sense) so that when used, for example, to connect a list of elements, the term “or” means one, some, or all of the elements in the list. Conjunctive language such as the phrase “at least one of X, Y, and Z,” unless specifically stated otherwise, is otherwise understood with the context as used in general to convey that an item, term, etc. may be either X, Y, Z, X and Y, X and Z, Y and Z, or X, Y, and Z. Thus, such conjunctive language is not generally intended to imply that certain embodiments require at least one of X, at least one of Y, and at least one of Z to each be present.

While various aspects and embodiments have been disclosed herein, other aspects and embodiments will be apparent to those skilled in the art. The various aspects and embodiments disclosed herein are for purposes of illustration and are not intended to be limiting, with the true scope and spirit being indicated by the following claims.

What is claimed is:

1. A method of managing titration using a management device for a patient implanted with a neurostimulator, comprising:

receiving, from the neurostimulator, data including a current titration schedule for the patient relating to stimulation being provided to the patient by the neurostimulator while the neurostimulator is in communication with the management device;

displaying a user interface on a display of the management device, the user interface including at least a portion of the current titration schedule displayed within a graph, the user interface allowing a user to modify a parameter of the titration;

receiving a selection of the parameter via the user interface;

based on the selection, generating a modified titration schedule for the patient; and

transmitting the modified titration schedule to the neurostimulator.

2. The method of claim 1, wherein receiving the data for the patient comprises receiving, from the neurostimulator, the data for the patient in real-time.

3. The method of claim 1, wherein the user interface allows a user to modify at least one of starting stimulation values, target stimulation values, or a period for the titration.

4. The method of claim 3, wherein the graph allows the user to modify the starting stimulation values by dragging a first marker along a path and modify the target stimulation values by dragging a second marker along a path.

5. The method of claim 1, wherein receiving the selection of the parameter comprises receiving the selection of the parameter from the user via the graph.

6. The method of claim 1, wherein the user interface further includes:

a first indicator reflecting that, based on the data, therapy is currently being provided to the patient; and

a second indicator reflecting that, based on the data, the patient is currently on the current titration schedule.

7. The method of claim 1, wherein the user interface further includes a field indicating when a next titration step will be applied to the patient.

8. The method of claim 1, wherein the stimulation provided by the neurostimulator is applied according to a duty cycle comprising an on period and an off period; and

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wherein the user interface further includes a stimulation cycle indicator configured to display whether, based on the data, the stimulation being applied to the patient is currently in the on period or the off period.

9. A management device for managing titration for a patient implanted with a neurostimulator, comprising:

a display;

a processor; and

a memory storing instructions that, when executed by the processor, cause the management device to:

receive, from the neurostimulator, data including a current titration schedule for the patient relating to stimulation being provided to the patient by the neurostimulator while the neurostimulator is in communication with the management device;

display, by the display, a user interface including at least a portion of the current titration schedule displayed within a graph, the user interface allowing a user to modify a parameter of the titration;

receive a selection of the parameter;

based on the selection, generate a modified titration schedule for the patient; and

transmit the modified titration schedule to the neurostimulator.

10. The management device of claim 9, wherein the instructions cause the management device to receive, from the neurostimulator, the data for the patient in real-time.

11. The management device of claim 9, wherein the user interface allows a user to modify at least one of starting stimulation values, target stimulation values, or a period for the titration.

12. The management device of claim 11, wherein the graph allows the user to modify the starting stimulation values by dragging a first marker along a path and modify the target stimulation values by dragging a second marker along the path.

13. The management device of claim 9, wherein the instructions cause the management device to receive the selection of the parameter from the user via the graph.

14. The management device of claim 9, wherein the user interface further includes:

a first indicator reflecting that, based on the data, therapy is currently being provided to the patient; and

a second indicator reflecting that, based on the data, the patient is currently on the current titration schedule.

15. The management device of claim 9, wherein the user interface further includes a field indicating when a next titration step will be applied to the patient.

16. The management device of claim 9, wherein the stimulation provided by the neurostimulator is applied according to a duty cycle comprising an on period and an off period; and

wherein the user interface further includes a stimulation cycle indicator configured to display whether, based on the data, the stimulation being applied to the patient is currently in the on period or the off period.

17. The management device of claim 9, wherein the user interface further includes an indicator configured to display a percentage of the currently selected target stimulation values to maximum stimulation values that the neurostimulator is able to provide to the patient.

18. A non-transitory computer-readable medium comprising instructions executable by a processor to:

receive, from a neurostimulator, data including a current titration schedule for a patient relating to stimulation being provided to the patient by the neurostimulator;

display a user interface on a display, the user interface including at least a portion of the current titration schedule displayed within a graph, the user interface allowing a user to modify a parameter of titration; receive a selection of the parameter; 5
based on the selection, generate a modified titration schedule for the patient; and
transmit the modified titration schedule to the neurostimulator.

19. The non-transitory computer-readable medium of claim 18, wherein the instructions are executable by the processor to receive, from the neurostimulator, the data for the patient in real-time. 10

20. The non-transitory computer-readable medium of claim 18, wherein the user interface further includes: 15
a first indicator reflecting that, based on the real-time data, therapy is currently being provided to the patient; and
a second indicator reflecting that, based on the real-time data, the patient is currently on the current titration schedule. 20

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